PLAGIARISMAI

Document integrity and similarity report

REVIEW 2 IEEE PAPER.docx

Submission ID	d3ffb66d-695a-4fc8-b1f1-27fdb39d9bab
Original file	REVIEW 2 IEEE PAPER.docx
Submission date	2026-11-10T07:36:23.804277+00:00
Report generated	2026-11-10T07:37:38+00:00
File size	2,652,357 bytes
Pages	Not available
Word count	4,463
Character count	35,000

Integrity overview

100.0%

Overall similarity reflects unique sentence-level direct and near-duplicate matches in the authorized source corpus. Semantic similarity is reported separately and is not added to this percentage.

Metric	Actual result
Matching sources	8
Matching passages	2472
Direct matches	1828
Near-duplicate matches	644
Corpus scope	available user document corpus

Match groups and applied filters

Match group	Result
Not cited or quoted	Unavailable
Missing quotations	Unavailable
Missing citation	Unavailable
Cited and quoted	Unavailable

Unavailable: citation and quotation analysis is not configured.

Applied filters

[]

Filters are recorded explicitly. No matching text is silently excluded by this report.

Top sources

#	Source	Type	Passages
1	ARPITHA_MEDARAMETLA_BACKEND_DEVELOPER__RESUME.pdf	Uploaded document	4
2	ARPITHA_MEDARAMETLA_BACKEND_DEVELOPER__RESUME.pdf	Uploaded document	4
3	ARPITHA_MEDARAMETLA_BACKEND_DEVELOPER__RESUME.pdf	Uploaded document	4
4	ARPITHA_MEDARAMETLA_BACKEND_DEVELOPER__RESUME.pdf	Uploaded document	4
5	REVIEW 2 IEEE PAPER.docx	Uploaded document	614
6	REVIEW 2 IEEE PAPER.docx	Uploaded document	614
7	REVIEW 2 IEEE PAPER.docx	Uploaded document	614
8	REVIEW 2 IEEE PAPER.docx	Uploaded document	614

Source coverage

Category	Availability	Sources
Internet sources	No source database configured	0
Publications	No source database configured	0
Submitted works	No source database configured	0
Uploaded documents	Configured	8

Detailed matching

1. Direct match | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

2. Direct match | Candidate sentence 1 | Source sentence 1

Submitted text	Source passage
Mrs .	Mrs .

3. Direct match | Candidate sentence 2 | Source sentence 2

Submitted text	Source passage
K .	K .

4. Direct match | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

5. Direct match | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

6. Direct match | Candidate sentence 5 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

7. Direct match | Candidate sentence 6 | Source sentence 6

Submitted text	Source passage
astakalanagamani78@gmail.com	astakalanagamani78@gmail.com

8. Direct match | Candidate sentence 7 | Source sentence 7

Submitted text	Source passage
P.	P.

9. Direct match | Candidate sentence 8 | Source sentence 8

Submitted text	Source passage
Srinath	Srinath

10. Direct match | Candidate sentence 9 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

11. Direct match | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

12. Direct match | Candidate sentence 11 | Source sentence 11

Submitted text	Source passage
MLR Institute of Technology	MLR Institute of Technology

13. Direct match | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

14. Direct match | Candidate sentence 13 | Source sentence 13

Submitted text	Source passage
23r21a05y0@mlrit.ac.in	23r21a05y0@mlrit.ac.in

15. Direct match | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

16. Direct match | Candidate sentence 15 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

17. Direct match | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

18. Direct match | Candidate sentence 17 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

19. Direct match | Candidate sentence 18 | Source sentence 18

Submitted text	Source passage
23r21a05t0@mlrit.ac.in	23r21a05t0@mlrit.ac.in

20. Direct match | Candidate sentence 19 | Source sentence 19

Submitted text	Source passage
V.	V.

21. Direct match | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

22. Direct match | Candidate sentence 21 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

23. Direct match | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

24. Direct match | Candidate sentence 23 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

25. Direct match | Candidate sentence 24 | Source sentence 24

Submitted text	Source passage
24r21a0532@mlrit.ac.in	24r21a0532@mlrit.ac.in

26. Direct match | Candidate sentence 25 | Source sentence 25

Submitted text	Source passage
ABSTRACT	ABSTRACT

27. Direct match | Candidate sentence 26 | Source sentence 26

Submitted text	Source passage
Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.	Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.

28. Direct match | Candidate sentence 27 | Source sentence 27

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.	StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.

29. Direct match | Candidate sentence 28 | Source sentence 28

Submitted text	Source passage
It combines message-based event ingestion, stream processing, and persistent state management.	It combines message-based event ingestion, stream processing, and persistent state management.

30. Direct match | Candidate sentence 29 | Source sentence 29

Submitted text	Source passage
The processing pipeline supports filtering, transformation, aggregation, and event-time operations.	The processing pipeline supports filtering, transformation, aggregation, and event-time operations.

31. Direct match | Candidate sentence 30 | Source sentence 30

Submitted text	Source passage
An LSM-tree-based state store manages persistent processing state.	An LSM-tree-based state store manages persistent processing state.

32. Direct match | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

33. Direct match | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

34. Direct match | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

35. Direct match | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

36. Direct match | Candidate sentence 35 | Source sentence 35

Submitted text	Source passage
The architecture is intended to support low-latency analytical queries and continuous data processing.	The architecture is intended to support low-latency analytical queries and continuous data processing.

37. Direct match | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

38. Direct match | Candidate sentence 37 | Source sentence 37

Submitted text	Source passage
StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.	StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.

39. Direct match | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

40. Direct match | Candidate sentence 39 | Source sentence 39

Submitted text	Source passage
The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.	The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.

41. Direct match | Candidate sentence 40 | Source sentence 40

Submitted text	Source passage
Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.	Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.

42. Direct match | Candidate sentence 41 | Source sentence 41

Submitted text	Source passage
INTRODUCTION	INTRODUCTION

43. Direct match | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

44. Direct match | Candidate sentence 43 | Source sentence 43

Submitted text	Source passage
However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.	However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.

45. Direct match | Candidate sentence 44 | Source sentence 44

Submitted text	Source passage
These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.	These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.

46. Direct match | Candidate sentence 45 | Source sentence 45

Submitted text	Source passage
Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].	Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].

47. Direct match | Candidate sentence 46 | Source sentence 46

Submitted text	Source passage
A streaming materialized-view engine can maintain precomputed query results as new events arrive.	A streaming materialized-view engine can maintain precomputed query results as new events arrive.

48. Direct match | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

49. Direct match | Candidate sentence 48 | Source sentence 48

Submitted text	Source passage
Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].	Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].

50. Direct match | Candidate sentence 49 | Source sentence 49

Submitted text	Source passage
This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.	This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.

51. Direct match | Candidate sentence 50 | Source sentence 50

Submitted text	Source passage
The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.	The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.

52. Direct match | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

53. Direct match | Candidate sentence 52 | Source sentence 52

Submitted text	Source passage
Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.	Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.

54. Direct match | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

55. Direct match | Candidate sentence 54 | Source sentence 227

Submitted text	Source passage
1.	1.

56. Direct match | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

57. Direct match | Candidate sentence 56 | Source sentence 56

Submitted text	Source passage
However, maintaining consistent analytical results under high event volumes remains a challenge.	However, maintaining consistent analytical results under high event volumes remains a challenge.

58. Direct match | Candidate sentence 57 | Source sentence 229

Submitted text	Source passage
2.	2.

59. Direct match | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

60. Direct match | Candidate sentence 59 | Source sentence 59

Submitted text	Source passage
However, complex aggregations and large-scale workloads can increase update overhead.	However, complex aggregations and large-scale workloads can increase update overhead.

61. Direct match | Candidate sentence 60 | Source sentence 231

Submitted text	Source passage
3.	3.

62. Direct match | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

63. Direct match | Candidate sentence 62 | Source sentence 62

Submitted text	Source passage
However, coordination, network communication, and fault recovery can introduce additional overhead.	However, coordination, network communication, and fault recovery can introduce additional overhead.

64. Direct match | Candidate sentence 63 | Source sentence 233

Submitted text	Source passage
4.	4.

65. Direct match | Candidate sentence 64 | Source sentence 64

Submitted text	Source passage
Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.	Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.

66. Direct match | Candidate sentence 65 | Source sentence 65

Submitted text	Source passage
However, selecting appropriate watermark policies can affect result completeness and processing latency.	However, selecting appropriate watermark policies can affect result completeness and processing latency.

67. Direct match | Candidate sentence 66 | Source sentence 235

Submitted text	Source passage
5.	5.

68. Direct match | Candidate sentence 67 | Source sentence 67

Submitted text	Source passage
Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.	Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.

69. Direct match | Candidate sentence 68 | Source sentence 68

Submitted text	Source passage
However, storage performance and recovery mechanisms can influence overall system efficiency.	However, storage performance and recovery mechanisms can influence overall system efficiency.

70. Direct match | Candidate sentence 69 | Source sentence 237

Submitted text	Source passage
6.	6.

71. Direct match | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

72. Direct match | Candidate sentence 71 | Source sentence 71

Submitted text	Source passage
However, log management and recovery procedures can increase storage and processing overhead.	However, log management and recovery procedures can increase storage and processing overhead.

73. Direct match | Candidate sentence 72 | Source sentence 239

Submitted text	Source passage
7.	7.

74. Direct match | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

75. Direct match | Candidate sentence 74 | Source sentence 74

Submitted text	Source passage
However, continuous index updates can increase memory consumption and computational costs.	However, continuous index updates can increase memory consumption and computational costs.

76. Direct match | Candidate sentence 75 | Source sentence 241

Submitted text	Source passage
8.	8.

77. Direct match | Candidate sentence 76 | Source sentence 76

Submitted text	Source passage
Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.	Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.

78. Direct match | Candidate sentence 77 | Source sentence 77

Submitted text	Source passage
However, maintaining reliable integrations requires ongoing API updates and webhook management.	However, maintaining reliable integrations requires ongoing API updates and webhook management.

79. Direct match | Candidate sentence 78 | Source sentence 243

Submitted text	Source passage
9.	9.

80. Direct match | Candidate sentence 79 | Source sentence 79

Submitted text	Source passage
AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.	AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.

81. Direct match | Candidate sentence 80 | Source sentence 80

Submitted text	Source passage
However, recommendations may become outdated when priorities and team workloads change.	However, recommendations may become outdated when priorities and team workloads change.

82. Direct match | Candidate sentence 81 | Source sentence 245

Submitted text	Source passage
10.	10.

83. Direct match | Candidate sentence 82 | Source sentence 82

Submitted text	Source passage
Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.	Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.

84. Direct match | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

85. Direct match | Candidate sentence 84 | Source sentence 247

Submitted text	Source passage
11.	11.

86. Direct match | Candidate sentence 85 | Source sentence 85

Submitted text	Source passage
Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.	Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.

87. Direct match | Candidate sentence 86 | Source sentence 86

Submitted text	Source passage
However, integrating documentation with daily development activities remains challenging.	However, integrating documentation with daily development activities remains challenging.

88. Direct match | Candidate sentence 87 | Source sentence 249

Submitted text	Source passage
12.	12.

89. Direct match | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

90. Direct match | Candidate sentence 89 | Source sentence 89

Submitted text	Source passage
However, organizations must address data privacy, access control, and data-residency requirements.	However, organizations must address data privacy, access control, and data-residency requirements.

91. Direct match | Candidate sentence 90 | Source sentence 251

Submitted text	Source passage
13.	13.

92. Direct match | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

93. Direct match | Candidate sentence 92 | Source sentence 92

Submitted text	Source passage
However, complex debugging tasks involving multiple codebases may require additional contextual information.	However, complex debugging tasks involving multiple codebases may require additional contextual information.

94. Direct match | Candidate sentence 93 | Source sentence 253

Submitted text	Source passage
14.	14.

95. Direct match | Candidate sentence 94 | Source sentence 94

Submitted text	Source passage
Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.	Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.

96. Direct match | Candidate sentence 95 | Source sentence 95

Submitted text	Source passage
However, high query volumes can increase memory consumption and affect write performance.	However, high query volumes can increase memory consumption and affect write performance.

97. Direct match | Candidate sentence 96 | Source sentence 255

Submitted text	Source passage
15.	15.

98. Direct match | Candidate sentence 97 | Source sentence 97

Submitted text	Source passage
Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.	Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.

99. Direct match | Candidate sentence 98 | Source sentence 98

Submitted text	Source passage
However, their effectiveness depends on the availability of accurate historical data.	However, their effectiveness depends on the availability of accurate historical data.

100. Direct match | Candidate sentence 99 | Source sentence 257

Submitted text	Source passage
16.	16.

101. Direct match | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

102. Direct match | Candidate sentence 101 | Source sentence 101

Submitted text	Source passage
However, limited historical activity can reduce recommendation accuracy for new team members.	However, limited historical activity can reduce recommendation accuracy for new team members.

103. Direct match | Candidate sentence 102 | Source sentence 102

Submitted text	Source passage
17.	17.

104. Direct match | Candidate sentence 103 | Source sentence 103

Submitted text	Source passage
Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.	Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.

105. Direct match | Candidate sentence 104 | Source sentence 104

Submitted text	Source passage
However, generated summaries may contain inaccuracies when project context or code versions are outdated.	However, generated summaries may contain inaccuracies when project context or code versions are outdated.

106. Direct match | Candidate sentence 105 | Source sentence 105

Submitted text	Source passage
18.	18.

107. Direct match | Candidate sentence 106 | Source sentence 106

Submitted text	Source passage
AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.	AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.

108. Direct match | Candidate sentence 107 | Source sentence 107

Submitted text	Source passage
However, integrating different permission models and data structures introduces administrative complexity.	However, integrating different permission models and data structures introduces administrative complexity.

109. Direct match | Candidate sentence 108 | Source sentence 108

Submitted text	Source passage
19.	19.

110. Direct match | Candidate sentence 109 | Source sentence 109

Submitted text	Source passage
Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.	Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.

111. Direct match | Candidate sentence 110 | Source sentence 110

Submitted text	Source passage
However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.	However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.

112. Direct match | Candidate sentence 111 | Source sentence 111

Submitted text	Source passage
20.	20.

113. Direct match | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

114. Direct match | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

115. Direct match | Candidate sentence 114 | Source sentence 114

Submitted text	Source passage
However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.	However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.

116. Direct match | Candidate sentence 115 | Source sentence 115

Submitted text	Source passage
Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.	Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.

117. Direct match | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

118. Direct match | Candidate sentence 117 | Source sentence 117

Submitted text	Source passage
Do not present unverified titles or numerical claims as published research.	Do not present unverified titles or numerical claims as published research.

119. Direct match | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

120. Direct match | Candidate sentence 119 | Source sentence 119

Submitted text	Source passage
Fragmented Data Processing	Fragmented Data Processing

121. Direct match | Candidate sentence 120 | Source sentence 120

Submitted text	Source passage
Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.	Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.

122. Direct match | Candidate sentence 121 | Source sentence 121

Submitted text	Source passage
Limited Incremental Computation	Limited Incremental Computation

123. Direct match | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

124. Direct match | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

125. Direct match | Candidate sentence 124 | Source sentence 124

Submitted text	Source passage
Inefficient Query Execution	Inefficient Query Execution

126. Direct match | Candidate sentence 125 | Source sentence 125

Submitted text	Source passage
Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.	Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.

127. Direct match | Candidate sentence 126 | Source sentence 126

Submitted text	Source passage
Complex Infrastructure Management	Complex Infrastructure Management

128. Direct match | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

129. Direct match | Candidate sentence 128 | Source sentence 128

Submitted text	Source passage
Limited Processing Visibility	Limited Processing Visibility

130. Direct match | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

131. Direct match | Candidate sentence 130 | Source sentence 130

Submitted text	Source passage
Challenges in State Recovery	Challenges in State Recovery

132. Direct match | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

133. Direct match | Candidate sentence 132 | Source sentence 132

Submitted text	Source passage
Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.	Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.

134. Direct match | Candidate sentence 133 | Source sentence 133

Submitted text	Source passage
Resource Utilization and Scalability	Resource Utilization and Scalability

135. Direct match | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

136. Direct match | Candidate sentence 135 | Source sentence 135

Submitted text	Source passage
Efficient resource utilization and scalable processing are essential for maintaining performance.	Efficient resource utilization and scalable processing are essential for maintaining performance.

137. Direct match | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

138. Direct match | Candidate sentence 137 | Source sentence 137

Submitted text	Source passage
Processing a continuously increasing volume of events while maintaining low latency is challenging.	Processing a continuously increasing volume of events while maintaining low latency is challenging.

139. Direct match | Candidate sentence 138 | Source sentence 138

Submitted text	Source passage
Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.	Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.

140. Direct match | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

141. Direct match | Candidate sentence 140 | Source sentence 140

Submitted text	Source passage
Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.	Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.

142. Direct match | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

143. Direct match | Candidate sentence 142 | Source sentence 142

Submitted text	Source passage
Limited Benchmarking of Self-Hosted Solutions	Limited Benchmarking of Self-Hosted Solutions

144. Direct match | Candidate sentence 143 | Source sentence 143

Submitted text	Source passage
The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.	The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.

145. Direct match | Candidate sentence 144 | Source sentence 144

Submitted text	Source passage
Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.	Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.

146. Direct match | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

147. Direct match | Candidate sentence 146 | Source sentence 146

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.	Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.

148. Direct match | Candidate sentence 147 | Source sentence 147

Submitted text	Source passage
Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.	Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.

149. Direct match | Candidate sentence 148 | Source sentence 148

Submitted text	Source passage
These limitations make it challenging to obtain up-to-date analytical results as new events arrive.	These limitations make it challenging to obtain up-to-date analytical results as new events arrive.

150. Direct match | Candidate sentence 149 | Source sentence 149

Submitted text	Source passage
Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.	Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.

151. Direct match | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

152. Direct match | Candidate sentence 151 | Source sentence 151

Submitted text	Source passage
It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.	It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.

153. Direct match | Candidate sentence 152 | Source sentence 152

Submitted text	Source passage
Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.	Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.

154. Direct match | Candidate sentence 153 | Source sentence 153

Submitted text	Source passage
By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.	By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.

155. Direct match | Candidate sentence 154 | Source sentence 154

Submitted text	Source passage
MOTIVATION	MOTIVATION

156. Direct match | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

157. Direct match | Candidate sentence 156 | Source sentence 156

Submitted text	Source passage
However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.	However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.

158. Direct match | Candidate sentence 157 | Source sentence 157

Submitted text	Source passage
As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.	As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.

159. Direct match | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

160. Direct match | Candidate sentence 159 | Source sentence 159

Submitted text	Source passage
StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.	StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.

161. Direct match | Candidate sentence 160 | Source sentence 160

Submitted text	Source passage
By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.	By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.

162. Direct match | Candidate sentence 161 | Source sentence 161

Submitted text	Source passage
Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.	Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.

163. Direct match | Candidate sentence 162 | Source sentence 162

Submitted text	Source passage
Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.	Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.

164. Direct match | Candidate sentence 163 | Source sentence 163

Submitted text	Source passage
StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.	StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.

165. Direct match | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

166. Direct match | Candidate sentence 165 | Source sentence 165

Submitted text	Source passage
Develop a Self-Hosted Streaming Engine	Develop a Self-Hosted Streaming Engine

167. Direct match | Candidate sentence 166 | Source sentence 166

Submitted text	Source passage
Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.	Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.

168. Direct match | Candidate sentence 167 | Source sentence 167

Submitted text	Source passage
The system aims to simplify real-time data processing through a unified architecture.	The system aims to simplify real-time data processing through a unified architecture.

169. Direct match | Candidate sentence 168 | Source sentence 168

Submitted text	Source passage
Implement Continuous Stream Processing	Implement Continuous Stream Processing

170. Direct match | Candidate sentence 169 | Source sentence 169

Submitted text	Source passage
Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.	Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.

171. Direct match | Candidate sentence 170 | Source sentence 170

Submitted text	Source passage
The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.	The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.

172. Direct match | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

173. Direct match | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

174. Direct match | Candidate sentence 173 | Source sentence 173

Submitted text	Source passage
This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.	This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.

175. Direct match | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

176. Direct match | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

177. Direct match | Candidate sentence 176 | Source sentence 176

Submitted text	Source passage
Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.	Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.

178. Direct match | Candidate sentence 177 | Source sentence 177

Submitted text	Source passage
Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.	Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.

179. Direct match | Candidate sentence 178 | Source sentence 178

Submitted text	Source passage
As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.	As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.

180. Direct match | Candidate sentence 179 | Source sentence 179

Submitted text	Source passage
Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.	Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.

181. Direct match | Candidate sentence 180 | Source sentence 180

Submitted text	Source passage
Periodic refreshes may prevent users from accessing the latest information immediately.	Periodic refreshes may prevent users from accessing the latest information immediately.

182. Direct match | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

183. Direct match | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

184. Direct match | Candidate sentence 183 | Source sentence 183

Submitted text	Source passage
StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.	StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.

185. Direct match | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

186. Direct match | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

187. Direct match | Candidate sentence 186 | Source sentence 186

Submitted text	Source passage
The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.	The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.

188. Direct match | Candidate sentence 187 | Source sentence 187

Submitted text	Source passage
Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.	Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.

189. Direct match | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

190. Direct match | Candidate sentence 189 | Source sentence 189

Submitted text	Source passage
StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.	StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.

191. Direct match | Candidate sentence 190 | Source sentence 190

Submitted text	Source passage
Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.	Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.

192. Direct match | Candidate sentence 191 | Source sentence 191

Submitted text	Source passage
The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results	The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results

193. Direct match | Candidate sentence 192 | Source sentence 192

Submitted text	Source passage
.	.

194. Direct match | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

195. Direct match | Candidate sentence 194 | Source sentence 194

Submitted text	Source passage
A persistent state store maintains the data required for processing and view updates.	A persistent state store maintains the data required for processing and view updates.

196. Direct match | Candidate sentence 195 | Source sentence 195

Submitted text	Source passage
Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.	Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.

197. Direct match | Candidate sentence 196 | Source sentence 196

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.	Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.

198. Direct match | Candidate sentence 197 | Source sentence 197

Submitted text	Source passage
The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.	The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.

199. Direct match | Candidate sentence 198 | Source sentence 198

Submitted text	Source passage
Backend APIs connect the dashboard with the processing engine and state store.	Backend APIs connect the dashboard with the processing engine and state store.

200. Direct match | Candidate sentence 199 | Source sentence 199

Submitted text	Source passage
REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.	REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.

201. Direct match | Candidate sentence 200 | Source sentence 200

Submitted text	Source passage
Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.	Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.

202. Direct match | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

203. Direct match | Candidate sentence 202 | Source sentence 202

Submitted text	Source passage
The following steps should be taken while using StreamView:	The following steps should be taken while using StreamView:

204. Direct match | Candidate sentence 203 | Source sentence 203

Submitted text	Source passage
Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.	Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.

205. Direct match | Candidate sentence 204 | Source sentence 204

Submitted text	Source passage
Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.	Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.

206. Direct match | Candidate sentence 205 | Source sentence 205

Submitted text	Source passage
Step 3: Ingest Events – Send events from a producer application to the configured message broker.	Step 3: Ingest Events – Send events from a producer application to the configured message broker.

207. Direct match | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

208. Direct match | Candidate sentence 207 | Source sentence 207

Submitted text	Source passage
Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.	Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.

209. Direct match | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

210. Direct match | Candidate sentence 209 | Source sentence 209

Submitted text	Source passage
The engine processes incoming events and maintains the corresponding results.	The engine processes incoming events and maintains the corresponding results.

211. Direct match | Candidate sentence 210 | Source sentence 210

Submitted text	Source passage
Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.	Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.

212. Direct match | Candidate sentence 211 | Source sentence 211

Submitted text	Source passage
Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.	Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.

213. Direct match | Candidate sentence 212 | Source sentence 212

Submitted text	Source passage
Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.	Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.

214. Direct match | Candidate sentence 213 | Source sentence 213

Submitted text	Source passage
Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.	Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.

215. Direct match | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

216. Direct match | Candidate sentence 215 | Source sentence 215

Submitted text	Source passage
FIG 8.1 Proposed workflow Diagram	FIG 8.1 Proposed workflow Diagram

217. Direct match | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

218. Direct match | Candidate sentence 217 | Source sentence 217

Submitted text	Source passage
Users can observe system performance and identify processing delays or potential bottlenecks.	Users can observe system performance and identify processing delays or potential bottlenecks.

219. Direct match | Candidate sentence 218 | Source sentence 218

Submitted text	Source passage
Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.	Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.

220. Direct match | Candidate sentence 219 | Source sentence 219

Submitted text	Source passage
The system applies the updated configurations based on its supported configuration and execution mechanisms.	The system applies the updated configurations based on its supported configuration and execution mechanisms.

221. Direct match | Candidate sentence 220 | Source sentence 220

Submitted text	Source passage
Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.	Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.

222. Direct match | Candidate sentence 221 | Source sentence 221

Submitted text	Source passage
These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.	These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.

223. Direct match | Candidate sentence 222 | Source sentence 222

Submitted text	Source passage
ALGORITHM	ALGORITHM

224. Direct match | Candidate sentence 223 | Source sentence 223

Submitted text	Source passage
The following steps describe the processing workflow of the StreamView application.	The following steps describe the processing workflow of the StreamView application.

225. Direct match | Candidate sentence 224 | Source sentence 224

Submitted text	Source passage
Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline	Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline

226. Direct match | Candidate sentence 225 | Source sentence 225

Submitted text	Source passage
Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)	Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)

227. Direct match | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

228. Direct match | Candidate sentence 227 | Source sentence 227

Submitted text	Source passage
1.	1.

229. Direct match | Candidate sentence 228 | Source sentence 228

Submitted text	Source passage
Initialize the message broker, stream-processing engine, persistent state store, and query interface.	Initialize the message broker, stream-processing engine, persistent state store, and query interface.

230. Direct match | Candidate sentence 229 | Source sentence 229

Submitted text	Source passage
2.	2.

231. Direct match | Candidate sentence 230 | Source sentence 230

Submitted text	Source passage
Configure the event stream and subscribe to the required message topic.	Configure the event stream and subscribe to the required message topic.

232. Direct match | Candidate sentence 231 | Source sentence 231

Submitted text	Source passage
3.	3.

233. Direct match | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

234. Direct match | Candidate sentence 233 | Source sentence 233

Submitted text	Source passage
4.	4.

235. Direct match | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

236. Direct match | Candidate sentence 235 | Source sentence 235

Submitted text	Source passage
5.	5.

237. Direct match | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

238. Direct match | Candidate sentence 237 | Source sentence 237

Submitted text	Source passage
6.	6.

239. Direct match | Candidate sentence 238 | Source sentence 238

Submitted text	Source passage
Apply the configured filtering and transformation operations to \(e\).	Apply the configured filtering and transformation operations to \(e\).

240. Direct match | Candidate sentence 239 | Source sentence 239

Submitted text	Source passage
7.	7.

241. Direct match | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

242. Direct match | Candidate sentence 241 | Source sentence 241

Submitted text	Source passage
8.	8.

243. Direct match | Candidate sentence 242 | Source sentence 242

Submitted text	Source passage
Identify the materialized views affected by the processed event.	Identify the materialized views affected by the processed event.

244. Direct match | Candidate sentence 243 | Source sentence 243

Submitted text	Source passage
9.	9.

245. Direct match | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

246. Direct match | Candidate sentence 245 | Source sentence 245

Submitted text	Source passage
10.	10.

247. Direct match | Candidate sentence 246 | Source sentence 246

Submitted text	Source passage
Persist the updated state and view data using the supported storage mechanism.	Persist the updated state and view data using the supported storage mechanism.

248. Direct match | Candidate sentence 247 | Source sentence 247

Submitted text	Source passage
11.	11.

249. Direct match | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

250. Direct match | Candidate sentence 249 | Source sentence 249

Submitted text	Source passage
12.	12.

251. Direct match | Candidate sentence 250 | Source sentence 250

Submitted text	Source passage
On receiving a query \(Q\) from the client interface:	On receiving a query \(Q\) from the client interface:

252. Direct match | Candidate sentence 251 | Source sentence 251

Submitted text	Source passage
13.	13.

253. Direct match | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

254. Direct match | Candidate sentence 253 | Source sentence 253

Submitted text	Source passage
14.	14.

255. Direct match | Candidate sentence 254 | Source sentence 254

Submitted text	Source passage
Return the query results \(R\) to the client interface.	Return the query results \(R\) to the client interface.

256. Direct match | Candidate sentence 255 | Source sentence 255

Submitted text	Source passage
15.	15.

257. Direct match | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

258. Direct match | Candidate sentence 257 | Source sentence 257

Submitted text	Source passage
16.	16.

259. Direct match | Candidate sentence 258 | Source sentence 258

Submitted text	Source passage
Repeat the process until the stream is stopped or the system is shut down.	Repeat the process until the stream is stopped or the system is shut down.

260. Direct match | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

261. Direct match | Candidate sentence 260 | Source sentence 260

Submitted text	Source passage
Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.	Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.

262. Direct match | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

263. Direct match | Candidate sentence 262 | Source sentence 262

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.	StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.

264. Direct match | Candidate sentence 263 | Source sentence 263

Submitted text	Source passage
The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.	The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.

265. Direct match | Candidate sentence 264 | Source sentence 264

Submitted text	Source passage
By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.	By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.

266. Direct match | Candidate sentence 265 | Source sentence 265

Submitted text	Source passage
From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.	From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.

267. Direct match | Candidate sentence 266 | Source sentence 266

Submitted text	Source passage
The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.	The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.

268. Direct match | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

269. Direct match | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

270. Direct match | Candidate sentence 269 | Source sentence 269

Submitted text	Source passage
Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.	Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.

271. Direct match | Candidate sentence 270 | Source sentence 270

Submitted text	Source passage
This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.	This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.

272. Direct match | Candidate sentence 271 | Source sentence 271

Submitted text	Source passage
The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.	The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.

273. Direct match | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

274. Direct match | Candidate sentence 273 | Source sentence 273

Submitted text	Source passage
Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.	Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.

275. Direct match | Candidate sentence 274 | Source sentence 274

Submitted text	Source passage
By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.	By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.

276. Direct match | Candidate sentence 275 | Source sentence 275

Submitted text	Source passage
StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.	StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.

277. Direct match | Candidate sentence 276 | Source sentence 276

Submitted text	Source passage
It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.	It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.

278. Direct match | Candidate sentence 277 | Source sentence 277

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing repeated computation.	Materialized views are updated incrementally as new events arrive, reducing repeated computation.

279. Direct match | Candidate sentence 278 | Source sentence 278

Submitted text	Source passage
The dashboard enables users to manage streams, execute queries, and monitor system performance.	The dashboard enables users to manage streams, execute queries, and monitor system performance.

280. Direct match | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

281. Direct match | Candidate sentence 280 | Source sentence 280

Submitted text	Source passage
The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.	The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.

282. Direct match | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

283. Direct match | Candidate sentence 282 | Source sentence 282

Submitted text	Source passage
The application backend manages business logic, API requests, stream configurations, and communication between components.	The application backend manages business logic, API requests, stream configurations, and communication between components.

284. Direct match | Candidate sentence 283 | Source sentence 283

Submitted text	Source passage
The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.	The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.

285. Direct match | Candidate sentence 284 | Source sentence 284

Submitted text	Source passage
The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.	The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.

286. Direct match | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

287. Direct match | Candidate sentence 286 | Source sentence 286

Submitted text	Source passage
These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.	These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.

288. Direct match | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

289. Direct match | Candidate sentence 288 | Source sentence 288

Submitted text	Source passage
RESULTS	RESULTS

290. Direct match | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

291. Direct match | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

292. Direct match | Candidate sentence 291 | Source sentence 291

Submitted text	Source passage
The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.	The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.

293. Direct match | Candidate sentence 292 | Source sentence 292

Submitted text	Source passage
The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.	The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.

294. Direct match | Candidate sentence 293 | Source sentence 293

Submitted text	Source passage
These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.	These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.

295. Direct match | Candidate sentence 294 | Source sentence 294

Submitted text	Source passage
The monitoring and event ingestion modules provide additional visibility into system operations.	The monitoring and event ingestion modules provide additional visibility into system operations.

296. Direct match | Candidate sentence 295 | Source sentence 295

Submitted text	Source passage
Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.	Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.

297. Direct match | Candidate sentence 296 | Source sentence 296

Submitted text	Source passage
The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.	The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.

298. Direct match | Candidate sentence 297 | Source sentence 297

Submitted text	Source passage
These measurements help evaluate system performance and identify potential processing bottlenecks.	These measurements help evaluate system performance and identify potential processing bottlenecks.

299. Direct match | Candidate sentence 298 | Source sentence 298

Submitted text	Source passage
Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.	Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.

300. Direct match | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

301. Direct match | Candidate sentence 300 | Source sentence 300

Submitted text	Source passage
FIG 12.1: Result	FIG 12.1: Result

302. Direct match | Candidate sentence 301 | Source sentence 301

Submitted text	Source passage
By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.	By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.

303. Direct match | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

304. Direct match | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

305. Direct match | Candidate sentence 304 | Source sentence 304

Submitted text	Source passage
In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.	In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.

306. Direct match | Candidate sentence 305 | Source sentence 305

Submitted text	Source passage
Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.	Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.

307. Direct match | Candidate sentence 306 | Source sentence 306

Submitted text	Source passage
Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.	Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.

308. Direct match | Candidate sentence 307 | Source sentence 307

Submitted text	Source passage
Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.	Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.

309. Direct match | Candidate sentence 308 | Source sentence 308

Submitted text	Source passage
Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.	Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.

310. Direct match | Candidate sentence 309 | Source sentence 309

Submitted text	Source passage
Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.	Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.

311. Direct match | Candidate sentence 310 | Source sentence 310

Submitted text	Source passage
Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.	Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.

312. Direct match | Candidate sentence 311 | Source sentence 311

Submitted text	Source passage
CONCLUSION	CONCLUSION

313. Direct match | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

314. Direct match | Candidate sentence 313 | Source sentence 313

Submitted text	Source passage
Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.	Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.

315. Direct match | Candidate sentence 314 | Source sentence 314

Submitted text	Source passage
The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.	The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.

316. Direct match | Candidate sentence 315 | Source sentence 315

Submitted text	Source passage
Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.	Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.

317. Direct match | Candidate sentence 316 | Source sentence 316

Submitted text	Source passage
This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.	This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.

318. Direct match | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

319. Direct match | Candidate sentence 318 | Source sentence 318

Submitted text	Source passage
REFERENCES	REFERENCES

320. Direct match | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

321. Direct match | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

322. Direct match | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

323. Direct match | Candidate sentence 322 | Source sentence 322

Submitted text	Source passage
Kalavri, and A.	Kalavri, and A.

324. Direct match | Candidate sentence 323 | Source sentence 323

Submitted text	Source passage
Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.	Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.

325. Direct match | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

326. Direct match | Candidate sentence 325 | Source sentence 325

Submitted text	Source passage
507–541, 2024.	507–541, 2024.

327. Direct match | Candidate sentence 326 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

328. Direct match | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

329. Direct match | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

330. Direct match | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

331. Direct match | Candidate sentence 330 | Source sentence 330

Submitted text	Source passage
Golab, and M.	Golab, and M.

332. Direct match | Candidate sentence 331 | Source sentence 331

Submitted text	Source passage
T.	T.

333. Direct match | Candidate sentence 332 | Source sentence 332

Submitted text	Source passage
Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.	Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.

334. Direct match | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

335. Direct match | Candidate sentence 334 | Source sentence 334

Submitted text	Source passage
55–67, 2017.	55–67, 2017.

336. Direct match | Candidate sentence 335 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

337. Direct match | Candidate sentence 336 | Source sentence 336

Submitted text	Source passage
Available: https://doi.org/10.1016/j.is.2017.07.002	Available: https://doi.org/10.1016/j.is.2017.07.002

338. Direct match | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

339. Direct match | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

340. Direct match | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

341. Direct match | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

342. Direct match | Candidate sentence 341 | Source sentence 341

Submitted text	Source passage
Ryzhyk, and V.	Ryzhyk, and V.

343. Direct match | Candidate sentence 342 | Source sentence 342

Submitted text	Source passage
Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.	Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.

344. Direct match | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

345. Direct match | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

346. Direct match | Candidate sentence 345 | Source sentence 345

Submitted text	Source passage
1601–1614, 2023.	1601–1614, 2023.

347. Direct match | Candidate sentence 346 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

348. Direct match | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

349. Direct match | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

350. Direct match | Candidate sentence 349 | Source sentence 349

Submitted text	Source passage
Gupta and I.	Gupta and I.

351. Direct match | Candidate sentence 350 | Source sentence 359

Submitted text	Source passage
S.	S.

352. Direct match | Candidate sentence 351 | Source sentence 351

Submitted text	Source passage
Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.	Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.

353. Direct match | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

354. Direct match | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

355. Direct match | Candidate sentence 354 | Source sentence 354

Submitted text	Source passage
3–18, 1995.	3–18, 1995.

356. Direct match | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

357. Direct match | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

358. Direct match | Candidate sentence 357 | Source sentence 359

Submitted text	Source passage
S.	S.

359. Direct match | Candidate sentence 358 | Source sentence 358

Submitted text	Source passage
Mumick, and V.	Mumick, and V.

360. Direct match | Candidate sentence 359 | Source sentence 359

Submitted text	Source passage
S.	S.

361. Direct match | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

362. Direct match | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

363. Direct match | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

364. Direct match | Candidate sentence 363 | Source sentence 363

Submitted text	Source passage
157–166, 1993.	157–166, 1993.

365. Direct match | Candidate sentence 364 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

366. Direct match | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

367. Direct match | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

368. Direct match | Candidate sentence 367 | Source sentence 420

Submitted text	Source passage
A.	A.

369. Direct match | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

370. Direct match | Candidate sentence 369 | Source sentence 369

Submitted text	Source passage
Srivastava, and S.	Srivastava, and S.

371. Direct match | Candidate sentence 370 | Source sentence 370

Submitted text	Source passage
Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.	Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.

372. Direct match | Candidate sentence 371 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

373. Direct match | Candidate sentence 372 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

374. Direct match | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

375. Direct match | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

376. Direct match | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

377. Direct match | Candidate sentence 376 | Source sentence 376

Submitted text	Source passage
Ceri and J.	Ceri and J.

378. Direct match | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

379. Direct match | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

380. Direct match | Candidate sentence 379 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

381. Direct match | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

382. Direct match | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

383. Direct match | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

384. Direct match | Candidate sentence 383 | Source sentence 420

Submitted text	Source passage
A.	A.

385. Direct match | Candidate sentence 384 | Source sentence 384

Submitted text	Source passage
Blakeley, P.-Å.	Blakeley, P.-Å.

386. Direct match | Candidate sentence 385 | Source sentence 385

Submitted text	Source passage
Larson, and F.	Larson, and F.

387. Direct match | Candidate sentence 386 | Source sentence 386

Submitted text	Source passage
W.	W.

388. Direct match | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

389. Direct match | Candidate sentence 388 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

390. Direct match | Candidate sentence 389 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

391. Direct match | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

392. Direct match | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

393. Direct match | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

394. Direct match | Candidate sentence 393 | Source sentence 393

Submitted text	Source passage
Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.	Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.

395. Direct match | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

396. Direct match | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

397. Direct match | Candidate sentence 396 | Source sentence 396

Submitted text	Source passage
1792–1803, 2015.	1792–1803, 2015.

398. Direct match | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

399. Direct match | Candidate sentence 398 | Source sentence 398

Submitted text	Source passage
Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.	Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.

400. Direct match | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

401. Direct match | Candidate sentence 400 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

402. Direct match | Candidate sentence 401 | Source sentence 401

Submitted text	Source passage
Very Large Data Bases (VLDB), 2013.	Very Large Data Bases (VLDB), 2013.

403. Direct match | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

404. Direct match | Candidate sentence 403 | Source sentence 403

Submitted text	Source passage
Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.	Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.

405. Direct match | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

406. Direct match | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

407. Direct match | Candidate sentence 406 | Source sentence 406

Submitted text	Source passage
28–38, 2015.	28–38, 2015.

408. Direct match | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

409. Direct match | Candidate sentence 408 | Source sentence 408

Submitted text	Source passage
Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.	Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.

410. Direct match | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

411. Direct match | Candidate sentence 410 | Source sentence 410

Submitted text	Source passage
Operating Systems Principles (SOSP), 2013, pp.	Operating Systems Principles (SOSP), 2013, pp.

412. Direct match | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

413. Direct match | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

414. Direct match | Candidate sentence 413 | Source sentence 413

Submitted text	Source passage
Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.	Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.

415. Direct match | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

416. Direct match | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

417. Direct match | Candidate sentence 416 | Source sentence 416

Submitted text	Source passage
3135–3147, 2021.	3135–3147, 2021.

418. Direct match | Candidate sentence 417 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

419. Direct match | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

420. Direct match | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

421. Direct match | Candidate sentence 420 | Source sentence 420

Submitted text	Source passage
A.	A.

422. Direct match | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

423. Direct match | Candidate sentence 422 | Source sentence 422

Submitted text	Source passage
M.	M.

424. Direct match | Candidate sentence 423 | Source sentence 423

Submitted text	Source passage
Hellerstein, and S.	Hellerstein, and S.

425. Direct match | Candidate sentence 424 | Source sentence 424

Submitted text	Source passage
Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.	Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.

426. Direct match | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

427. Direct match | Candidate sentence 426 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

428. Direct match | Candidate sentence 427 | Source sentence 427

Submitted text	Source passage
Data Engineering (ICDE), 2003.	Data Engineering (ICDE), 2003.

429. Direct match | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

430. Direct match | Candidate sentence 429 | Source sentence 429

Submitted text	Source passage
Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.	Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.

431. Direct match | Candidate sentence 430 | Source sentence 430

Submitted text	Source passage
1st Biennial Conf.	1st Biennial Conf.

432. Direct match | Candidate sentence 431 | Source sentence 431

Submitted text	Source passage
Innovative Data Systems Research (CIDR), 2003.	Innovative Data Systems Research (CIDR), 2003.

433. Direct match | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

434. Direct match | Candidate sentence 433 | Source sentence 433

Submitted text	Source passage
J.	J.

435. Direct match | Candidate sentence 434 | Source sentence 434

Submitted text	Source passage
Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.	Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.

436. Direct match | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

437. Direct match | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

438. Direct match | Candidate sentence 437 | Source sentence 437

Submitted text	Source passage
120–139, 2003.	120–139, 2003.

439. Direct match | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

440. Direct match | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

441. Direct match | Candidate sentence 440 | Source sentence 440

Submitted text	Source passage
Babu, and J.	Babu, and J.

442. Direct match | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

443. Direct match | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

444. Direct match | Candidate sentence 443 | Source sentence 443

Submitted text	Source passage
121–142, 2006.	121–142, 2006.

445. Direct match | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

446. Direct match | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

447. Direct match | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

448. Direct match | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

449. Direct match | Candidate sentence 448 | Source sentence 448

Submitted text	Source passage
42–47, 2005.	42–47, 2005.

450. Direct match | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

451. Direct match | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

452. Direct match | Candidate sentence 451 | Source sentence 451

Submitted text	Source passage
4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.	4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.

453. Direct match | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

454. Direct match | Candidate sentence 453 | Source sentence 453

Submitted text	Source passage
Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.	Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.

455. Direct match | Candidate sentence 454 | Source sentence 454

Submitted text	Source passage
34, article 39, 2025.	34, article 39, 2025.

456. Direct match | Candidate sentence 455 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

457. Direct match | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

458. Direct match | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

459. Direct match | Candidate sentence 1 | Source sentence 1

Submitted text	Source passage
Mrs .	Mrs .

460. Direct match | Candidate sentence 2 | Source sentence 2

Submitted text	Source passage
K .	K .

461. Direct match | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

462. Direct match | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

463. Direct match | Candidate sentence 5 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

464. Direct match | Candidate sentence 6 | Source sentence 6

Submitted text	Source passage
astakalanagamani78@gmail.com	astakalanagamani78@gmail.com

465. Direct match | Candidate sentence 7 | Source sentence 7

Submitted text	Source passage
P.	P.

466. Direct match | Candidate sentence 8 | Source sentence 8

Submitted text	Source passage
Srinath	Srinath

467. Direct match | Candidate sentence 9 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

468. Direct match | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

469. Direct match | Candidate sentence 11 | Source sentence 11

Submitted text	Source passage
MLR Institute of Technology	MLR Institute of Technology

470. Direct match | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

471. Direct match | Candidate sentence 13 | Source sentence 13

Submitted text	Source passage
23r21a05y0@mlrit.ac.in	23r21a05y0@mlrit.ac.in

472. Direct match | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

473. Direct match | Candidate sentence 15 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

474. Direct match | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

475. Direct match | Candidate sentence 17 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

476. Direct match | Candidate sentence 18 | Source sentence 18

Submitted text	Source passage
23r21a05t0@mlrit.ac.in	23r21a05t0@mlrit.ac.in

477. Direct match | Candidate sentence 19 | Source sentence 19

Submitted text	Source passage
V.	V.

478. Direct match | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

479. Direct match | Candidate sentence 21 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

480. Direct match | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

481. Direct match | Candidate sentence 23 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

482. Direct match | Candidate sentence 24 | Source sentence 24

Submitted text	Source passage
24r21a0532@mlrit.ac.in	24r21a0532@mlrit.ac.in

483. Direct match | Candidate sentence 25 | Source sentence 25

Submitted text	Source passage
ABSTRACT	ABSTRACT

484. Direct match | Candidate sentence 26 | Source sentence 26

Submitted text	Source passage
Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.	Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.

485. Direct match | Candidate sentence 27 | Source sentence 27

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.	StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.

486. Direct match | Candidate sentence 28 | Source sentence 28

Submitted text	Source passage
It combines message-based event ingestion, stream processing, and persistent state management.	It combines message-based event ingestion, stream processing, and persistent state management.

487. Direct match | Candidate sentence 29 | Source sentence 29

Submitted text	Source passage
The processing pipeline supports filtering, transformation, aggregation, and event-time operations.	The processing pipeline supports filtering, transformation, aggregation, and event-time operations.

488. Direct match | Candidate sentence 30 | Source sentence 30

Submitted text	Source passage
An LSM-tree-based state store manages persistent processing state.	An LSM-tree-based state store manages persistent processing state.

489. Direct match | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

490. Direct match | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

491. Direct match | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

492. Direct match | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

493. Direct match | Candidate sentence 35 | Source sentence 35

Submitted text	Source passage
The architecture is intended to support low-latency analytical queries and continuous data processing.	The architecture is intended to support low-latency analytical queries and continuous data processing.

494. Direct match | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

495. Direct match | Candidate sentence 37 | Source sentence 37

Submitted text	Source passage
StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.	StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.

496. Direct match | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

497. Direct match | Candidate sentence 39 | Source sentence 39

Submitted text	Source passage
The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.	The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.

498. Direct match | Candidate sentence 40 | Source sentence 40

Submitted text	Source passage
Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.	Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.

499. Direct match | Candidate sentence 41 | Source sentence 41

Submitted text	Source passage
INTRODUCTION	INTRODUCTION

500. Direct match | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

501. Direct match | Candidate sentence 43 | Source sentence 43

Submitted text	Source passage
However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.	However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.

502. Direct match | Candidate sentence 44 | Source sentence 44

Submitted text	Source passage
These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.	These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.

503. Direct match | Candidate sentence 45 | Source sentence 45

Submitted text	Source passage
Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].	Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].

504. Direct match | Candidate sentence 46 | Source sentence 46

Submitted text	Source passage
A streaming materialized-view engine can maintain precomputed query results as new events arrive.	A streaming materialized-view engine can maintain precomputed query results as new events arrive.

505. Direct match | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

506. Direct match | Candidate sentence 48 | Source sentence 48

Submitted text	Source passage
Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].	Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].

507. Direct match | Candidate sentence 49 | Source sentence 49

Submitted text	Source passage
This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.	This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.

508. Direct match | Candidate sentence 50 | Source sentence 50

Submitted text	Source passage
The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.	The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.

509. Direct match | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

510. Direct match | Candidate sentence 52 | Source sentence 52

Submitted text	Source passage
Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.	Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.

511. Direct match | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

512. Direct match | Candidate sentence 54 | Source sentence 227

Submitted text	Source passage
1.	1.

513. Direct match | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

514. Direct match | Candidate sentence 56 | Source sentence 56

Submitted text	Source passage
However, maintaining consistent analytical results under high event volumes remains a challenge.	However, maintaining consistent analytical results under high event volumes remains a challenge.

515. Direct match | Candidate sentence 57 | Source sentence 229

Submitted text	Source passage
2.	2.

516. Direct match | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

517. Direct match | Candidate sentence 59 | Source sentence 59

Submitted text	Source passage
However, complex aggregations and large-scale workloads can increase update overhead.	However, complex aggregations and large-scale workloads can increase update overhead.

518. Direct match | Candidate sentence 60 | Source sentence 231

Submitted text	Source passage
3.	3.

519. Direct match | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

520. Direct match | Candidate sentence 62 | Source sentence 62

Submitted text	Source passage
However, coordination, network communication, and fault recovery can introduce additional overhead.	However, coordination, network communication, and fault recovery can introduce additional overhead.

521. Direct match | Candidate sentence 63 | Source sentence 233

Submitted text	Source passage
4.	4.

522. Direct match | Candidate sentence 64 | Source sentence 64

Submitted text	Source passage
Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.	Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.

523. Direct match | Candidate sentence 65 | Source sentence 65

Submitted text	Source passage
However, selecting appropriate watermark policies can affect result completeness and processing latency.	However, selecting appropriate watermark policies can affect result completeness and processing latency.

524. Direct match | Candidate sentence 66 | Source sentence 235

Submitted text	Source passage
5.	5.

525. Direct match | Candidate sentence 67 | Source sentence 67

Submitted text	Source passage
Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.	Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.

526. Direct match | Candidate sentence 68 | Source sentence 68

Submitted text	Source passage
However, storage performance and recovery mechanisms can influence overall system efficiency.	However, storage performance and recovery mechanisms can influence overall system efficiency.

527. Direct match | Candidate sentence 69 | Source sentence 237

Submitted text	Source passage
6.	6.

528. Direct match | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

529. Direct match | Candidate sentence 71 | Source sentence 71

Submitted text	Source passage
However, log management and recovery procedures can increase storage and processing overhead.	However, log management and recovery procedures can increase storage and processing overhead.

530. Direct match | Candidate sentence 72 | Source sentence 239

Submitted text	Source passage
7.	7.

531. Direct match | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

532. Direct match | Candidate sentence 74 | Source sentence 74

Submitted text	Source passage
However, continuous index updates can increase memory consumption and computational costs.	However, continuous index updates can increase memory consumption and computational costs.

533. Direct match | Candidate sentence 75 | Source sentence 241

Submitted text	Source passage
8.	8.

534. Direct match | Candidate sentence 76 | Source sentence 76

Submitted text	Source passage
Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.	Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.

535. Direct match | Candidate sentence 77 | Source sentence 77

Submitted text	Source passage
However, maintaining reliable integrations requires ongoing API updates and webhook management.	However, maintaining reliable integrations requires ongoing API updates and webhook management.

536. Direct match | Candidate sentence 78 | Source sentence 243

Submitted text	Source passage
9.	9.

537. Direct match | Candidate sentence 79 | Source sentence 79

Submitted text	Source passage
AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.	AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.

538. Direct match | Candidate sentence 80 | Source sentence 80

Submitted text	Source passage
However, recommendations may become outdated when priorities and team workloads change.	However, recommendations may become outdated when priorities and team workloads change.

539. Direct match | Candidate sentence 81 | Source sentence 245

Submitted text	Source passage
10.	10.

540. Direct match | Candidate sentence 82 | Source sentence 82

Submitted text	Source passage
Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.	Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.

541. Direct match | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

542. Direct match | Candidate sentence 84 | Source sentence 247

Submitted text	Source passage
11.	11.

543. Direct match | Candidate sentence 85 | Source sentence 85

Submitted text	Source passage
Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.	Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.

544. Direct match | Candidate sentence 86 | Source sentence 86

Submitted text	Source passage
However, integrating documentation with daily development activities remains challenging.	However, integrating documentation with daily development activities remains challenging.

545. Direct match | Candidate sentence 87 | Source sentence 249

Submitted text	Source passage
12.	12.

546. Direct match | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

547. Direct match | Candidate sentence 89 | Source sentence 89

Submitted text	Source passage
However, organizations must address data privacy, access control, and data-residency requirements.	However, organizations must address data privacy, access control, and data-residency requirements.

548. Direct match | Candidate sentence 90 | Source sentence 251

Submitted text	Source passage
13.	13.

549. Direct match | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

550. Direct match | Candidate sentence 92 | Source sentence 92

Submitted text	Source passage
However, complex debugging tasks involving multiple codebases may require additional contextual information.	However, complex debugging tasks involving multiple codebases may require additional contextual information.

551. Direct match | Candidate sentence 93 | Source sentence 253

Submitted text	Source passage
14.	14.

552. Direct match | Candidate sentence 94 | Source sentence 94

Submitted text	Source passage
Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.	Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.

553. Direct match | Candidate sentence 95 | Source sentence 95

Submitted text	Source passage
However, high query volumes can increase memory consumption and affect write performance.	However, high query volumes can increase memory consumption and affect write performance.

554. Direct match | Candidate sentence 96 | Source sentence 255

Submitted text	Source passage
15.	15.

555. Direct match | Candidate sentence 97 | Source sentence 97

Submitted text	Source passage
Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.	Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.

556. Direct match | Candidate sentence 98 | Source sentence 98

Submitted text	Source passage
However, their effectiveness depends on the availability of accurate historical data.	However, their effectiveness depends on the availability of accurate historical data.

557. Direct match | Candidate sentence 99 | Source sentence 257

Submitted text	Source passage
16.	16.

558. Direct match | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

559. Direct match | Candidate sentence 101 | Source sentence 101

Submitted text	Source passage
However, limited historical activity can reduce recommendation accuracy for new team members.	However, limited historical activity can reduce recommendation accuracy for new team members.

560. Direct match | Candidate sentence 102 | Source sentence 102

Submitted text	Source passage
17.	17.

561. Direct match | Candidate sentence 103 | Source sentence 103

Submitted text	Source passage
Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.	Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.

562. Direct match | Candidate sentence 104 | Source sentence 104

Submitted text	Source passage
However, generated summaries may contain inaccuracies when project context or code versions are outdated.	However, generated summaries may contain inaccuracies when project context or code versions are outdated.

563. Direct match | Candidate sentence 105 | Source sentence 105

Submitted text	Source passage
18.	18.

564. Direct match | Candidate sentence 106 | Source sentence 106

Submitted text	Source passage
AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.	AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.

565. Direct match | Candidate sentence 107 | Source sentence 107

Submitted text	Source passage
However, integrating different permission models and data structures introduces administrative complexity.	However, integrating different permission models and data structures introduces administrative complexity.

566. Direct match | Candidate sentence 108 | Source sentence 108

Submitted text	Source passage
19.	19.

567. Direct match | Candidate sentence 109 | Source sentence 109

Submitted text	Source passage
Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.	Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.

568. Direct match | Candidate sentence 110 | Source sentence 110

Submitted text	Source passage
However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.	However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.

569. Direct match | Candidate sentence 111 | Source sentence 111

Submitted text	Source passage
20.	20.

570. Direct match | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

571. Direct match | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

572. Direct match | Candidate sentence 114 | Source sentence 114

Submitted text	Source passage
However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.	However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.

573. Direct match | Candidate sentence 115 | Source sentence 115

Submitted text	Source passage
Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.	Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.

574. Direct match | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

575. Direct match | Candidate sentence 117 | Source sentence 117

Submitted text	Source passage
Do not present unverified titles or numerical claims as published research.	Do not present unverified titles or numerical claims as published research.

576. Direct match | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

577. Direct match | Candidate sentence 119 | Source sentence 119

Submitted text	Source passage
Fragmented Data Processing	Fragmented Data Processing

578. Direct match | Candidate sentence 120 | Source sentence 120

Submitted text	Source passage
Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.	Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.

579. Direct match | Candidate sentence 121 | Source sentence 121

Submitted text	Source passage
Limited Incremental Computation	Limited Incremental Computation

580. Direct match | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

581. Direct match | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

582. Direct match | Candidate sentence 124 | Source sentence 124

Submitted text	Source passage
Inefficient Query Execution	Inefficient Query Execution

583. Direct match | Candidate sentence 125 | Source sentence 125

Submitted text	Source passage
Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.	Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.

584. Direct match | Candidate sentence 126 | Source sentence 126

Submitted text	Source passage
Complex Infrastructure Management	Complex Infrastructure Management

585. Direct match | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

586. Direct match | Candidate sentence 128 | Source sentence 128

Submitted text	Source passage
Limited Processing Visibility	Limited Processing Visibility

587. Direct match | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

588. Direct match | Candidate sentence 130 | Source sentence 130

Submitted text	Source passage
Challenges in State Recovery	Challenges in State Recovery

589. Direct match | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

590. Direct match | Candidate sentence 132 | Source sentence 132

Submitted text	Source passage
Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.	Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.

591. Direct match | Candidate sentence 133 | Source sentence 133

Submitted text	Source passage
Resource Utilization and Scalability	Resource Utilization and Scalability

592. Direct match | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

593. Direct match | Candidate sentence 135 | Source sentence 135

Submitted text	Source passage
Efficient resource utilization and scalable processing are essential for maintaining performance.	Efficient resource utilization and scalable processing are essential for maintaining performance.

594. Direct match | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

595. Direct match | Candidate sentence 137 | Source sentence 137

Submitted text	Source passage
Processing a continuously increasing volume of events while maintaining low latency is challenging.	Processing a continuously increasing volume of events while maintaining low latency is challenging.

596. Direct match | Candidate sentence 138 | Source sentence 138

Submitted text	Source passage
Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.	Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.

597. Direct match | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

598. Direct match | Candidate sentence 140 | Source sentence 140

Submitted text	Source passage
Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.	Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.

599. Direct match | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

600. Direct match | Candidate sentence 142 | Source sentence 142

Submitted text	Source passage
Limited Benchmarking of Self-Hosted Solutions	Limited Benchmarking of Self-Hosted Solutions

601. Direct match | Candidate sentence 143 | Source sentence 143

Submitted text	Source passage
The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.	The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.

602. Direct match | Candidate sentence 144 | Source sentence 144

Submitted text	Source passage
Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.	Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.

603. Direct match | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

604. Direct match | Candidate sentence 146 | Source sentence 146

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.	Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.

605. Direct match | Candidate sentence 147 | Source sentence 147

Submitted text	Source passage
Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.	Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.

606. Direct match | Candidate sentence 148 | Source sentence 148

Submitted text	Source passage
These limitations make it challenging to obtain up-to-date analytical results as new events arrive.	These limitations make it challenging to obtain up-to-date analytical results as new events arrive.

607. Direct match | Candidate sentence 149 | Source sentence 149

Submitted text	Source passage
Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.	Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.

608. Direct match | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

609. Direct match | Candidate sentence 151 | Source sentence 151

Submitted text	Source passage
It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.	It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.

610. Direct match | Candidate sentence 152 | Source sentence 152

Submitted text	Source passage
Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.	Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.

611. Direct match | Candidate sentence 153 | Source sentence 153

Submitted text	Source passage
By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.	By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.

612. Direct match | Candidate sentence 154 | Source sentence 154

Submitted text	Source passage
MOTIVATION	MOTIVATION

613. Direct match | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

614. Direct match | Candidate sentence 156 | Source sentence 156

Submitted text	Source passage
However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.	However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.

615. Direct match | Candidate sentence 157 | Source sentence 157

Submitted text	Source passage
As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.	As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.

616. Direct match | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

617. Direct match | Candidate sentence 159 | Source sentence 159

Submitted text	Source passage
StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.	StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.

618. Direct match | Candidate sentence 160 | Source sentence 160

Submitted text	Source passage
By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.	By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.

619. Direct match | Candidate sentence 161 | Source sentence 161

Submitted text	Source passage
Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.	Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.

620. Direct match | Candidate sentence 162 | Source sentence 162

Submitted text	Source passage
Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.	Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.

621. Direct match | Candidate sentence 163 | Source sentence 163

Submitted text	Source passage
StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.	StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.

622. Direct match | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

623. Direct match | Candidate sentence 165 | Source sentence 165

Submitted text	Source passage
Develop a Self-Hosted Streaming Engine	Develop a Self-Hosted Streaming Engine

624. Direct match | Candidate sentence 166 | Source sentence 166

Submitted text	Source passage
Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.	Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.

625. Direct match | Candidate sentence 167 | Source sentence 167

Submitted text	Source passage
The system aims to simplify real-time data processing through a unified architecture.	The system aims to simplify real-time data processing through a unified architecture.

626. Direct match | Candidate sentence 168 | Source sentence 168

Submitted text	Source passage
Implement Continuous Stream Processing	Implement Continuous Stream Processing

627. Direct match | Candidate sentence 169 | Source sentence 169

Submitted text	Source passage
Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.	Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.

628. Direct match | Candidate sentence 170 | Source sentence 170

Submitted text	Source passage
The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.	The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.

629. Direct match | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

630. Direct match | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

631. Direct match | Candidate sentence 173 | Source sentence 173

Submitted text	Source passage
This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.	This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.

632. Direct match | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

633. Direct match | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

634. Direct match | Candidate sentence 176 | Source sentence 176

Submitted text	Source passage
Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.	Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.

635. Direct match | Candidate sentence 177 | Source sentence 177

Submitted text	Source passage
Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.	Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.

636. Direct match | Candidate sentence 178 | Source sentence 178

Submitted text	Source passage
As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.	As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.

637. Direct match | Candidate sentence 179 | Source sentence 179

Submitted text	Source passage
Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.	Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.

638. Direct match | Candidate sentence 180 | Source sentence 180

Submitted text	Source passage
Periodic refreshes may prevent users from accessing the latest information immediately.	Periodic refreshes may prevent users from accessing the latest information immediately.

639. Direct match | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

640. Direct match | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

641. Direct match | Candidate sentence 183 | Source sentence 183

Submitted text	Source passage
StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.	StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.

642. Direct match | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

643. Direct match | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

644. Direct match | Candidate sentence 186 | Source sentence 186

Submitted text	Source passage
The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.	The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.

645. Direct match | Candidate sentence 187 | Source sentence 187

Submitted text	Source passage
Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.	Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.

646. Direct match | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

647. Direct match | Candidate sentence 189 | Source sentence 189

Submitted text	Source passage
StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.	StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.

648. Direct match | Candidate sentence 190 | Source sentence 190

Submitted text	Source passage
Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.	Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.

649. Direct match | Candidate sentence 191 | Source sentence 191

Submitted text	Source passage
The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results	The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results

650. Direct match | Candidate sentence 192 | Source sentence 192

Submitted text	Source passage
.	.

651. Direct match | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

652. Direct match | Candidate sentence 194 | Source sentence 194

Submitted text	Source passage
A persistent state store maintains the data required for processing and view updates.	A persistent state store maintains the data required for processing and view updates.

653. Direct match | Candidate sentence 195 | Source sentence 195

Submitted text	Source passage
Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.	Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.

654. Direct match | Candidate sentence 196 | Source sentence 196

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.	Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.

655. Direct match | Candidate sentence 197 | Source sentence 197

Submitted text	Source passage
The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.	The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.

656. Direct match | Candidate sentence 198 | Source sentence 198

Submitted text	Source passage
Backend APIs connect the dashboard with the processing engine and state store.	Backend APIs connect the dashboard with the processing engine and state store.

657. Direct match | Candidate sentence 199 | Source sentence 199

Submitted text	Source passage
REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.	REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.

658. Direct match | Candidate sentence 200 | Source sentence 200

Submitted text	Source passage
Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.	Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.

659. Direct match | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

660. Direct match | Candidate sentence 202 | Source sentence 202

Submitted text	Source passage
The following steps should be taken while using StreamView:	The following steps should be taken while using StreamView:

661. Direct match | Candidate sentence 203 | Source sentence 203

Submitted text	Source passage
Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.	Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.

662. Direct match | Candidate sentence 204 | Source sentence 204

Submitted text	Source passage
Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.	Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.

663. Direct match | Candidate sentence 205 | Source sentence 205

Submitted text	Source passage
Step 3: Ingest Events – Send events from a producer application to the configured message broker.	Step 3: Ingest Events – Send events from a producer application to the configured message broker.

664. Direct match | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

665. Direct match | Candidate sentence 207 | Source sentence 207

Submitted text	Source passage
Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.	Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.

666. Direct match | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

667. Direct match | Candidate sentence 209 | Source sentence 209

Submitted text	Source passage
The engine processes incoming events and maintains the corresponding results.	The engine processes incoming events and maintains the corresponding results.

668. Direct match | Candidate sentence 210 | Source sentence 210

Submitted text	Source passage
Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.	Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.

669. Direct match | Candidate sentence 211 | Source sentence 211

Submitted text	Source passage
Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.	Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.

670. Direct match | Candidate sentence 212 | Source sentence 212

Submitted text	Source passage
Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.	Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.

671. Direct match | Candidate sentence 213 | Source sentence 213

Submitted text	Source passage
Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.	Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.

672. Direct match | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

673. Direct match | Candidate sentence 215 | Source sentence 215

Submitted text	Source passage
FIG 8.1 Proposed workflow Diagram	FIG 8.1 Proposed workflow Diagram

674. Direct match | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

675. Direct match | Candidate sentence 217 | Source sentence 217

Submitted text	Source passage
Users can observe system performance and identify processing delays or potential bottlenecks.	Users can observe system performance and identify processing delays or potential bottlenecks.

676. Direct match | Candidate sentence 218 | Source sentence 218

Submitted text	Source passage
Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.	Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.

677. Direct match | Candidate sentence 219 | Source sentence 219

Submitted text	Source passage
The system applies the updated configurations based on its supported configuration and execution mechanisms.	The system applies the updated configurations based on its supported configuration and execution mechanisms.

678. Direct match | Candidate sentence 220 | Source sentence 220

Submitted text	Source passage
Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.	Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.

679. Direct match | Candidate sentence 221 | Source sentence 221

Submitted text	Source passage
These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.	These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.

680. Direct match | Candidate sentence 222 | Source sentence 222

Submitted text	Source passage
ALGORITHM	ALGORITHM

681. Direct match | Candidate sentence 223 | Source sentence 223

Submitted text	Source passage
The following steps describe the processing workflow of the StreamView application.	The following steps describe the processing workflow of the StreamView application.

682. Direct match | Candidate sentence 224 | Source sentence 224

Submitted text	Source passage
Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline	Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline

683. Direct match | Candidate sentence 225 | Source sentence 225

Submitted text	Source passage
Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)	Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)

684. Direct match | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

685. Direct match | Candidate sentence 227 | Source sentence 227

Submitted text	Source passage
1.	1.

686. Direct match | Candidate sentence 228 | Source sentence 228

Submitted text	Source passage
Initialize the message broker, stream-processing engine, persistent state store, and query interface.	Initialize the message broker, stream-processing engine, persistent state store, and query interface.

687. Direct match | Candidate sentence 229 | Source sentence 229

Submitted text	Source passage
2.	2.

688. Direct match | Candidate sentence 230 | Source sentence 230

Submitted text	Source passage
Configure the event stream and subscribe to the required message topic.	Configure the event stream and subscribe to the required message topic.

689. Direct match | Candidate sentence 231 | Source sentence 231

Submitted text	Source passage
3.	3.

690. Direct match | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

691. Direct match | Candidate sentence 233 | Source sentence 233

Submitted text	Source passage
4.	4.

692. Direct match | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

693. Direct match | Candidate sentence 235 | Source sentence 235

Submitted text	Source passage
5.	5.

694. Direct match | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

695. Direct match | Candidate sentence 237 | Source sentence 237

Submitted text	Source passage
6.	6.

696. Direct match | Candidate sentence 238 | Source sentence 238

Submitted text	Source passage
Apply the configured filtering and transformation operations to \(e\).	Apply the configured filtering and transformation operations to \(e\).

697. Direct match | Candidate sentence 239 | Source sentence 239

Submitted text	Source passage
7.	7.

698. Direct match | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

699. Direct match | Candidate sentence 241 | Source sentence 241

Submitted text	Source passage
8.	8.

700. Direct match | Candidate sentence 242 | Source sentence 242

Submitted text	Source passage
Identify the materialized views affected by the processed event.	Identify the materialized views affected by the processed event.

701. Direct match | Candidate sentence 243 | Source sentence 243

Submitted text	Source passage
9.	9.

702. Direct match | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

703. Direct match | Candidate sentence 245 | Source sentence 245

Submitted text	Source passage
10.	10.

704. Direct match | Candidate sentence 246 | Source sentence 246

Submitted text	Source passage
Persist the updated state and view data using the supported storage mechanism.	Persist the updated state and view data using the supported storage mechanism.

705. Direct match | Candidate sentence 247 | Source sentence 247

Submitted text	Source passage
11.	11.

706. Direct match | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

707. Direct match | Candidate sentence 249 | Source sentence 249

Submitted text	Source passage
12.	12.

708. Direct match | Candidate sentence 250 | Source sentence 250

Submitted text	Source passage
On receiving a query \(Q\) from the client interface:	On receiving a query \(Q\) from the client interface:

709. Direct match | Candidate sentence 251 | Source sentence 251

Submitted text	Source passage
13.	13.

710. Direct match | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

711. Direct match | Candidate sentence 253 | Source sentence 253

Submitted text	Source passage
14.	14.

712. Direct match | Candidate sentence 254 | Source sentence 254

Submitted text	Source passage
Return the query results \(R\) to the client interface.	Return the query results \(R\) to the client interface.

713. Direct match | Candidate sentence 255 | Source sentence 255

Submitted text	Source passage
15.	15.

714. Direct match | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

715. Direct match | Candidate sentence 257 | Source sentence 257

Submitted text	Source passage
16.	16.

716. Direct match | Candidate sentence 258 | Source sentence 258

Submitted text	Source passage
Repeat the process until the stream is stopped or the system is shut down.	Repeat the process until the stream is stopped or the system is shut down.

717. Direct match | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

718. Direct match | Candidate sentence 260 | Source sentence 260

Submitted text	Source passage
Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.	Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.

719. Direct match | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

720. Direct match | Candidate sentence 262 | Source sentence 262

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.	StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.

721. Direct match | Candidate sentence 263 | Source sentence 263

Submitted text	Source passage
The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.	The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.

722. Direct match | Candidate sentence 264 | Source sentence 264

Submitted text	Source passage
By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.	By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.

723. Direct match | Candidate sentence 265 | Source sentence 265

Submitted text	Source passage
From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.	From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.

724. Direct match | Candidate sentence 266 | Source sentence 266

Submitted text	Source passage
The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.	The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.

725. Direct match | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

726. Direct match | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

727. Direct match | Candidate sentence 269 | Source sentence 269

Submitted text	Source passage
Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.	Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.

728. Direct match | Candidate sentence 270 | Source sentence 270

Submitted text	Source passage
This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.	This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.

729. Direct match | Candidate sentence 271 | Source sentence 271

Submitted text	Source passage
The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.	The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.

730. Direct match | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

731. Direct match | Candidate sentence 273 | Source sentence 273

Submitted text	Source passage
Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.	Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.

732. Direct match | Candidate sentence 274 | Source sentence 274

Submitted text	Source passage
By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.	By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.

733. Direct match | Candidate sentence 275 | Source sentence 275

Submitted text	Source passage
StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.	StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.

734. Direct match | Candidate sentence 276 | Source sentence 276

Submitted text	Source passage
It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.	It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.

735. Direct match | Candidate sentence 277 | Source sentence 277

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing repeated computation.	Materialized views are updated incrementally as new events arrive, reducing repeated computation.

736. Direct match | Candidate sentence 278 | Source sentence 278

Submitted text	Source passage
The dashboard enables users to manage streams, execute queries, and monitor system performance.	The dashboard enables users to manage streams, execute queries, and monitor system performance.

737. Direct match | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

738. Direct match | Candidate sentence 280 | Source sentence 280

Submitted text	Source passage
The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.	The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.

739. Direct match | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

740. Direct match | Candidate sentence 282 | Source sentence 282

Submitted text	Source passage
The application backend manages business logic, API requests, stream configurations, and communication between components.	The application backend manages business logic, API requests, stream configurations, and communication between components.

741. Direct match | Candidate sentence 283 | Source sentence 283

Submitted text	Source passage
The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.	The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.

742. Direct match | Candidate sentence 284 | Source sentence 284

Submitted text	Source passage
The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.	The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.

743. Direct match | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

744. Direct match | Candidate sentence 286 | Source sentence 286

Submitted text	Source passage
These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.	These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.

745. Direct match | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

746. Direct match | Candidate sentence 288 | Source sentence 288

Submitted text	Source passage
RESULTS	RESULTS

747. Direct match | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

748. Direct match | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

749. Direct match | Candidate sentence 291 | Source sentence 291

Submitted text	Source passage
The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.	The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.

750. Direct match | Candidate sentence 292 | Source sentence 292

Submitted text	Source passage
The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.	The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.

751. Direct match | Candidate sentence 293 | Source sentence 293

Submitted text	Source passage
These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.	These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.

752. Direct match | Candidate sentence 294 | Source sentence 294

Submitted text	Source passage
The monitoring and event ingestion modules provide additional visibility into system operations.	The monitoring and event ingestion modules provide additional visibility into system operations.

753. Direct match | Candidate sentence 295 | Source sentence 295

Submitted text	Source passage
Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.	Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.

754. Direct match | Candidate sentence 296 | Source sentence 296

Submitted text	Source passage
The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.	The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.

755. Direct match | Candidate sentence 297 | Source sentence 297

Submitted text	Source passage
These measurements help evaluate system performance and identify potential processing bottlenecks.	These measurements help evaluate system performance and identify potential processing bottlenecks.

756. Direct match | Candidate sentence 298 | Source sentence 298

Submitted text	Source passage
Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.	Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.

757. Direct match | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

758. Direct match | Candidate sentence 300 | Source sentence 300

Submitted text	Source passage
FIG 12.1: Result	FIG 12.1: Result

759. Direct match | Candidate sentence 301 | Source sentence 301

Submitted text	Source passage
By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.	By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.

760. Direct match | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

761. Direct match | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

762. Direct match | Candidate sentence 304 | Source sentence 304

Submitted text	Source passage
In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.	In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.

763. Direct match | Candidate sentence 305 | Source sentence 305

Submitted text	Source passage
Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.	Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.

764. Direct match | Candidate sentence 306 | Source sentence 306

Submitted text	Source passage
Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.	Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.

765. Direct match | Candidate sentence 307 | Source sentence 307

Submitted text	Source passage
Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.	Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.

766. Direct match | Candidate sentence 308 | Source sentence 308

Submitted text	Source passage
Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.	Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.

767. Direct match | Candidate sentence 309 | Source sentence 309

Submitted text	Source passage
Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.	Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.

768. Direct match | Candidate sentence 310 | Source sentence 310

Submitted text	Source passage
Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.	Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.

769. Direct match | Candidate sentence 311 | Source sentence 311

Submitted text	Source passage
CONCLUSION	CONCLUSION

770. Direct match | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

771. Direct match | Candidate sentence 313 | Source sentence 313

Submitted text	Source passage
Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.	Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.

772. Direct match | Candidate sentence 314 | Source sentence 314

Submitted text	Source passage
The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.	The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.

773. Direct match | Candidate sentence 315 | Source sentence 315

Submitted text	Source passage
Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.	Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.

774. Direct match | Candidate sentence 316 | Source sentence 316

Submitted text	Source passage
This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.	This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.

775. Direct match | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

776. Direct match | Candidate sentence 318 | Source sentence 318

Submitted text	Source passage
REFERENCES	REFERENCES

777. Direct match | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

778. Direct match | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

779. Direct match | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

780. Direct match | Candidate sentence 322 | Source sentence 322

Submitted text	Source passage
Kalavri, and A.	Kalavri, and A.

781. Direct match | Candidate sentence 323 | Source sentence 323

Submitted text	Source passage
Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.	Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.

782. Direct match | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

783. Direct match | Candidate sentence 325 | Source sentence 325

Submitted text	Source passage
507–541, 2024.	507–541, 2024.

784. Direct match | Candidate sentence 326 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

785. Direct match | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

786. Direct match | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

787. Direct match | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

788. Direct match | Candidate sentence 330 | Source sentence 330

Submitted text	Source passage
Golab, and M.	Golab, and M.

789. Direct match | Candidate sentence 331 | Source sentence 331

Submitted text	Source passage
T.	T.

790. Direct match | Candidate sentence 332 | Source sentence 332

Submitted text	Source passage
Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.	Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.

791. Direct match | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

792. Direct match | Candidate sentence 334 | Source sentence 334

Submitted text	Source passage
55–67, 2017.	55–67, 2017.

793. Direct match | Candidate sentence 335 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

794. Direct match | Candidate sentence 336 | Source sentence 336

Submitted text	Source passage
Available: https://doi.org/10.1016/j.is.2017.07.002	Available: https://doi.org/10.1016/j.is.2017.07.002

795. Direct match | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

796. Direct match | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

797. Direct match | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

798. Direct match | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

799. Direct match | Candidate sentence 341 | Source sentence 341

Submitted text	Source passage
Ryzhyk, and V.	Ryzhyk, and V.

800. Direct match | Candidate sentence 342 | Source sentence 342

Submitted text	Source passage
Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.	Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.

801. Direct match | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

802. Direct match | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

803. Direct match | Candidate sentence 345 | Source sentence 345

Submitted text	Source passage
1601–1614, 2023.	1601–1614, 2023.

804. Direct match | Candidate sentence 346 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

805. Direct match | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

806. Direct match | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

807. Direct match | Candidate sentence 349 | Source sentence 349

Submitted text	Source passage
Gupta and I.	Gupta and I.

808. Direct match | Candidate sentence 350 | Source sentence 359

Submitted text	Source passage
S.	S.

809. Direct match | Candidate sentence 351 | Source sentence 351

Submitted text	Source passage
Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.	Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.

810. Direct match | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

811. Direct match | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

812. Direct match | Candidate sentence 354 | Source sentence 354

Submitted text	Source passage
3–18, 1995.	3–18, 1995.

813. Direct match | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

814. Direct match | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

815. Direct match | Candidate sentence 357 | Source sentence 359

Submitted text	Source passage
S.	S.

816. Direct match | Candidate sentence 358 | Source sentence 358

Submitted text	Source passage
Mumick, and V.	Mumick, and V.

817. Direct match | Candidate sentence 359 | Source sentence 359

Submitted text	Source passage
S.	S.

818. Direct match | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

819. Direct match | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

820. Direct match | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

821. Direct match | Candidate sentence 363 | Source sentence 363

Submitted text	Source passage
157–166, 1993.	157–166, 1993.

822. Direct match | Candidate sentence 364 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

823. Direct match | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

824. Direct match | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

825. Direct match | Candidate sentence 367 | Source sentence 420

Submitted text	Source passage
A.	A.

826. Direct match | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

827. Direct match | Candidate sentence 369 | Source sentence 369

Submitted text	Source passage
Srivastava, and S.	Srivastava, and S.

828. Direct match | Candidate sentence 370 | Source sentence 370

Submitted text	Source passage
Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.	Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.

829. Direct match | Candidate sentence 371 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

830. Direct match | Candidate sentence 372 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

831. Direct match | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

832. Direct match | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

833. Direct match | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

834. Direct match | Candidate sentence 376 | Source sentence 376

Submitted text	Source passage
Ceri and J.	Ceri and J.

835. Direct match | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

836. Direct match | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

837. Direct match | Candidate sentence 379 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

838. Direct match | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

839. Direct match | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

840. Direct match | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

841. Direct match | Candidate sentence 383 | Source sentence 420

Submitted text	Source passage
A.	A.

842. Direct match | Candidate sentence 384 | Source sentence 384

Submitted text	Source passage
Blakeley, P.-Å.	Blakeley, P.-Å.

843. Direct match | Candidate sentence 385 | Source sentence 385

Submitted text	Source passage
Larson, and F.	Larson, and F.

844. Direct match | Candidate sentence 386 | Source sentence 386

Submitted text	Source passage
W.	W.

845. Direct match | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

846. Direct match | Candidate sentence 388 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

847. Direct match | Candidate sentence 389 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

848. Direct match | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

849. Direct match | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

850. Direct match | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

851. Direct match | Candidate sentence 393 | Source sentence 393

Submitted text	Source passage
Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.	Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.

852. Direct match | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

853. Direct match | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

854. Direct match | Candidate sentence 396 | Source sentence 396

Submitted text	Source passage
1792–1803, 2015.	1792–1803, 2015.

855. Direct match | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

856. Direct match | Candidate sentence 398 | Source sentence 398

Submitted text	Source passage
Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.	Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.

857. Direct match | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

858. Direct match | Candidate sentence 400 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

859. Direct match | Candidate sentence 401 | Source sentence 401

Submitted text	Source passage
Very Large Data Bases (VLDB), 2013.	Very Large Data Bases (VLDB), 2013.

860. Direct match | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

861. Direct match | Candidate sentence 403 | Source sentence 403

Submitted text	Source passage
Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.	Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.

862. Direct match | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

863. Direct match | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

864. Direct match | Candidate sentence 406 | Source sentence 406

Submitted text	Source passage
28–38, 2015.	28–38, 2015.

865. Direct match | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

866. Direct match | Candidate sentence 408 | Source sentence 408

Submitted text	Source passage
Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.	Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.

867. Direct match | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

868. Direct match | Candidate sentence 410 | Source sentence 410

Submitted text	Source passage
Operating Systems Principles (SOSP), 2013, pp.	Operating Systems Principles (SOSP), 2013, pp.

869. Direct match | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

870. Direct match | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

871. Direct match | Candidate sentence 413 | Source sentence 413

Submitted text	Source passage
Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.	Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.

872. Direct match | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

873. Direct match | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

874. Direct match | Candidate sentence 416 | Source sentence 416

Submitted text	Source passage
3135–3147, 2021.	3135–3147, 2021.

875. Direct match | Candidate sentence 417 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

876. Direct match | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

877. Direct match | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

878. Direct match | Candidate sentence 420 | Source sentence 420

Submitted text	Source passage
A.	A.

879. Direct match | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

880. Direct match | Candidate sentence 422 | Source sentence 422

Submitted text	Source passage
M.	M.

881. Direct match | Candidate sentence 423 | Source sentence 423

Submitted text	Source passage
Hellerstein, and S.	Hellerstein, and S.

882. Direct match | Candidate sentence 424 | Source sentence 424

Submitted text	Source passage
Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.	Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.

883. Direct match | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

884. Direct match | Candidate sentence 426 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

885. Direct match | Candidate sentence 427 | Source sentence 427

Submitted text	Source passage
Data Engineering (ICDE), 2003.	Data Engineering (ICDE), 2003.

886. Direct match | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

887. Direct match | Candidate sentence 429 | Source sentence 429

Submitted text	Source passage
Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.	Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.

888. Direct match | Candidate sentence 430 | Source sentence 430

Submitted text	Source passage
1st Biennial Conf.	1st Biennial Conf.

889. Direct match | Candidate sentence 431 | Source sentence 431

Submitted text	Source passage
Innovative Data Systems Research (CIDR), 2003.	Innovative Data Systems Research (CIDR), 2003.

890. Direct match | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

891. Direct match | Candidate sentence 433 | Source sentence 433

Submitted text	Source passage
J.	J.

892. Direct match | Candidate sentence 434 | Source sentence 434

Submitted text	Source passage
Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.	Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.

893. Direct match | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

894. Direct match | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

895. Direct match | Candidate sentence 437 | Source sentence 437

Submitted text	Source passage
120–139, 2003.	120–139, 2003.

896. Direct match | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

897. Direct match | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

898. Direct match | Candidate sentence 440 | Source sentence 440

Submitted text	Source passage
Babu, and J.	Babu, and J.

899. Direct match | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

900. Direct match | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

901. Direct match | Candidate sentence 443 | Source sentence 443

Submitted text	Source passage
121–142, 2006.	121–142, 2006.

902. Direct match | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

903. Direct match | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

904. Direct match | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

905. Direct match | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

906. Direct match | Candidate sentence 448 | Source sentence 448

Submitted text	Source passage
42–47, 2005.	42–47, 2005.

907. Direct match | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

908. Direct match | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

909. Direct match | Candidate sentence 451 | Source sentence 451

Submitted text	Source passage
4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.	4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.

910. Direct match | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

911. Direct match | Candidate sentence 453 | Source sentence 453

Submitted text	Source passage
Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.	Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.

912. Direct match | Candidate sentence 454 | Source sentence 454

Submitted text	Source passage
34, article 39, 2025.	34, article 39, 2025.

913. Direct match | Candidate sentence 455 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

914. Direct match | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

915. Direct match | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

916. Direct match | Candidate sentence 1 | Source sentence 1

Submitted text	Source passage
Mrs .	Mrs .

917. Direct match | Candidate sentence 2 | Source sentence 2

Submitted text	Source passage
K .	K .

918. Direct match | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

919. Direct match | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

920. Direct match | Candidate sentence 5 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

921. Direct match | Candidate sentence 6 | Source sentence 6

Submitted text	Source passage
astakalanagamani78@gmail.com	astakalanagamani78@gmail.com

922. Direct match | Candidate sentence 7 | Source sentence 7

Submitted text	Source passage
P.	P.

923. Direct match | Candidate sentence 8 | Source sentence 8

Submitted text	Source passage
Srinath	Srinath

924. Direct match | Candidate sentence 9 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

925. Direct match | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

926. Direct match | Candidate sentence 11 | Source sentence 11

Submitted text	Source passage
MLR Institute of Technology	MLR Institute of Technology

927. Direct match | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

928. Direct match | Candidate sentence 13 | Source sentence 13

Submitted text	Source passage
23r21a05y0@mlrit.ac.in	23r21a05y0@mlrit.ac.in

929. Direct match | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

930. Direct match | Candidate sentence 15 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

931. Direct match | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

932. Direct match | Candidate sentence 17 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

933. Direct match | Candidate sentence 18 | Source sentence 18

Submitted text	Source passage
23r21a05t0@mlrit.ac.in	23r21a05t0@mlrit.ac.in

934. Direct match | Candidate sentence 19 | Source sentence 19

Submitted text	Source passage
V.	V.

935. Direct match | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

936. Direct match | Candidate sentence 21 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

937. Direct match | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

938. Direct match | Candidate sentence 23 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

939. Direct match | Candidate sentence 24 | Source sentence 24

Submitted text	Source passage
24r21a0532@mlrit.ac.in	24r21a0532@mlrit.ac.in

940. Direct match | Candidate sentence 25 | Source sentence 25

Submitted text	Source passage
ABSTRACT	ABSTRACT

941. Direct match | Candidate sentence 26 | Source sentence 26

Submitted text	Source passage
Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.	Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.

942. Direct match | Candidate sentence 27 | Source sentence 27

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.	StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.

943. Direct match | Candidate sentence 28 | Source sentence 28

Submitted text	Source passage
It combines message-based event ingestion, stream processing, and persistent state management.	It combines message-based event ingestion, stream processing, and persistent state management.

944. Direct match | Candidate sentence 29 | Source sentence 29

Submitted text	Source passage
The processing pipeline supports filtering, transformation, aggregation, and event-time operations.	The processing pipeline supports filtering, transformation, aggregation, and event-time operations.

945. Direct match | Candidate sentence 30 | Source sentence 30

Submitted text	Source passage
An LSM-tree-based state store manages persistent processing state.	An LSM-tree-based state store manages persistent processing state.

946. Direct match | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

947. Direct match | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

948. Direct match | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

949. Direct match | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

950. Direct match | Candidate sentence 35 | Source sentence 35

Submitted text	Source passage
The architecture is intended to support low-latency analytical queries and continuous data processing.	The architecture is intended to support low-latency analytical queries and continuous data processing.

951. Direct match | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

952. Direct match | Candidate sentence 37 | Source sentence 37

Submitted text	Source passage
StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.	StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.

953. Direct match | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

954. Direct match | Candidate sentence 39 | Source sentence 39

Submitted text	Source passage
The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.	The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.

955. Direct match | Candidate sentence 40 | Source sentence 40

Submitted text	Source passage
Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.	Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.

956. Direct match | Candidate sentence 41 | Source sentence 41

Submitted text	Source passage
INTRODUCTION	INTRODUCTION

957. Direct match | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

958. Direct match | Candidate sentence 43 | Source sentence 43

Submitted text	Source passage
However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.	However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.

959. Direct match | Candidate sentence 44 | Source sentence 44

Submitted text	Source passage
These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.	These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.

960. Direct match | Candidate sentence 45 | Source sentence 45

Submitted text	Source passage
Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].	Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].

961. Direct match | Candidate sentence 46 | Source sentence 46

Submitted text	Source passage
A streaming materialized-view engine can maintain precomputed query results as new events arrive.	A streaming materialized-view engine can maintain precomputed query results as new events arrive.

962. Direct match | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

963. Direct match | Candidate sentence 48 | Source sentence 48

Submitted text	Source passage
Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].	Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].

964. Direct match | Candidate sentence 49 | Source sentence 49

Submitted text	Source passage
This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.	This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.

965. Direct match | Candidate sentence 50 | Source sentence 50

Submitted text	Source passage
The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.	The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.

966. Direct match | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

967. Direct match | Candidate sentence 52 | Source sentence 52

Submitted text	Source passage
Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.	Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.

968. Direct match | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

969. Direct match | Candidate sentence 54 | Source sentence 227

Submitted text	Source passage
1.	1.

970. Direct match | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

971. Direct match | Candidate sentence 56 | Source sentence 56

Submitted text	Source passage
However, maintaining consistent analytical results under high event volumes remains a challenge.	However, maintaining consistent analytical results under high event volumes remains a challenge.

972. Direct match | Candidate sentence 57 | Source sentence 229

Submitted text	Source passage
2.	2.

973. Direct match | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

974. Direct match | Candidate sentence 59 | Source sentence 59

Submitted text	Source passage
However, complex aggregations and large-scale workloads can increase update overhead.	However, complex aggregations and large-scale workloads can increase update overhead.

975. Direct match | Candidate sentence 60 | Source sentence 231

Submitted text	Source passage
3.	3.

976. Direct match | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

977. Direct match | Candidate sentence 62 | Source sentence 62

Submitted text	Source passage
However, coordination, network communication, and fault recovery can introduce additional overhead.	However, coordination, network communication, and fault recovery can introduce additional overhead.

978. Direct match | Candidate sentence 63 | Source sentence 233

Submitted text	Source passage
4.	4.

979. Direct match | Candidate sentence 64 | Source sentence 64

Submitted text	Source passage
Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.	Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.

980. Direct match | Candidate sentence 65 | Source sentence 65

Submitted text	Source passage
However, selecting appropriate watermark policies can affect result completeness and processing latency.	However, selecting appropriate watermark policies can affect result completeness and processing latency.

981. Direct match | Candidate sentence 66 | Source sentence 235

Submitted text	Source passage
5.	5.

982. Direct match | Candidate sentence 67 | Source sentence 67

Submitted text	Source passage
Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.	Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.

983. Direct match | Candidate sentence 68 | Source sentence 68

Submitted text	Source passage
However, storage performance and recovery mechanisms can influence overall system efficiency.	However, storage performance and recovery mechanisms can influence overall system efficiency.

984. Direct match | Candidate sentence 69 | Source sentence 237

Submitted text	Source passage
6.	6.

985. Direct match | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

986. Direct match | Candidate sentence 71 | Source sentence 71

Submitted text	Source passage
However, log management and recovery procedures can increase storage and processing overhead.	However, log management and recovery procedures can increase storage and processing overhead.

987. Direct match | Candidate sentence 72 | Source sentence 239

Submitted text	Source passage
7.	7.

988. Direct match | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

989. Direct match | Candidate sentence 74 | Source sentence 74

Submitted text	Source passage
However, continuous index updates can increase memory consumption and computational costs.	However, continuous index updates can increase memory consumption and computational costs.

990. Direct match | Candidate sentence 75 | Source sentence 241

Submitted text	Source passage
8.	8.

991. Direct match | Candidate sentence 76 | Source sentence 76

Submitted text	Source passage
Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.	Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.

992. Direct match | Candidate sentence 77 | Source sentence 77

Submitted text	Source passage
However, maintaining reliable integrations requires ongoing API updates and webhook management.	However, maintaining reliable integrations requires ongoing API updates and webhook management.

993. Direct match | Candidate sentence 78 | Source sentence 243

Submitted text	Source passage
9.	9.

994. Direct match | Candidate sentence 79 | Source sentence 79

Submitted text	Source passage
AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.	AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.

995. Direct match | Candidate sentence 80 | Source sentence 80

Submitted text	Source passage
However, recommendations may become outdated when priorities and team workloads change.	However, recommendations may become outdated when priorities and team workloads change.

996. Direct match | Candidate sentence 81 | Source sentence 245

Submitted text	Source passage
10.	10.

997. Direct match | Candidate sentence 82 | Source sentence 82

Submitted text	Source passage
Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.	Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.

998. Direct match | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

999. Direct match | Candidate sentence 84 | Source sentence 247

Submitted text	Source passage
11.	11.

1000. Direct match | Candidate sentence 85 | Source sentence 85

Submitted text	Source passage
Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.	Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.

1001. Direct match | Candidate sentence 86 | Source sentence 86

Submitted text	Source passage
However, integrating documentation with daily development activities remains challenging.	However, integrating documentation with daily development activities remains challenging.

1002. Direct match | Candidate sentence 87 | Source sentence 249

Submitted text	Source passage
12.	12.

1003. Direct match | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

1004. Direct match | Candidate sentence 89 | Source sentence 89

Submitted text	Source passage
However, organizations must address data privacy, access control, and data-residency requirements.	However, organizations must address data privacy, access control, and data-residency requirements.

1005. Direct match | Candidate sentence 90 | Source sentence 251

Submitted text	Source passage
13.	13.

1006. Direct match | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

1007. Direct match | Candidate sentence 92 | Source sentence 92

Submitted text	Source passage
However, complex debugging tasks involving multiple codebases may require additional contextual information.	However, complex debugging tasks involving multiple codebases may require additional contextual information.

1008. Direct match | Candidate sentence 93 | Source sentence 253

Submitted text	Source passage
14.	14.

1009. Direct match | Candidate sentence 94 | Source sentence 94

Submitted text	Source passage
Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.	Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.

1010. Direct match | Candidate sentence 95 | Source sentence 95

Submitted text	Source passage
However, high query volumes can increase memory consumption and affect write performance.	However, high query volumes can increase memory consumption and affect write performance.

1011. Direct match | Candidate sentence 96 | Source sentence 255

Submitted text	Source passage
15.	15.

1012. Direct match | Candidate sentence 97 | Source sentence 97

Submitted text	Source passage
Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.	Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.

1013. Direct match | Candidate sentence 98 | Source sentence 98

Submitted text	Source passage
However, their effectiveness depends on the availability of accurate historical data.	However, their effectiveness depends on the availability of accurate historical data.

1014. Direct match | Candidate sentence 99 | Source sentence 257

Submitted text	Source passage
16.	16.

1015. Direct match | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

1016. Direct match | Candidate sentence 101 | Source sentence 101

Submitted text	Source passage
However, limited historical activity can reduce recommendation accuracy for new team members.	However, limited historical activity can reduce recommendation accuracy for new team members.

1017. Direct match | Candidate sentence 102 | Source sentence 102

Submitted text	Source passage
17.	17.

1018. Direct match | Candidate sentence 103 | Source sentence 103

Submitted text	Source passage
Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.	Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.

1019. Direct match | Candidate sentence 104 | Source sentence 104

Submitted text	Source passage
However, generated summaries may contain inaccuracies when project context or code versions are outdated.	However, generated summaries may contain inaccuracies when project context or code versions are outdated.

1020. Direct match | Candidate sentence 105 | Source sentence 105

Submitted text	Source passage
18.	18.

1021. Direct match | Candidate sentence 106 | Source sentence 106

Submitted text	Source passage
AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.	AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.

1022. Direct match | Candidate sentence 107 | Source sentence 107

Submitted text	Source passage
However, integrating different permission models and data structures introduces administrative complexity.	However, integrating different permission models and data structures introduces administrative complexity.

1023. Direct match | Candidate sentence 108 | Source sentence 108

Submitted text	Source passage
19.	19.

1024. Direct match | Candidate sentence 109 | Source sentence 109

Submitted text	Source passage
Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.	Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.

1025. Direct match | Candidate sentence 110 | Source sentence 110

Submitted text	Source passage
However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.	However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.

1026. Direct match | Candidate sentence 111 | Source sentence 111

Submitted text	Source passage
20.	20.

1027. Direct match | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

1028. Direct match | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

1029. Direct match | Candidate sentence 114 | Source sentence 114

Submitted text	Source passage
However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.	However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.

1030. Direct match | Candidate sentence 115 | Source sentence 115

Submitted text	Source passage
Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.	Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.

1031. Direct match | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

1032. Direct match | Candidate sentence 117 | Source sentence 117

Submitted text	Source passage
Do not present unverified titles or numerical claims as published research.	Do not present unverified titles or numerical claims as published research.

1033. Direct match | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

1034. Direct match | Candidate sentence 119 | Source sentence 119

Submitted text	Source passage
Fragmented Data Processing	Fragmented Data Processing

1035. Direct match | Candidate sentence 120 | Source sentence 120

Submitted text	Source passage
Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.	Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.

1036. Direct match | Candidate sentence 121 | Source sentence 121

Submitted text	Source passage
Limited Incremental Computation	Limited Incremental Computation

1037. Direct match | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

1038. Direct match | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

1039. Direct match | Candidate sentence 124 | Source sentence 124

Submitted text	Source passage
Inefficient Query Execution	Inefficient Query Execution

1040. Direct match | Candidate sentence 125 | Source sentence 125

Submitted text	Source passage
Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.	Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.

1041. Direct match | Candidate sentence 126 | Source sentence 126

Submitted text	Source passage
Complex Infrastructure Management	Complex Infrastructure Management

1042. Direct match | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

1043. Direct match | Candidate sentence 128 | Source sentence 128

Submitted text	Source passage
Limited Processing Visibility	Limited Processing Visibility

1044. Direct match | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

1045. Direct match | Candidate sentence 130 | Source sentence 130

Submitted text	Source passage
Challenges in State Recovery	Challenges in State Recovery

1046. Direct match | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

1047. Direct match | Candidate sentence 132 | Source sentence 132

Submitted text	Source passage
Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.	Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.

1048. Direct match | Candidate sentence 133 | Source sentence 133

Submitted text	Source passage
Resource Utilization and Scalability	Resource Utilization and Scalability

1049. Direct match | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

1050. Direct match | Candidate sentence 135 | Source sentence 135

Submitted text	Source passage
Efficient resource utilization and scalable processing are essential for maintaining performance.	Efficient resource utilization and scalable processing are essential for maintaining performance.

1051. Direct match | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

1052. Direct match | Candidate sentence 137 | Source sentence 137

Submitted text	Source passage
Processing a continuously increasing volume of events while maintaining low latency is challenging.	Processing a continuously increasing volume of events while maintaining low latency is challenging.

1053. Direct match | Candidate sentence 138 | Source sentence 138

Submitted text	Source passage
Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.	Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.

1054. Direct match | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

1055. Direct match | Candidate sentence 140 | Source sentence 140

Submitted text	Source passage
Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.	Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.

1056. Direct match | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

1057. Direct match | Candidate sentence 142 | Source sentence 142

Submitted text	Source passage
Limited Benchmarking of Self-Hosted Solutions	Limited Benchmarking of Self-Hosted Solutions

1058. Direct match | Candidate sentence 143 | Source sentence 143

Submitted text	Source passage
The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.	The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.

1059. Direct match | Candidate sentence 144 | Source sentence 144

Submitted text	Source passage
Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.	Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.

1060. Direct match | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

1061. Direct match | Candidate sentence 146 | Source sentence 146

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.	Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.

1062. Direct match | Candidate sentence 147 | Source sentence 147

Submitted text	Source passage
Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.	Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.

1063. Direct match | Candidate sentence 148 | Source sentence 148

Submitted text	Source passage
These limitations make it challenging to obtain up-to-date analytical results as new events arrive.	These limitations make it challenging to obtain up-to-date analytical results as new events arrive.

1064. Direct match | Candidate sentence 149 | Source sentence 149

Submitted text	Source passage
Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.	Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.

1065. Direct match | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

1066. Direct match | Candidate sentence 151 | Source sentence 151

Submitted text	Source passage
It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.	It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.

1067. Direct match | Candidate sentence 152 | Source sentence 152

Submitted text	Source passage
Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.	Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.

1068. Direct match | Candidate sentence 153 | Source sentence 153

Submitted text	Source passage
By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.	By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.

1069. Direct match | Candidate sentence 154 | Source sentence 154

Submitted text	Source passage
MOTIVATION	MOTIVATION

1070. Direct match | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

1071. Direct match | Candidate sentence 156 | Source sentence 156

Submitted text	Source passage
However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.	However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.

1072. Direct match | Candidate sentence 157 | Source sentence 157

Submitted text	Source passage
As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.	As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.

1073. Direct match | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

1074. Direct match | Candidate sentence 159 | Source sentence 159

Submitted text	Source passage
StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.	StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.

1075. Direct match | Candidate sentence 160 | Source sentence 160

Submitted text	Source passage
By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.	By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.

1076. Direct match | Candidate sentence 161 | Source sentence 161

Submitted text	Source passage
Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.	Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.

1077. Direct match | Candidate sentence 162 | Source sentence 162

Submitted text	Source passage
Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.	Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.

1078. Direct match | Candidate sentence 163 | Source sentence 163

Submitted text	Source passage
StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.	StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.

1079. Direct match | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

1080. Direct match | Candidate sentence 165 | Source sentence 165

Submitted text	Source passage
Develop a Self-Hosted Streaming Engine	Develop a Self-Hosted Streaming Engine

1081. Direct match | Candidate sentence 166 | Source sentence 166

Submitted text	Source passage
Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.	Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.

1082. Direct match | Candidate sentence 167 | Source sentence 167

Submitted text	Source passage
The system aims to simplify real-time data processing through a unified architecture.	The system aims to simplify real-time data processing through a unified architecture.

1083. Direct match | Candidate sentence 168 | Source sentence 168

Submitted text	Source passage
Implement Continuous Stream Processing	Implement Continuous Stream Processing

1084. Direct match | Candidate sentence 169 | Source sentence 169

Submitted text	Source passage
Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.	Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.

1085. Direct match | Candidate sentence 170 | Source sentence 170

Submitted text	Source passage
The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.	The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.

1086. Direct match | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

1087. Direct match | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

1088. Direct match | Candidate sentence 173 | Source sentence 173

Submitted text	Source passage
This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.	This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.

1089. Direct match | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

1090. Direct match | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

1091. Direct match | Candidate sentence 176 | Source sentence 176

Submitted text	Source passage
Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.	Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.

1092. Direct match | Candidate sentence 177 | Source sentence 177

Submitted text	Source passage
Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.	Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.

1093. Direct match | Candidate sentence 178 | Source sentence 178

Submitted text	Source passage
As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.	As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.

1094. Direct match | Candidate sentence 179 | Source sentence 179

Submitted text	Source passage
Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.	Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.

1095. Direct match | Candidate sentence 180 | Source sentence 180

Submitted text	Source passage
Periodic refreshes may prevent users from accessing the latest information immediately.	Periodic refreshes may prevent users from accessing the latest information immediately.

1096. Direct match | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

1097. Direct match | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

1098. Direct match | Candidate sentence 183 | Source sentence 183

Submitted text	Source passage
StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.	StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.

1099. Direct match | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

1100. Direct match | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

1101. Direct match | Candidate sentence 186 | Source sentence 186

Submitted text	Source passage
The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.	The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.

1102. Direct match | Candidate sentence 187 | Source sentence 187

Submitted text	Source passage
Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.	Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.

1103. Direct match | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

1104. Direct match | Candidate sentence 189 | Source sentence 189

Submitted text	Source passage
StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.	StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.

1105. Direct match | Candidate sentence 190 | Source sentence 190

Submitted text	Source passage
Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.	Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.

1106. Direct match | Candidate sentence 191 | Source sentence 191

Submitted text	Source passage
The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results	The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results

1107. Direct match | Candidate sentence 192 | Source sentence 192

Submitted text	Source passage
.	.

1108. Direct match | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

1109. Direct match | Candidate sentence 194 | Source sentence 194

Submitted text	Source passage
A persistent state store maintains the data required for processing and view updates.	A persistent state store maintains the data required for processing and view updates.

1110. Direct match | Candidate sentence 195 | Source sentence 195

Submitted text	Source passage
Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.	Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.

1111. Direct match | Candidate sentence 196 | Source sentence 196

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.	Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.

1112. Direct match | Candidate sentence 197 | Source sentence 197

Submitted text	Source passage
The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.	The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.

1113. Direct match | Candidate sentence 198 | Source sentence 198

Submitted text	Source passage
Backend APIs connect the dashboard with the processing engine and state store.	Backend APIs connect the dashboard with the processing engine and state store.

1114. Direct match | Candidate sentence 199 | Source sentence 199

Submitted text	Source passage
REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.	REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.

1115. Direct match | Candidate sentence 200 | Source sentence 200

Submitted text	Source passage
Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.	Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.

1116. Direct match | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

1117. Direct match | Candidate sentence 202 | Source sentence 202

Submitted text	Source passage
The following steps should be taken while using StreamView:	The following steps should be taken while using StreamView:

1118. Direct match | Candidate sentence 203 | Source sentence 203

Submitted text	Source passage
Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.	Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.

1119. Direct match | Candidate sentence 204 | Source sentence 204

Submitted text	Source passage
Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.	Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.

1120. Direct match | Candidate sentence 205 | Source sentence 205

Submitted text	Source passage
Step 3: Ingest Events – Send events from a producer application to the configured message broker.	Step 3: Ingest Events – Send events from a producer application to the configured message broker.

1121. Direct match | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

1122. Direct match | Candidate sentence 207 | Source sentence 207

Submitted text	Source passage
Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.	Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.

1123. Direct match | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

1124. Direct match | Candidate sentence 209 | Source sentence 209

Submitted text	Source passage
The engine processes incoming events and maintains the corresponding results.	The engine processes incoming events and maintains the corresponding results.

1125. Direct match | Candidate sentence 210 | Source sentence 210

Submitted text	Source passage
Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.	Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.

1126. Direct match | Candidate sentence 211 | Source sentence 211

Submitted text	Source passage
Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.	Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.

1127. Direct match | Candidate sentence 212 | Source sentence 212

Submitted text	Source passage
Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.	Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.

1128. Direct match | Candidate sentence 213 | Source sentence 213

Submitted text	Source passage
Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.	Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.

1129. Direct match | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

1130. Direct match | Candidate sentence 215 | Source sentence 215

Submitted text	Source passage
FIG 8.1 Proposed workflow Diagram	FIG 8.1 Proposed workflow Diagram

1131. Direct match | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

1132. Direct match | Candidate sentence 217 | Source sentence 217

Submitted text	Source passage
Users can observe system performance and identify processing delays or potential bottlenecks.	Users can observe system performance and identify processing delays or potential bottlenecks.

1133. Direct match | Candidate sentence 218 | Source sentence 218

Submitted text	Source passage
Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.	Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.

1134. Direct match | Candidate sentence 219 | Source sentence 219

Submitted text	Source passage
The system applies the updated configurations based on its supported configuration and execution mechanisms.	The system applies the updated configurations based on its supported configuration and execution mechanisms.

1135. Direct match | Candidate sentence 220 | Source sentence 220

Submitted text	Source passage
Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.	Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.

1136. Direct match | Candidate sentence 221 | Source sentence 221

Submitted text	Source passage
These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.	These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.

1137. Direct match | Candidate sentence 222 | Source sentence 222

Submitted text	Source passage
ALGORITHM	ALGORITHM

1138. Direct match | Candidate sentence 223 | Source sentence 223

Submitted text	Source passage
The following steps describe the processing workflow of the StreamView application.	The following steps describe the processing workflow of the StreamView application.

1139. Direct match | Candidate sentence 224 | Source sentence 224

Submitted text	Source passage
Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline	Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline

1140. Direct match | Candidate sentence 225 | Source sentence 225

Submitted text	Source passage
Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)	Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)

1141. Direct match | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

1142. Direct match | Candidate sentence 227 | Source sentence 227

Submitted text	Source passage
1.	1.

1143. Direct match | Candidate sentence 228 | Source sentence 228

Submitted text	Source passage
Initialize the message broker, stream-processing engine, persistent state store, and query interface.	Initialize the message broker, stream-processing engine, persistent state store, and query interface.

1144. Direct match | Candidate sentence 229 | Source sentence 229

Submitted text	Source passage
2.	2.

1145. Direct match | Candidate sentence 230 | Source sentence 230

Submitted text	Source passage
Configure the event stream and subscribe to the required message topic.	Configure the event stream and subscribe to the required message topic.

1146. Direct match | Candidate sentence 231 | Source sentence 231

Submitted text	Source passage
3.	3.

1147. Direct match | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

1148. Direct match | Candidate sentence 233 | Source sentence 233

Submitted text	Source passage
4.	4.

1149. Direct match | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

1150. Direct match | Candidate sentence 235 | Source sentence 235

Submitted text	Source passage
5.	5.

1151. Direct match | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

1152. Direct match | Candidate sentence 237 | Source sentence 237

Submitted text	Source passage
6.	6.

1153. Direct match | Candidate sentence 238 | Source sentence 238

Submitted text	Source passage
Apply the configured filtering and transformation operations to \(e\).	Apply the configured filtering and transformation operations to \(e\).

1154. Direct match | Candidate sentence 239 | Source sentence 239

Submitted text	Source passage
7.	7.

1155. Direct match | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

1156. Direct match | Candidate sentence 241 | Source sentence 241

Submitted text	Source passage
8.	8.

1157. Direct match | Candidate sentence 242 | Source sentence 242

Submitted text	Source passage
Identify the materialized views affected by the processed event.	Identify the materialized views affected by the processed event.

1158. Direct match | Candidate sentence 243 | Source sentence 243

Submitted text	Source passage
9.	9.

1159. Direct match | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

1160. Direct match | Candidate sentence 245 | Source sentence 245

Submitted text	Source passage
10.	10.

1161. Direct match | Candidate sentence 246 | Source sentence 246

Submitted text	Source passage
Persist the updated state and view data using the supported storage mechanism.	Persist the updated state and view data using the supported storage mechanism.

1162. Direct match | Candidate sentence 247 | Source sentence 247

Submitted text	Source passage
11.	11.

1163. Direct match | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

1164. Direct match | Candidate sentence 249 | Source sentence 249

Submitted text	Source passage
12.	12.

1165. Direct match | Candidate sentence 250 | Source sentence 250

Submitted text	Source passage
On receiving a query \(Q\) from the client interface:	On receiving a query \(Q\) from the client interface:

1166. Direct match | Candidate sentence 251 | Source sentence 251

Submitted text	Source passage
13.	13.

1167. Direct match | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

1168. Direct match | Candidate sentence 253 | Source sentence 253

Submitted text	Source passage
14.	14.

1169. Direct match | Candidate sentence 254 | Source sentence 254

Submitted text	Source passage
Return the query results \(R\) to the client interface.	Return the query results \(R\) to the client interface.

1170. Direct match | Candidate sentence 255 | Source sentence 255

Submitted text	Source passage
15.	15.

1171. Direct match | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

1172. Direct match | Candidate sentence 257 | Source sentence 257

Submitted text	Source passage
16.	16.

1173. Direct match | Candidate sentence 258 | Source sentence 258

Submitted text	Source passage
Repeat the process until the stream is stopped or the system is shut down.	Repeat the process until the stream is stopped or the system is shut down.

1174. Direct match | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

1175. Direct match | Candidate sentence 260 | Source sentence 260

Submitted text	Source passage
Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.	Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.

1176. Direct match | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

1177. Direct match | Candidate sentence 262 | Source sentence 262

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.	StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.

1178. Direct match | Candidate sentence 263 | Source sentence 263

Submitted text	Source passage
The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.	The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.

1179. Direct match | Candidate sentence 264 | Source sentence 264

Submitted text	Source passage
By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.	By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.

1180. Direct match | Candidate sentence 265 | Source sentence 265

Submitted text	Source passage
From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.	From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.

1181. Direct match | Candidate sentence 266 | Source sentence 266

Submitted text	Source passage
The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.	The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.

1182. Direct match | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

1183. Direct match | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

1184. Direct match | Candidate sentence 269 | Source sentence 269

Submitted text	Source passage
Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.	Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.

1185. Direct match | Candidate sentence 270 | Source sentence 270

Submitted text	Source passage
This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.	This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.

1186. Direct match | Candidate sentence 271 | Source sentence 271

Submitted text	Source passage
The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.	The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.

1187. Direct match | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

1188. Direct match | Candidate sentence 273 | Source sentence 273

Submitted text	Source passage
Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.	Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.

1189. Direct match | Candidate sentence 274 | Source sentence 274

Submitted text	Source passage
By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.	By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.

1190. Direct match | Candidate sentence 275 | Source sentence 275

Submitted text	Source passage
StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.	StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.

1191. Direct match | Candidate sentence 276 | Source sentence 276

Submitted text	Source passage
It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.	It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.

1192. Direct match | Candidate sentence 277 | Source sentence 277

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing repeated computation.	Materialized views are updated incrementally as new events arrive, reducing repeated computation.

1193. Direct match | Candidate sentence 278 | Source sentence 278

Submitted text	Source passage
The dashboard enables users to manage streams, execute queries, and monitor system performance.	The dashboard enables users to manage streams, execute queries, and monitor system performance.

1194. Direct match | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

1195. Direct match | Candidate sentence 280 | Source sentence 280

Submitted text	Source passage
The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.	The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.

1196. Direct match | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

1197. Direct match | Candidate sentence 282 | Source sentence 282

Submitted text	Source passage
The application backend manages business logic, API requests, stream configurations, and communication between components.	The application backend manages business logic, API requests, stream configurations, and communication between components.

1198. Direct match | Candidate sentence 283 | Source sentence 283

Submitted text	Source passage
The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.	The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.

1199. Direct match | Candidate sentence 284 | Source sentence 284

Submitted text	Source passage
The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.	The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.

1200. Direct match | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

1201. Direct match | Candidate sentence 286 | Source sentence 286

Submitted text	Source passage
These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.	These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.

1202. Direct match | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

1203. Direct match | Candidate sentence 288 | Source sentence 288

Submitted text	Source passage
RESULTS	RESULTS

1204. Direct match | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

1205. Direct match | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

1206. Direct match | Candidate sentence 291 | Source sentence 291

Submitted text	Source passage
The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.	The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.

1207. Direct match | Candidate sentence 292 | Source sentence 292

Submitted text	Source passage
The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.	The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.

1208. Direct match | Candidate sentence 293 | Source sentence 293

Submitted text	Source passage
These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.	These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.

1209. Direct match | Candidate sentence 294 | Source sentence 294

Submitted text	Source passage
The monitoring and event ingestion modules provide additional visibility into system operations.	The monitoring and event ingestion modules provide additional visibility into system operations.

1210. Direct match | Candidate sentence 295 | Source sentence 295

Submitted text	Source passage
Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.	Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.

1211. Direct match | Candidate sentence 296 | Source sentence 296

Submitted text	Source passage
The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.	The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.

1212. Direct match | Candidate sentence 297 | Source sentence 297

Submitted text	Source passage
These measurements help evaluate system performance and identify potential processing bottlenecks.	These measurements help evaluate system performance and identify potential processing bottlenecks.

1213. Direct match | Candidate sentence 298 | Source sentence 298

Submitted text	Source passage
Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.	Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.

1214. Direct match | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

1215. Direct match | Candidate sentence 300 | Source sentence 300

Submitted text	Source passage
FIG 12.1: Result	FIG 12.1: Result

1216. Direct match | Candidate sentence 301 | Source sentence 301

Submitted text	Source passage
By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.	By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.

1217. Direct match | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

1218. Direct match | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

1219. Direct match | Candidate sentence 304 | Source sentence 304

Submitted text	Source passage
In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.	In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.

1220. Direct match | Candidate sentence 305 | Source sentence 305

Submitted text	Source passage
Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.	Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.

1221. Direct match | Candidate sentence 306 | Source sentence 306

Submitted text	Source passage
Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.	Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.

1222. Direct match | Candidate sentence 307 | Source sentence 307

Submitted text	Source passage
Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.	Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.

1223. Direct match | Candidate sentence 308 | Source sentence 308

Submitted text	Source passage
Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.	Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.

1224. Direct match | Candidate sentence 309 | Source sentence 309

Submitted text	Source passage
Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.	Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.

1225. Direct match | Candidate sentence 310 | Source sentence 310

Submitted text	Source passage
Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.	Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.

1226. Direct match | Candidate sentence 311 | Source sentence 311

Submitted text	Source passage
CONCLUSION	CONCLUSION

1227. Direct match | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

1228. Direct match | Candidate sentence 313 | Source sentence 313

Submitted text	Source passage
Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.	Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.

1229. Direct match | Candidate sentence 314 | Source sentence 314

Submitted text	Source passage
The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.	The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.

1230. Direct match | Candidate sentence 315 | Source sentence 315

Submitted text	Source passage
Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.	Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.

1231. Direct match | Candidate sentence 316 | Source sentence 316

Submitted text	Source passage
This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.	This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.

1232. Direct match | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

1233. Direct match | Candidate sentence 318 | Source sentence 318

Submitted text	Source passage
REFERENCES	REFERENCES

1234. Direct match | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

1235. Direct match | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

1236. Direct match | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

1237. Direct match | Candidate sentence 322 | Source sentence 322

Submitted text	Source passage
Kalavri, and A.	Kalavri, and A.

1238. Direct match | Candidate sentence 323 | Source sentence 323

Submitted text	Source passage
Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.	Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.

1239. Direct match | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

1240. Direct match | Candidate sentence 325 | Source sentence 325

Submitted text	Source passage
507–541, 2024.	507–541, 2024.

1241. Direct match | Candidate sentence 326 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1242. Direct match | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

1243. Direct match | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

1244. Direct match | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

1245. Direct match | Candidate sentence 330 | Source sentence 330

Submitted text	Source passage
Golab, and M.	Golab, and M.

1246. Direct match | Candidate sentence 331 | Source sentence 331

Submitted text	Source passage
T.	T.

1247. Direct match | Candidate sentence 332 | Source sentence 332

Submitted text	Source passage
Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.	Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.

1248. Direct match | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

1249. Direct match | Candidate sentence 334 | Source sentence 334

Submitted text	Source passage
55–67, 2017.	55–67, 2017.

1250. Direct match | Candidate sentence 335 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1251. Direct match | Candidate sentence 336 | Source sentence 336

Submitted text	Source passage
Available: https://doi.org/10.1016/j.is.2017.07.002	Available: https://doi.org/10.1016/j.is.2017.07.002

1252. Direct match | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

1253. Direct match | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

1254. Direct match | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

1255. Direct match | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

1256. Direct match | Candidate sentence 341 | Source sentence 341

Submitted text	Source passage
Ryzhyk, and V.	Ryzhyk, and V.

1257. Direct match | Candidate sentence 342 | Source sentence 342

Submitted text	Source passage
Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.	Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.

1258. Direct match | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

1259. Direct match | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

1260. Direct match | Candidate sentence 345 | Source sentence 345

Submitted text	Source passage
1601–1614, 2023.	1601–1614, 2023.

1261. Direct match | Candidate sentence 346 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1262. Direct match | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

1263. Direct match | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

1264. Direct match | Candidate sentence 349 | Source sentence 349

Submitted text	Source passage
Gupta and I.	Gupta and I.

1265. Direct match | Candidate sentence 350 | Source sentence 359

Submitted text	Source passage
S.	S.

1266. Direct match | Candidate sentence 351 | Source sentence 351

Submitted text	Source passage
Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.	Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.

1267. Direct match | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

1268. Direct match | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1269. Direct match | Candidate sentence 354 | Source sentence 354

Submitted text	Source passage
3–18, 1995.	3–18, 1995.

1270. Direct match | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

1271. Direct match | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

1272. Direct match | Candidate sentence 357 | Source sentence 359

Submitted text	Source passage
S.	S.

1273. Direct match | Candidate sentence 358 | Source sentence 358

Submitted text	Source passage
Mumick, and V.	Mumick, and V.

1274. Direct match | Candidate sentence 359 | Source sentence 359

Submitted text	Source passage
S.	S.

1275. Direct match | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

1276. Direct match | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

1277. Direct match | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1278. Direct match | Candidate sentence 363 | Source sentence 363

Submitted text	Source passage
157–166, 1993.	157–166, 1993.

1279. Direct match | Candidate sentence 364 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1280. Direct match | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

1281. Direct match | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

1282. Direct match | Candidate sentence 367 | Source sentence 420

Submitted text	Source passage
A.	A.

1283. Direct match | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

1284. Direct match | Candidate sentence 369 | Source sentence 369

Submitted text	Source passage
Srivastava, and S.	Srivastava, and S.

1285. Direct match | Candidate sentence 370 | Source sentence 370

Submitted text	Source passage
Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.	Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.

1286. Direct match | Candidate sentence 371 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

1287. Direct match | Candidate sentence 372 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1288. Direct match | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

1289. Direct match | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

1290. Direct match | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

1291. Direct match | Candidate sentence 376 | Source sentence 376

Submitted text	Source passage
Ceri and J.	Ceri and J.

1292. Direct match | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

1293. Direct match | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

1294. Direct match | Candidate sentence 379 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1295. Direct match | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

1296. Direct match | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

1297. Direct match | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

1298. Direct match | Candidate sentence 383 | Source sentence 420

Submitted text	Source passage
A.	A.

1299. Direct match | Candidate sentence 384 | Source sentence 384

Submitted text	Source passage
Blakeley, P.-Å.	Blakeley, P.-Å.

1300. Direct match | Candidate sentence 385 | Source sentence 385

Submitted text	Source passage
Larson, and F.	Larson, and F.

1301. Direct match | Candidate sentence 386 | Source sentence 386

Submitted text	Source passage
W.	W.

1302. Direct match | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

1303. Direct match | Candidate sentence 388 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

1304. Direct match | Candidate sentence 389 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1305. Direct match | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

1306. Direct match | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

1307. Direct match | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

1308. Direct match | Candidate sentence 393 | Source sentence 393

Submitted text	Source passage
Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.	Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.

1309. Direct match | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

1310. Direct match | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

1311. Direct match | Candidate sentence 396 | Source sentence 396

Submitted text	Source passage
1792–1803, 2015.	1792–1803, 2015.

1312. Direct match | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

1313. Direct match | Candidate sentence 398 | Source sentence 398

Submitted text	Source passage
Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.	Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.

1314. Direct match | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

1315. Direct match | Candidate sentence 400 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1316. Direct match | Candidate sentence 401 | Source sentence 401

Submitted text	Source passage
Very Large Data Bases (VLDB), 2013.	Very Large Data Bases (VLDB), 2013.

1317. Direct match | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

1318. Direct match | Candidate sentence 403 | Source sentence 403

Submitted text	Source passage
Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.	Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.

1319. Direct match | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

1320. Direct match | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

1321. Direct match | Candidate sentence 406 | Source sentence 406

Submitted text	Source passage
28–38, 2015.	28–38, 2015.

1322. Direct match | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

1323. Direct match | Candidate sentence 408 | Source sentence 408

Submitted text	Source passage
Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.	Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.

1324. Direct match | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

1325. Direct match | Candidate sentence 410 | Source sentence 410

Submitted text	Source passage
Operating Systems Principles (SOSP), 2013, pp.	Operating Systems Principles (SOSP), 2013, pp.

1326. Direct match | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

1327. Direct match | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

1328. Direct match | Candidate sentence 413 | Source sentence 413

Submitted text	Source passage
Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.	Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.

1329. Direct match | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

1330. Direct match | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

1331. Direct match | Candidate sentence 416 | Source sentence 416

Submitted text	Source passage
3135–3147, 2021.	3135–3147, 2021.

1332. Direct match | Candidate sentence 417 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1333. Direct match | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

1334. Direct match | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

1335. Direct match | Candidate sentence 420 | Source sentence 420

Submitted text	Source passage
A.	A.

1336. Direct match | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

1337. Direct match | Candidate sentence 422 | Source sentence 422

Submitted text	Source passage
M.	M.

1338. Direct match | Candidate sentence 423 | Source sentence 423

Submitted text	Source passage
Hellerstein, and S.	Hellerstein, and S.

1339. Direct match | Candidate sentence 424 | Source sentence 424

Submitted text	Source passage
Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.	Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.

1340. Direct match | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

1341. Direct match | Candidate sentence 426 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1342. Direct match | Candidate sentence 427 | Source sentence 427

Submitted text	Source passage
Data Engineering (ICDE), 2003.	Data Engineering (ICDE), 2003.

1343. Direct match | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

1344. Direct match | Candidate sentence 429 | Source sentence 429

Submitted text	Source passage
Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.	Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.

1345. Direct match | Candidate sentence 430 | Source sentence 430

Submitted text	Source passage
1st Biennial Conf.	1st Biennial Conf.

1346. Direct match | Candidate sentence 431 | Source sentence 431

Submitted text	Source passage
Innovative Data Systems Research (CIDR), 2003.	Innovative Data Systems Research (CIDR), 2003.

1347. Direct match | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

1348. Direct match | Candidate sentence 433 | Source sentence 433

Submitted text	Source passage
J.	J.

1349. Direct match | Candidate sentence 434 | Source sentence 434

Submitted text	Source passage
Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.	Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.

1350. Direct match | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

1351. Direct match | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1352. Direct match | Candidate sentence 437 | Source sentence 437

Submitted text	Source passage
120–139, 2003.	120–139, 2003.

1353. Direct match | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

1354. Direct match | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

1355. Direct match | Candidate sentence 440 | Source sentence 440

Submitted text	Source passage
Babu, and J.	Babu, and J.

1356. Direct match | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

1357. Direct match | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

1358. Direct match | Candidate sentence 443 | Source sentence 443

Submitted text	Source passage
121–142, 2006.	121–142, 2006.

1359. Direct match | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

1360. Direct match | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

1361. Direct match | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

1362. Direct match | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

1363. Direct match | Candidate sentence 448 | Source sentence 448

Submitted text	Source passage
42–47, 2005.	42–47, 2005.

1364. Direct match | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

1365. Direct match | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

1366. Direct match | Candidate sentence 451 | Source sentence 451

Submitted text	Source passage
4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.	4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.

1367. Direct match | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

1368. Direct match | Candidate sentence 453 | Source sentence 453

Submitted text	Source passage
Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.	Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.

1369. Direct match | Candidate sentence 454 | Source sentence 454

Submitted text	Source passage
34, article 39, 2025.	34, article 39, 2025.

1370. Direct match | Candidate sentence 455 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1371. Direct match | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

1372. Direct match | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

1373. Direct match | Candidate sentence 1 | Source sentence 1

Submitted text	Source passage
Mrs .	Mrs .

1374. Direct match | Candidate sentence 2 | Source sentence 2

Submitted text	Source passage
K .	K .

1375. Direct match | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

1376. Direct match | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1377. Direct match | Candidate sentence 5 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

1378. Direct match | Candidate sentence 6 | Source sentence 6

Submitted text	Source passage
astakalanagamani78@gmail.com	astakalanagamani78@gmail.com

1379. Direct match | Candidate sentence 7 | Source sentence 7

Submitted text	Source passage
P.	P.

1380. Direct match | Candidate sentence 8 | Source sentence 8

Submitted text	Source passage
Srinath	Srinath

1381. Direct match | Candidate sentence 9 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

1382. Direct match | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1383. Direct match | Candidate sentence 11 | Source sentence 11

Submitted text	Source passage
MLR Institute of Technology	MLR Institute of Technology

1384. Direct match | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

1385. Direct match | Candidate sentence 13 | Source sentence 13

Submitted text	Source passage
23r21a05y0@mlrit.ac.in	23r21a05y0@mlrit.ac.in

1386. Direct match | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

1387. Direct match | Candidate sentence 15 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

1388. Direct match | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1389. Direct match | Candidate sentence 17 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

1390. Direct match | Candidate sentence 18 | Source sentence 18

Submitted text	Source passage
23r21a05t0@mlrit.ac.in	23r21a05t0@mlrit.ac.in

1391. Direct match | Candidate sentence 19 | Source sentence 19

Submitted text	Source passage
V.	V.

1392. Direct match | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

1393. Direct match | Candidate sentence 21 | Source sentence 21

Submitted text	Source passage
Dept.	Dept.

1394. Direct match | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1395. Direct match | Candidate sentence 23 | Source sentence 23

Submitted text	Source passage
MLR Institute of Technology Hyderabad, India.	MLR Institute of Technology Hyderabad, India.

1396. Direct match | Candidate sentence 24 | Source sentence 24

Submitted text	Source passage
24r21a0532@mlrit.ac.in	24r21a0532@mlrit.ac.in

1397. Direct match | Candidate sentence 25 | Source sentence 25

Submitted text	Source passage
ABSTRACT	ABSTRACT

1398. Direct match | Candidate sentence 26 | Source sentence 26

Submitted text	Source passage
Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.	Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results.

1399. Direct match | Candidate sentence 27 | Source sentence 27

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.	StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services.

1400. Direct match | Candidate sentence 28 | Source sentence 28

Submitted text	Source passage
It combines message-based event ingestion, stream processing, and persistent state management.	It combines message-based event ingestion, stream processing, and persistent state management.

1401. Direct match | Candidate sentence 29 | Source sentence 29

Submitted text	Source passage
The processing pipeline supports filtering, transformation, aggregation, and event-time operations.	The processing pipeline supports filtering, transformation, aggregation, and event-time operations.

1402. Direct match | Candidate sentence 30 | Source sentence 30

Submitted text	Source passage
An LSM-tree-based state store manages persistent processing state.	An LSM-tree-based state store manages persistent processing state.

1403. Direct match | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

1404. Direct match | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

1405. Direct match | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

1406. Direct match | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

1407. Direct match | Candidate sentence 35 | Source sentence 35

Submitted text	Source passage
The architecture is intended to support low-latency analytical queries and continuous data processing.	The architecture is intended to support low-latency analytical queries and continuous data processing.

1408. Direct match | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

1409. Direct match | Candidate sentence 37 | Source sentence 37

Submitted text	Source passage
StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.	StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources.

1410. Direct match | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

1411. Direct match | Candidate sentence 39 | Source sentence 39

Submitted text	Source passage
The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.	The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure.

1412. Direct match | Candidate sentence 40 | Source sentence 40

Submitted text	Source passage
Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.	Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems.

1413. Direct match | Candidate sentence 41 | Source sentence 41

Submitted text	Source passage
INTRODUCTION	INTRODUCTION

1414. Direct match | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

1415. Direct match | Candidate sentence 43 | Source sentence 43

Submitted text	Source passage
However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.	However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead.

1416. Direct match | Candidate sentence 44 | Source sentence 44

Submitted text	Source passage
These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.	These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation.

1417. Direct match | Candidate sentence 45 | Source sentence 45

Submitted text	Source passage
Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].	Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3].

1418. Direct match | Candidate sentence 46 | Source sentence 46

Submitted text	Source passage
A streaming materialized-view engine can maintain precomputed query results as new events arrive.	A streaming materialized-view engine can maintain precomputed query results as new events arrive.

1419. Direct match | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

1420. Direct match | Candidate sentence 48 | Source sentence 48

Submitted text	Source passage
Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].	Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6].

1421. Direct match | Candidate sentence 49 | Source sentence 49

Submitted text	Source passage
This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.	This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance.

1422. Direct match | Candidate sentence 50 | Source sentence 50

Submitted text	Source passage
The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.	The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics.

1423. Direct match | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

1424. Direct match | Candidate sentence 52 | Source sentence 52

Submitted text	Source passage
Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.	Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads.

1425. Direct match | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

1426. Direct match | Candidate sentence 54 | Source sentence 227

Submitted text	Source passage
1.	1.

1427. Direct match | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

1428. Direct match | Candidate sentence 56 | Source sentence 56

Submitted text	Source passage
However, maintaining consistent analytical results under high event volumes remains a challenge.	However, maintaining consistent analytical results under high event volumes remains a challenge.

1429. Direct match | Candidate sentence 57 | Source sentence 229

Submitted text	Source passage
2.	2.

1430. Direct match | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

1431. Direct match | Candidate sentence 59 | Source sentence 59

Submitted text	Source passage
However, complex aggregations and large-scale workloads can increase update overhead.	However, complex aggregations and large-scale workloads can increase update overhead.

1432. Direct match | Candidate sentence 60 | Source sentence 231

Submitted text	Source passage
3.	3.

1433. Direct match | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

1434. Direct match | Candidate sentence 62 | Source sentence 62

Submitted text	Source passage
However, coordination, network communication, and fault recovery can introduce additional overhead.	However, coordination, network communication, and fault recovery can introduce additional overhead.

1435. Direct match | Candidate sentence 63 | Source sentence 233

Submitted text	Source passage
4.	4.

1436. Direct match | Candidate sentence 64 | Source sentence 64

Submitted text	Source passage
Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.	Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events.

1437. Direct match | Candidate sentence 65 | Source sentence 65

Submitted text	Source passage
However, selecting appropriate watermark policies can affect result completeness and processing latency.	However, selecting appropriate watermark policies can affect result completeness and processing latency.

1438. Direct match | Candidate sentence 66 | Source sentence 235

Submitted text	Source passage
5.	5.

1439. Direct match | Candidate sentence 67 | Source sentence 67

Submitted text	Source passage
Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.	Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures.

1440. Direct match | Candidate sentence 68 | Source sentence 68

Submitted text	Source passage
However, storage performance and recovery mechanisms can influence overall system efficiency.	However, storage performance and recovery mechanisms can influence overall system efficiency.

1441. Direct match | Candidate sentence 69 | Source sentence 237

Submitted text	Source passage
6.	6.

1442. Direct match | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

1443. Direct match | Candidate sentence 71 | Source sentence 71

Submitted text	Source passage
However, log management and recovery procedures can increase storage and processing overhead.	However, log management and recovery procedures can increase storage and processing overhead.

1444. Direct match | Candidate sentence 72 | Source sentence 239

Submitted text	Source passage
7.	7.

1445. Direct match | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

1446. Direct match | Candidate sentence 74 | Source sentence 74

Submitted text	Source passage
However, continuous index updates can increase memory consumption and computational costs.	However, continuous index updates can increase memory consumption and computational costs.

1447. Direct match | Candidate sentence 75 | Source sentence 241

Submitted text	Source passage
8.	8.

1448. Direct match | Candidate sentence 76 | Source sentence 76

Submitted text	Source passage
Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.	Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching.

1449. Direct match | Candidate sentence 77 | Source sentence 77

Submitted text	Source passage
However, maintaining reliable integrations requires ongoing API updates and webhook management.	However, maintaining reliable integrations requires ongoing API updates and webhook management.

1450. Direct match | Candidate sentence 78 | Source sentence 243

Submitted text	Source passage
9.	9.

1451. Direct match | Candidate sentence 79 | Source sentence 79

Submitted text	Source passage
AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.	AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns.

1452. Direct match | Candidate sentence 80 | Source sentence 80

Submitted text	Source passage
However, recommendations may become outdated when priorities and team workloads change.	However, recommendations may become outdated when priorities and team workloads change.

1453. Direct match | Candidate sentence 81 | Source sentence 245

Submitted text	Source passage
10.	10.

1454. Direct match | Candidate sentence 82 | Source sentence 82

Submitted text	Source passage
Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.	Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions.

1455. Direct match | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

1456. Direct match | Candidate sentence 84 | Source sentence 247

Submitted text	Source passage
11.	11.

1457. Direct match | Candidate sentence 85 | Source sentence 85

Submitted text	Source passage
Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.	Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows.

1458. Direct match | Candidate sentence 86 | Source sentence 86

Submitted text	Source passage
However, integrating documentation with daily development activities remains challenging.	However, integrating documentation with daily development activities remains challenging.

1459. Direct match | Candidate sentence 87 | Source sentence 249

Submitted text	Source passage
12.	12.

1460. Direct match | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

1461. Direct match | Candidate sentence 89 | Source sentence 89

Submitted text	Source passage
However, organizations must address data privacy, access control, and data-residency requirements.	However, organizations must address data privacy, access control, and data-residency requirements.

1462. Direct match | Candidate sentence 90 | Source sentence 251

Submitted text	Source passage
13.	13.

1463. Direct match | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

1464. Direct match | Candidate sentence 92 | Source sentence 92

Submitted text	Source passage
However, complex debugging tasks involving multiple codebases may require additional contextual information.	However, complex debugging tasks involving multiple codebases may require additional contextual information.

1465. Direct match | Candidate sentence 93 | Source sentence 253

Submitted text	Source passage
14.	14.

1466. Direct match | Candidate sentence 94 | Source sentence 94

Submitted text	Source passage
Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.	Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords.

1467. Direct match | Candidate sentence 95 | Source sentence 95

Submitted text	Source passage
However, high query volumes can increase memory consumption and affect write performance.	However, high query volumes can increase memory consumption and affect write performance.

1468. Direct match | Candidate sentence 96 | Source sentence 255

Submitted text	Source passage
15.	15.

1469. Direct match | Candidate sentence 97 | Source sentence 97

Submitted text	Source passage
Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.	Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and workflow automation.

1470. Direct match | Candidate sentence 98 | Source sentence 98

Submitted text	Source passage
However, their effectiveness depends on the availability of accurate historical data.	However, their effectiveness depends on the availability of accurate historical data.

1471. Direct match | Candidate sentence 99 | Source sentence 257

Submitted text	Source passage
16.	16.

1472. Direct match | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

1473. Direct match | Candidate sentence 101 | Source sentence 101

Submitted text	Source passage
However, limited historical activity can reduce recommendation accuracy for new team members.	However, limited historical activity can reduce recommendation accuracy for new team members.

1474. Direct match | Candidate sentence 102 | Source sentence 102

Submitted text	Source passage
17.	17.

1475. Direct match | Candidate sentence 103 | Source sentence 103

Submitted text	Source passage
Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.	Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions.

1476. Direct match | Candidate sentence 104 | Source sentence 104

Submitted text	Source passage
However, generated summaries may contain inaccuracies when project context or code versions are outdated.	However, generated summaries may contain inaccuracies when project context or code versions are outdated.

1477. Direct match | Candidate sentence 105 | Source sentence 105

Submitted text	Source passage
18.	18.

1478. Direct match | Candidate sentence 106 | Source sentence 106

Submitted text	Source passage
AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.	AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility.

1479. Direct match | Candidate sentence 107 | Source sentence 107

Submitted text	Source passage
However, integrating different permission models and data structures introduces administrative complexity.	However, integrating different permission models and data structures introduces administrative complexity.

1480. Direct match | Candidate sentence 108 | Source sentence 108

Submitted text	Source passage
19.	19.

1481. Direct match | Candidate sentence 109 | Source sentence 109

Submitted text	Source passage
Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.	Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries.

1482. Direct match | Candidate sentence 110 | Source sentence 110

Submitted text	Source passage
However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.	However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging.

1483. Direct match | Candidate sentence 111 | Source sentence 111

Submitted text	Source passage
20.	20.

1484. Direct match | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

1485. Direct match | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

1486. Direct match | Candidate sentence 114 | Source sentence 114

Submitted text	Source passage
However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.	However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources.

1487. Direct match | Candidate sentence 115 | Source sentence 115

Submitted text	Source passage
Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.	Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications.

1488. Direct match | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

1489. Direct match | Candidate sentence 117 | Source sentence 117

Submitted text	Source passage
Do not present unverified titles or numerical claims as published research.	Do not present unverified titles or numerical claims as published research.

1490. Direct match | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

1491. Direct match | Candidate sentence 119 | Source sentence 119

Submitted text	Source passage
Fragmented Data Processing	Fragmented Data Processing

1492. Direct match | Candidate sentence 120 | Source sentence 120

Submitted text	Source passage
Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.	Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently.

1493. Direct match | Candidate sentence 121 | Source sentence 121

Submitted text	Source passage
Limited Incremental Computation	Limited Incremental Computation

1494. Direct match | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

1495. Direct match | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

1496. Direct match | Candidate sentence 124 | Source sentence 124

Submitted text	Source passage
Inefficient Query Execution	Inefficient Query Execution

1497. Direct match | Candidate sentence 125 | Source sentence 125

Submitted text	Source passage
Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.	Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets.

1498. Direct match | Candidate sentence 126 | Source sentence 126

Submitted text	Source passage
Complex Infrastructure Management	Complex Infrastructure Management

1499. Direct match | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

1500. Direct match | Candidate sentence 128 | Source sentence 128

Submitted text	Source passage
Limited Processing Visibility	Limited Processing Visibility

1501. Direct match | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

1502. Direct match | Candidate sentence 130 | Source sentence 130

Submitted text	Source passage
Challenges in State Recovery	Challenges in State Recovery

1503. Direct match | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

1504. Direct match | Candidate sentence 132 | Source sentence 132

Submitted text	Source passage
Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.	Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity.

1505. Direct match | Candidate sentence 133 | Source sentence 133

Submitted text	Source passage
Resource Utilization and Scalability	Resource Utilization and Scalability

1506. Direct match | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

1507. Direct match | Candidate sentence 135 | Source sentence 135

Submitted text	Source passage
Efficient resource utilization and scalable processing are essential for maintaining performance.	Efficient resource utilization and scalable processing are essential for maintaining performance.

1508. Direct match | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

1509. Direct match | Candidate sentence 137 | Source sentence 137

Submitted text	Source passage
Processing a continuously increasing volume of events while maintaining low latency is challenging.	Processing a continuously increasing volume of events while maintaining low latency is challenging.

1510. Direct match | Candidate sentence 138 | Source sentence 138

Submitted text	Source passage
Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.	Efficient ingestion and processing mechanisms are required to handle high-throughput workloads.

1511. Direct match | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

1512. Direct match | Candidate sentence 140 | Source sentence 140

Submitted text	Source passage
Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.	Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results.

1513. Direct match | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

1514. Direct match | Candidate sentence 142 | Source sentence 142

Submitted text	Source passage
Limited Benchmarking of Self-Hosted Solutions	Limited Benchmarking of Self-Hosted Solutions

1515. Direct match | Candidate sentence 143 | Source sentence 143

Submitted text	Source passage
The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.	The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations.

1516. Direct match | Candidate sentence 144 | Source sentence 144

Submitted text	Source passage
Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.	Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations.

1517. Direct match | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

1518. Direct match | Candidate sentence 146 | Source sentence 146

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.	Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access.

1519. Direct match | Candidate sentence 147 | Source sentence 147

Submitted text	Source passage
Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.	Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead.

1520. Direct match | Candidate sentence 148 | Source sentence 148

Submitted text	Source passage
These limitations make it challenging to obtain up-to-date analytical results as new events arrive.	These limitations make it challenging to obtain up-to-date analytical results as new events arrive.

1521. Direct match | Candidate sentence 149 | Source sentence 149

Submitted text	Source passage
Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.	Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs.

1522. Direct match | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

1523. Direct match | Candidate sentence 151 | Source sentence 151

Submitted text	Source passage
It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.	It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes.

1524. Direct match | Candidate sentence 152 | Source sentence 152

Submitted text	Source passage
Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.	Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation.

1525. Direct match | Candidate sentence 153 | Source sentence 153

Submitted text	Source passage
By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.	By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization.

1526. Direct match | Candidate sentence 154 | Source sentence 154

Submitted text	Source passage
MOTIVATION	MOTIVATION

1527. Direct match | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

1528. Direct match | Candidate sentence 156 | Source sentence 156

Submitted text	Source passage
However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.	However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption.

1529. Direct match | Candidate sentence 157 | Source sentence 157

Submitted text	Source passage
As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.	As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important.

1530. Direct match | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

1531. Direct match | Candidate sentence 159 | Source sentence 159

Submitted text	Source passage
StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.	StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance.

1532. Direct match | Candidate sentence 160 | Source sentence 160

Submitted text	Source passage
By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.	By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead.

1533. Direct match | Candidate sentence 161 | Source sentence 161

Submitted text	Source passage
Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.	Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance.

1534. Direct match | Candidate sentence 162 | Source sentence 162

Submitted text	Source passage
Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.	Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs.

1535. Direct match | Candidate sentence 163 | Source sentence 163

Submitted text	Source passage
StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.	StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services.

1536. Direct match | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

1537. Direct match | Candidate sentence 165 | Source sentence 165

Submitted text	Source passage
Develop a Self-Hosted Streaming Engine	Develop a Self-Hosted Streaming Engine

1538. Direct match | Candidate sentence 166 | Source sentence 166

Submitted text	Source passage
Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.	Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying.

1539. Direct match | Candidate sentence 167 | Source sentence 167

Submitted text	Source passage
The system aims to simplify real-time data processing through a unified architecture.	The system aims to simplify real-time data processing through a unified architecture.

1540. Direct match | Candidate sentence 168 | Source sentence 168

Submitted text	Source passage
Implement Continuous Stream Processing	Implement Continuous Stream Processing

1541. Direct match | Candidate sentence 169 | Source sentence 169

Submitted text	Source passage
Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.	Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation.

1542. Direct match | Candidate sentence 170 | Source sentence 170

Submitted text	Source passage
The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.	The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches.

1543. Direct match | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

1544. Direct match | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

1545. Direct match | Candidate sentence 173 | Source sentence 173

Submitted text	Source passage
This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.	This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets.

1546. Direct match | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

1547. Direct match | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

1548. Direct match | Candidate sentence 176 | Source sentence 176

Submitted text	Source passage
Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.	Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events.

1549. Direct match | Candidate sentence 177 | Source sentence 177

Submitted text	Source passage
Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.	Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results.

1550. Direct match | Candidate sentence 178 | Source sentence 178

Submitted text	Source passage
As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.	As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption.

1551. Direct match | Candidate sentence 179 | Source sentence 179

Submitted text	Source passage
Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.	Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult.

1552. Direct match | Candidate sentence 180 | Source sentence 180

Submitted text	Source passage
Periodic refreshes may prevent users from accessing the latest information immediately.	Periodic refreshes may prevent users from accessing the latest information immediately.

1553. Direct match | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

1554. Direct match | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

1555. Direct match | Candidate sentence 183 | Source sentence 183

Submitted text	Source passage
StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.	StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance.

1556. Direct match | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

1557. Direct match | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

1558. Direct match | Candidate sentence 186 | Source sentence 186

Submitted text	Source passage
The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.	The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub.

1559. Direct match | Candidate sentence 187 | Source sentence 187

Submitted text	Source passage
Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.	Every tool is great on its own, but they all require a separate set of notifications that clutter up the user’s time and confuse the context.

1560. Direct match | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

1561. Direct match | Candidate sentence 189 | Source sentence 189

Submitted text	Source passage
StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.	StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform.

1562. Direct match | Candidate sentence 190 | Source sentence 190

Submitted text	Source passage
Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.	Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion.

1563. Direct match | Candidate sentence 191 | Source sentence 191

Submitted text	Source passage
The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results	The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results

1564. Direct match | Candidate sentence 192 | Source sentence 192

Submitted text	Source passage
.	.

1565. Direct match | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

1566. Direct match | Candidate sentence 194 | Source sentence 194

Submitted text	Source passage
A persistent state store maintains the data required for processing and view updates.	A persistent state store maintains the data required for processing and view updates.

1567. Direct match | Candidate sentence 195 | Source sentence 195

Submitted text	Source passage
Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.	Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events.

1568. Direct match | Candidate sentence 196 | Source sentence 196

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.	Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset.

1569. Direct match | Candidate sentence 197 | Source sentence 197

Submitted text	Source passage
The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.	The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity.

1570. Direct match | Candidate sentence 198 | Source sentence 198

Submitted text	Source passage
Backend APIs connect the dashboard with the processing engine and state store.	Backend APIs connect the dashboard with the processing engine and state store.

1571. Direct match | Candidate sentence 199 | Source sentence 199

Submitted text	Source passage
REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.	REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status.

1572. Direct match | Candidate sentence 200 | Source sentence 200

Submitted text	Source passage
Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.	Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data.

1573. Direct match | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

1574. Direct match | Candidate sentence 202 | Source sentence 202

Submitted text	Source passage
The following steps should be taken while using StreamView:	The following steps should be taken while using StreamView:

1575. Direct match | Candidate sentence 203 | Source sentence 203

Submitted text	Source passage
Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.	Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance.

1576. Direct match | Candidate sentence 204 | Source sentence 204

Submitted text	Source passage
Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.	Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings.

1577. Direct match | Candidate sentence 205 | Source sentence 205

Submitted text	Source passage
Step 3: Ingest Events – Send events from a producer application to the configured message broker.	Step 3: Ingest Events – Send events from a producer application to the configured message broker.

1578. Direct match | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

1579. Direct match | Candidate sentence 207 | Source sentence 207

Submitted text	Source passage
Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.	Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements.

1580. Direct match | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

1581. Direct match | Candidate sentence 209 | Source sentence 209

Submitted text	Source passage
The engine processes incoming events and maintains the corresponding results.	The engine processes incoming events and maintains the corresponding results.

1582. Direct match | Candidate sentence 210 | Source sentence 210

Submitted text	Source passage
Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.	Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset.

1583. Direct match | Candidate sentence 211 | Source sentence 211

Submitted text	Source passage
Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.	Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics.

1584. Direct match | Candidate sentence 212 | Source sentence 212

Submitted text	Source passage
Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.	Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure.

1585. Direct match | Candidate sentence 213 | Source sentence 213

Submitted text	Source passage
Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.	Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness.

1586. Direct match | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

1587. Direct match | Candidate sentence 215 | Source sentence 215

Submitted text	Source passage
FIG 8.1 Proposed workflow Diagram	FIG 8.1 Proposed workflow Diagram

1588. Direct match | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

1589. Direct match | Candidate sentence 217 | Source sentence 217

Submitted text	Source passage
Users can observe system performance and identify processing delays or potential bottlenecks.	Users can observe system performance and identify processing delays or potential bottlenecks.

1590. Direct match | Candidate sentence 218 | Source sentence 218

Submitted text	Source passage
Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.	Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements.

1591. Direct match | Candidate sentence 219 | Source sentence 219

Submitted text	Source passage
The system applies the updated configurations based on its supported configuration and execution mechanisms.	The system applies the updated configurations based on its supported configuration and execution mechanisms.

1592. Direct match | Candidate sentence 220 | Source sentence 220

Submitted text	Source passage
Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.	Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption.

1593. Direct match | Candidate sentence 221 | Source sentence 221

Submitted text	Source passage
These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.	These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization.

1594. Direct match | Candidate sentence 222 | Source sentence 222

Submitted text	Source passage
ALGORITHM	ALGORITHM

1595. Direct match | Candidate sentence 223 | Source sentence 223

Submitted text	Source passage
The following steps describe the processing workflow of the StreamView application.	The following steps describe the processing workflow of the StreamView application.

1596. Direct match | Candidate sentence 224 | Source sentence 224

Submitted text	Source passage
Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline	Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline

1597. Direct match | Candidate sentence 225 | Source sentence 225

Submitted text	Source passage
Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)	Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\)

1598. Direct match | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

1599. Direct match | Candidate sentence 227 | Source sentence 227

Submitted text	Source passage
1.	1.

1600. Direct match | Candidate sentence 228 | Source sentence 228

Submitted text	Source passage
Initialize the message broker, stream-processing engine, persistent state store, and query interface.	Initialize the message broker, stream-processing engine, persistent state store, and query interface.

1601. Direct match | Candidate sentence 229 | Source sentence 229

Submitted text	Source passage
2.	2.

1602. Direct match | Candidate sentence 230 | Source sentence 230

Submitted text	Source passage
Configure the event stream and subscribe to the required message topic.	Configure the event stream and subscribe to the required message topic.

1603. Direct match | Candidate sentence 231 | Source sentence 231

Submitted text	Source passage
3.	3.

1604. Direct match | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

1605. Direct match | Candidate sentence 233 | Source sentence 233

Submitted text	Source passage
4.	4.

1606. Direct match | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

1607. Direct match | Candidate sentence 235 | Source sentence 235

Submitted text	Source passage
5.	5.

1608. Direct match | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

1609. Direct match | Candidate sentence 237 | Source sentence 237

Submitted text	Source passage
6.	6.

1610. Direct match | Candidate sentence 238 | Source sentence 238

Submitted text	Source passage
Apply the configured filtering and transformation operations to \(e\).	Apply the configured filtering and transformation operations to \(e\).

1611. Direct match | Candidate sentence 239 | Source sentence 239

Submitted text	Source passage
7.	7.

1612. Direct match | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

1613. Direct match | Candidate sentence 241 | Source sentence 241

Submitted text	Source passage
8.	8.

1614. Direct match | Candidate sentence 242 | Source sentence 242

Submitted text	Source passage
Identify the materialized views affected by the processed event.	Identify the materialized views affected by the processed event.

1615. Direct match | Candidate sentence 243 | Source sentence 243

Submitted text	Source passage
9.	9.

1616. Direct match | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

1617. Direct match | Candidate sentence 245 | Source sentence 245

Submitted text	Source passage
10.	10.

1618. Direct match | Candidate sentence 246 | Source sentence 246

Submitted text	Source passage
Persist the updated state and view data using the supported storage mechanism.	Persist the updated state and view data using the supported storage mechanism.

1619. Direct match | Candidate sentence 247 | Source sentence 247

Submitted text	Source passage
11.	11.

1620. Direct match | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

1621. Direct match | Candidate sentence 249 | Source sentence 249

Submitted text	Source passage
12.	12.

1622. Direct match | Candidate sentence 250 | Source sentence 250

Submitted text	Source passage
On receiving a query \(Q\) from the client interface:	On receiving a query \(Q\) from the client interface:

1623. Direct match | Candidate sentence 251 | Source sentence 251

Submitted text	Source passage
13.	13.

1624. Direct match | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

1625. Direct match | Candidate sentence 253 | Source sentence 253

Submitted text	Source passage
14.	14.

1626. Direct match | Candidate sentence 254 | Source sentence 254

Submitted text	Source passage
Return the query results \(R\) to the client interface.	Return the query results \(R\) to the client interface.

1627. Direct match | Candidate sentence 255 | Source sentence 255

Submitted text	Source passage
15.	15.

1628. Direct match | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

1629. Direct match | Candidate sentence 257 | Source sentence 257

Submitted text	Source passage
16.	16.

1630. Direct match | Candidate sentence 258 | Source sentence 258

Submitted text	Source passage
Repeat the process until the stream is stopped or the system is shut down.	Repeat the process until the stream is stopped or the system is shut down.

1631. Direct match | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

1632. Direct match | Candidate sentence 260 | Source sentence 260

Submitted text	Source passage
Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.	Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation.

1633. Direct match | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

1634. Direct match | Candidate sentence 262 | Source sentence 262

Submitted text	Source passage
StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.	StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results.

1635. Direct match | Candidate sentence 263 | Source sentence 263

Submitted text	Source passage
The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.	The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface.

1636. Direct match | Candidate sentence 264 | Source sentence 264

Submitted text	Source passage
By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.	By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services.

1637. Direct match | Candidate sentence 265 | Source sentence 265

Submitted text	Source passage
From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.	From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results.

1638. Direct match | Candidate sentence 266 | Source sentence 266

Submitted text	Source passage
The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.	The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset.

1639. Direct match | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

1640. Direct match | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

1641. Direct match | Candidate sentence 269 | Source sentence 269

Submitted text	Source passage
Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.	Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization.

1642. Direct match | Candidate sentence 270 | Source sentence 270

Submitted text	Source passage
This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.	This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics.

1643. Direct match | Candidate sentence 271 | Source sentence 271

Submitted text	Source passage
The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.	The system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset.

1644. Direct match | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

1645. Direct match | Candidate sentence 273 | Source sentence 273

Submitted text	Source passage
Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.	Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization.

1646. Direct match | Candidate sentence 274 | Source sentence 274

Submitted text	Source passage
By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.	By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs.

1647. Direct match | Candidate sentence 275 | Source sentence 275

Submitted text	Source passage
StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.	StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results.

1648. Direct match | Candidate sentence 276 | Source sentence 276

Submitted text	Source passage
It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.	It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation.

1649. Direct match | Candidate sentence 277 | Source sentence 277

Submitted text	Source passage
Materialized views are updated incrementally as new events arrive, reducing repeated computation.	Materialized views are updated incrementally as new events arrive, reducing repeated computation.

1650. Direct match | Candidate sentence 278 | Source sentence 278

Submitted text	Source passage
The dashboard enables users to manage streams, execute queries, and monitor system performance.	The dashboard enables users to manage streams, execute queries, and monitor system performance.

1651. Direct match | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

1652. Direct match | Candidate sentence 280 | Source sentence 280

Submitted text	Source passage
The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.	The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring.

1653. Direct match | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

1654. Direct match | Candidate sentence 282 | Source sentence 282

Submitted text	Source passage
The application backend manages business logic, API requests, stream configurations, and communication between components.	The application backend manages business logic, API requests, stream configurations, and communication between components.

1655. Direct match | Candidate sentence 283 | Source sentence 283

Submitted text	Source passage
The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.	The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation.

1656. Direct match | Candidate sentence 284 | Source sentence 284

Submitted text	Source passage
The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.	The persistent storage layer maintains processing state and materialized-view results for efficient retrieval.

1657. Direct match | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

1658. Direct match | Candidate sentence 286 | Source sentence 286

Submitted text	Source passage
These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.	These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment.

1659. Direct match | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

1660. Direct match | Candidate sentence 288 | Source sentence 288

Submitted text	Source passage
RESULTS	RESULTS

1661. Direct match | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

1662. Direct match | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

1663. Direct match | Candidate sentence 291 | Source sentence 291

Submitted text	Source passage
The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.	The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information.

1664. Direct match | Candidate sentence 292 | Source sentence 292

Submitted text	Source passage
The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.	The query interface allows users to retrieve analytical results from materialized views using SQL-like queries.

1665. Direct match | Candidate sentence 293 | Source sentence 293

Submitted text	Source passage
These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.	These components work together to provide a unified environment for managing continuous data processing and accessing analytical information.

1666. Direct match | Candidate sentence 294 | Source sentence 294

Submitted text	Source passage
The monitoring and event ingestion modules provide additional visibility into system operations.	The monitoring and event ingestion modules provide additional visibility into system operations.

1667. Direct match | Candidate sentence 295 | Source sentence 295

Submitted text	Source passage
Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.	Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated.

1668. Direct match | Candidate sentence 296 | Source sentence 296

Submitted text	Source passage
The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.	The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization.

1669. Direct match | Candidate sentence 297 | Source sentence 297

Submitted text	Source passage
These measurements help evaluate system performance and identify potential processing bottlenecks.	These measurements help evaluate system performance and identify potential processing bottlenecks.

1670. Direct match | Candidate sentence 298 | Source sentence 298

Submitted text	Source passage
Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.	Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance.

1671. Direct match | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

1672. Direct match | Candidate sentence 300 | Source sentence 300

Submitted text	Source passage
FIG 12.1: Result	FIG 12.1: Result

1673. Direct match | Candidate sentence 301 | Source sentence 301

Submitted text	Source passage
By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.	By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge.

1674. Direct match | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

1675. Direct match | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

1676. Direct match | Candidate sentence 304 | Source sentence 304

Submitted text	Source passage
In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.	In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads.

1677. Direct match | Candidate sentence 305 | Source sentence 305

Submitted text	Source passage
Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.	Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency.

1678. Direct match | Candidate sentence 306 | Source sentence 306

Submitted text	Source passage
Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.	Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability.

1679. Direct match | Candidate sentence 307 | Source sentence 307

Submitted text	Source passage
Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.	Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks.

1680. Direct match | Candidate sentence 308 | Source sentence 308

Submitted text	Source passage
Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.	Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments.

1681. Direct match | Candidate sentence 309 | Source sentence 309

Submitted text	Source passage
Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.	Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection.

1682. Direct match | Candidate sentence 310 | Source sentence 310

Submitted text	Source passage
Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.	Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform.

1683. Direct match | Candidate sentence 311 | Source sentence 311

Submitted text	Source passage
CONCLUSION	CONCLUSION

1684. Direct match | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

1685. Direct match | Candidate sentence 313 | Source sentence 313

Submitted text	Source passage
Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.	Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives.

1686. Direct match | Candidate sentence 314 | Source sentence 314

Submitted text	Source passage
The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.	The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval.

1687. Direct match | Candidate sentence 315 | Source sentence 315

Submitted text	Source passage
Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.	Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance.

1688. Direct match | Candidate sentence 316 | Source sentence 316

Submitted text	Source passage
This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.	This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results.

1689. Direct match | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

1690. Direct match | Candidate sentence 318 | Source sentence 318

Submitted text	Source passage
REFERENCES	REFERENCES

1691. Direct match | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

1692. Direct match | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

1693. Direct match | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

1694. Direct match | Candidate sentence 322 | Source sentence 322

Submitted text	Source passage
Kalavri, and A.	Kalavri, and A.

1695. Direct match | Candidate sentence 323 | Source sentence 323

Submitted text	Source passage
Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.	Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol.

1696. Direct match | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

1697. Direct match | Candidate sentence 325 | Source sentence 325

Submitted text	Source passage
507–541, 2024.	507–541, 2024.

1698. Direct match | Candidate sentence 326 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1699. Direct match | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

1700. Direct match | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

1701. Direct match | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

1702. Direct match | Candidate sentence 330 | Source sentence 330

Submitted text	Source passage
Golab, and M.	Golab, and M.

1703. Direct match | Candidate sentence 331 | Source sentence 331

Submitted text	Source passage
T.	T.

1704. Direct match | Candidate sentence 332 | Source sentence 332

Submitted text	Source passage
Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.	Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol.

1705. Direct match | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

1706. Direct match | Candidate sentence 334 | Source sentence 334

Submitted text	Source passage
55–67, 2017.	55–67, 2017.

1707. Direct match | Candidate sentence 335 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1708. Direct match | Candidate sentence 336 | Source sentence 336

Submitted text	Source passage
Available: https://doi.org/10.1016/j.is.2017.07.002	Available: https://doi.org/10.1016/j.is.2017.07.002

1709. Direct match | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

1710. Direct match | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

1711. Direct match | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

1712. Direct match | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

1713. Direct match | Candidate sentence 341 | Source sentence 341

Submitted text	Source passage
Ryzhyk, and V.	Ryzhyk, and V.

1714. Direct match | Candidate sentence 342 | Source sentence 342

Submitted text	Source passage
Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.	Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol.

1715. Direct match | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

1716. Direct match | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

1717. Direct match | Candidate sentence 345 | Source sentence 345

Submitted text	Source passage
1601–1614, 2023.	1601–1614, 2023.

1718. Direct match | Candidate sentence 346 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1719. Direct match | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

1720. Direct match | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

1721. Direct match | Candidate sentence 349 | Source sentence 349

Submitted text	Source passage
Gupta and I.	Gupta and I.

1722. Direct match | Candidate sentence 350 | Source sentence 359

Submitted text	Source passage
S.	S.

1723. Direct match | Candidate sentence 351 | Source sentence 351

Submitted text	Source passage
Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.	Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol.

1724. Direct match | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

1725. Direct match | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1726. Direct match | Candidate sentence 354 | Source sentence 354

Submitted text	Source passage
3–18, 1995.	3–18, 1995.

1727. Direct match | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

1728. Direct match | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

1729. Direct match | Candidate sentence 357 | Source sentence 359

Submitted text	Source passage
S.	S.

1730. Direct match | Candidate sentence 358 | Source sentence 358

Submitted text	Source passage
Mumick, and V.	Mumick, and V.

1731. Direct match | Candidate sentence 359 | Source sentence 359

Submitted text	Source passage
S.	S.

1732. Direct match | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

1733. Direct match | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

1734. Direct match | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1735. Direct match | Candidate sentence 363 | Source sentence 363

Submitted text	Source passage
157–166, 1993.	157–166, 1993.

1736. Direct match | Candidate sentence 364 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1737. Direct match | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

1738. Direct match | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

1739. Direct match | Candidate sentence 367 | Source sentence 420

Submitted text	Source passage
A.	A.

1740. Direct match | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

1741. Direct match | Candidate sentence 369 | Source sentence 369

Submitted text	Source passage
Srivastava, and S.	Srivastava, and S.

1742. Direct match | Candidate sentence 370 | Source sentence 370

Submitted text	Source passage
Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.	Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc.

1743. Direct match | Candidate sentence 371 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

1744. Direct match | Candidate sentence 372 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1745. Direct match | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

1746. Direct match | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

1747. Direct match | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

1748. Direct match | Candidate sentence 376 | Source sentence 376

Submitted text	Source passage
Ceri and J.	Ceri and J.

1749. Direct match | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

1750. Direct match | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

1751. Direct match | Candidate sentence 379 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1752. Direct match | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

1753. Direct match | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

1754. Direct match | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

1755. Direct match | Candidate sentence 383 | Source sentence 420

Submitted text	Source passage
A.	A.

1756. Direct match | Candidate sentence 384 | Source sentence 384

Submitted text	Source passage
Blakeley, P.-Å.	Blakeley, P.-Å.

1757. Direct match | Candidate sentence 385 | Source sentence 385

Submitted text	Source passage
Larson, and F.	Larson, and F.

1758. Direct match | Candidate sentence 386 | Source sentence 386

Submitted text	Source passage
W.	W.

1759. Direct match | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

1760. Direct match | Candidate sentence 388 | Source sentence 388

Submitted text	Source passage
ACM SIGMOD Int.	ACM SIGMOD Int.

1761. Direct match | Candidate sentence 389 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1762. Direct match | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

1763. Direct match | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

1764. Direct match | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

1765. Direct match | Candidate sentence 393 | Source sentence 393

Submitted text	Source passage
Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.	Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol.

1766. Direct match | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

1767. Direct match | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

1768. Direct match | Candidate sentence 396 | Source sentence 396

Submitted text	Source passage
1792–1803, 2015.	1792–1803, 2015.

1769. Direct match | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

1770. Direct match | Candidate sentence 398 | Source sentence 398

Submitted text	Source passage
Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.	Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc.

1771. Direct match | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

1772. Direct match | Candidate sentence 400 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1773. Direct match | Candidate sentence 401 | Source sentence 401

Submitted text	Source passage
Very Large Data Bases (VLDB), 2013.	Very Large Data Bases (VLDB), 2013.

1774. Direct match | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

1775. Direct match | Candidate sentence 403 | Source sentence 403

Submitted text	Source passage
Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.	Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol.

1776. Direct match | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

1777. Direct match | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

1778. Direct match | Candidate sentence 406 | Source sentence 406

Submitted text	Source passage
28–38, 2015.	28–38, 2015.

1779. Direct match | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

1780. Direct match | Candidate sentence 408 | Source sentence 408

Submitted text	Source passage
Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.	Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc.

1781. Direct match | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

1782. Direct match | Candidate sentence 410 | Source sentence 410

Submitted text	Source passage
Operating Systems Principles (SOSP), 2013, pp.	Operating Systems Principles (SOSP), 2013, pp.

1783. Direct match | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

1784. Direct match | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

1785. Direct match | Candidate sentence 413 | Source sentence 413

Submitted text	Source passage
Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.	Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol.

1786. Direct match | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

1787. Direct match | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

1788. Direct match | Candidate sentence 416 | Source sentence 416

Submitted text	Source passage
3135–3147, 2021.	3135–3147, 2021.

1789. Direct match | Candidate sentence 417 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1790. Direct match | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

1791. Direct match | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

1792. Direct match | Candidate sentence 420 | Source sentence 420

Submitted text	Source passage
A.	A.

1793. Direct match | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

1794. Direct match | Candidate sentence 422 | Source sentence 422

Submitted text	Source passage
M.	M.

1795. Direct match | Candidate sentence 423 | Source sentence 423

Submitted text	Source passage
Hellerstein, and S.	Hellerstein, and S.

1796. Direct match | Candidate sentence 424 | Source sentence 424

Submitted text	Source passage
Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.	Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc.

1797. Direct match | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

1798. Direct match | Candidate sentence 426 | Source sentence 426

Submitted text	Source passage
Conf.	Conf.

1799. Direct match | Candidate sentence 427 | Source sentence 427

Submitted text	Source passage
Data Engineering (ICDE), 2003.	Data Engineering (ICDE), 2003.

1800. Direct match | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

1801. Direct match | Candidate sentence 429 | Source sentence 429

Submitted text	Source passage
Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.	Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc.

1802. Direct match | Candidate sentence 430 | Source sentence 430

Submitted text	Source passage
1st Biennial Conf.	1st Biennial Conf.

1803. Direct match | Candidate sentence 431 | Source sentence 431

Submitted text	Source passage
Innovative Data Systems Research (CIDR), 2003.	Innovative Data Systems Research (CIDR), 2003.

1804. Direct match | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

1805. Direct match | Candidate sentence 433 | Source sentence 433

Submitted text	Source passage
J.	J.

1806. Direct match | Candidate sentence 434 | Source sentence 434

Submitted text	Source passage
Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.	Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol.

1807. Direct match | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

1808. Direct match | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1809. Direct match | Candidate sentence 437 | Source sentence 437

Submitted text	Source passage
120–139, 2003.	120–139, 2003.

1810. Direct match | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

1811. Direct match | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

1812. Direct match | Candidate sentence 440 | Source sentence 440

Submitted text	Source passage
Babu, and J.	Babu, and J.

1813. Direct match | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

1814. Direct match | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

1815. Direct match | Candidate sentence 443 | Source sentence 443

Submitted text	Source passage
121–142, 2006.	121–142, 2006.

1816. Direct match | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

1817. Direct match | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

1818. Direct match | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

1819. Direct match | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

1820. Direct match | Candidate sentence 448 | Source sentence 448

Submitted text	Source passage
42–47, 2005.	42–47, 2005.

1821. Direct match | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

1822. Direct match | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

1823. Direct match | Candidate sentence 451 | Source sentence 451

Submitted text	Source passage
4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.	4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012.

1824. Direct match | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

1825. Direct match | Candidate sentence 453 | Source sentence 453

Submitted text	Source passage
Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.	Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol.

1826. Direct match | Candidate sentence 454 | Source sentence 454

Submitted text	Source passage
34, article 39, 2025.	34, article 39, 2025.

1827. Direct match | Candidate sentence 455 | Source sentence 455

Submitted text	Source passage
[Online].	[Online].

1828. Direct match | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

1829. Near duplicate | Candidate sentence 4 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1830. Near duplicate | Candidate sentence 10 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1831. Near duplicate | Candidate sentence 16 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1832. Near duplicate | Candidate sentence 22 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1833. Near duplicate | Candidate sentence 4 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1834. Near duplicate | Candidate sentence 10 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1835. Near duplicate | Candidate sentence 16 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1836. Near duplicate | Candidate sentence 22 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1837. Near duplicate | Candidate sentence 4 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1838. Near duplicate | Candidate sentence 10 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1839. Near duplicate | Candidate sentence 16 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1840. Near duplicate | Candidate sentence 22 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1841. Near duplicate | Candidate sentence 4 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1842. Near duplicate | Candidate sentence 10 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1843. Near duplicate | Candidate sentence 16 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1844. Near duplicate | Candidate sentence 22 | Source sentence 13

Submitted text	Source passage
of Computer Science and Engineering	Bachelor of Technology (B.Tech) in Computer Science and Engineering

1845. Near duplicate | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

1846. Near duplicate | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

1847. Near duplicate | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1848. Near duplicate | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1849. Near duplicate | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

1850. Near duplicate | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

1851. Near duplicate | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1852. Near duplicate | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

1853. Near duplicate | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

1854. Near duplicate | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

1855. Near duplicate | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

1856. Near duplicate | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

1857. Near duplicate | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

1858. Near duplicate | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

1859. Near duplicate | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

1860. Near duplicate | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

1861. Near duplicate | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

1862. Near duplicate | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

1863. Near duplicate | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

1864. Near duplicate | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

1865. Near duplicate | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

1866. Near duplicate | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

1867. Near duplicate | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

1868. Near duplicate | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

1869. Near duplicate | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

1870. Near duplicate | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

1871. Near duplicate | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

1872. Near duplicate | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

1873. Near duplicate | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

1874. Near duplicate | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

1875. Near duplicate | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

1876. Near duplicate | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

1877. Near duplicate | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

1878. Near duplicate | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

1879. Near duplicate | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

1880. Near duplicate | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

1881. Near duplicate | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

1882. Near duplicate | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

1883. Near duplicate | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

1884. Near duplicate | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

1885. Near duplicate | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

1886. Near duplicate | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

1887. Near duplicate | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

1888. Near duplicate | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

1889. Near duplicate | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

1890. Near duplicate | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

1891. Near duplicate | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

1892. Near duplicate | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

1893. Near duplicate | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

1894. Near duplicate | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

1895. Near duplicate | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

1896. Near duplicate | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

1897. Near duplicate | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

1898. Near duplicate | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

1899. Near duplicate | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

1900. Near duplicate | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

1901. Near duplicate | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

1902. Near duplicate | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

1903. Near duplicate | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

1904. Near duplicate | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

1905. Near duplicate | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

1906. Near duplicate | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

1907. Near duplicate | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

1908. Near duplicate | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

1909. Near duplicate | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

1910. Near duplicate | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

1911. Near duplicate | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

1912. Near duplicate | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

1913. Near duplicate | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

1914. Near duplicate | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

1915. Near duplicate | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

1916. Near duplicate | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

1917. Near duplicate | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

1918. Near duplicate | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

1919. Near duplicate | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

1920. Near duplicate | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

1921. Near duplicate | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

1922. Near duplicate | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

1923. Near duplicate | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

1924. Near duplicate | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

1925. Near duplicate | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

1926. Near duplicate | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

1927. Near duplicate | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

1928. Near duplicate | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

1929. Near duplicate | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

1930. Near duplicate | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

1931. Near duplicate | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

1932. Near duplicate | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

1933. Near duplicate | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

1934. Near duplicate | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

1935. Near duplicate | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

1936. Near duplicate | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

1937. Near duplicate | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

1938. Near duplicate | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

1939. Near duplicate | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

1940. Near duplicate | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

1941. Near duplicate | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

1942. Near duplicate | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

1943. Near duplicate | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

1944. Near duplicate | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

1945. Near duplicate | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

1946. Near duplicate | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

1947. Near duplicate | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

1948. Near duplicate | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1949. Near duplicate | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

1950. Near duplicate | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

1951. Near duplicate | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

1952. Near duplicate | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

1953. Near duplicate | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1954. Near duplicate | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

1955. Near duplicate | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

1956. Near duplicate | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

1957. Near duplicate | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

1958. Near duplicate | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

1959. Near duplicate | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

1960. Near duplicate | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

1961. Near duplicate | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

1962. Near duplicate | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

1963. Near duplicate | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

1964. Near duplicate | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

1965. Near duplicate | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

1966. Near duplicate | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

1967. Near duplicate | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

1968. Near duplicate | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

1969. Near duplicate | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

1970. Near duplicate | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

1971. Near duplicate | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

1972. Near duplicate | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

1973. Near duplicate | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

1974. Near duplicate | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

1975. Near duplicate | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

1976. Near duplicate | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

1977. Near duplicate | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

1978. Near duplicate | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

1979. Near duplicate | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

1980. Near duplicate | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

1981. Near duplicate | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

1982. Near duplicate | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

1983. Near duplicate | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

1984. Near duplicate | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

1985. Near duplicate | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

1986. Near duplicate | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

1987. Near duplicate | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

1988. Near duplicate | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

1989. Near duplicate | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

1990. Near duplicate | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

1991. Near duplicate | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

1992. Near duplicate | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

1993. Near duplicate | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

1994. Near duplicate | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

1995. Near duplicate | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

1996. Near duplicate | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

1997. Near duplicate | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

1998. Near duplicate | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

1999. Near duplicate | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

2000. Near duplicate | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

2001. Near duplicate | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

2002. Near duplicate | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

2003. Near duplicate | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

2004. Near duplicate | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2005. Near duplicate | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2006. Near duplicate | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

2007. Near duplicate | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

2008. Near duplicate | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2009. Near duplicate | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

2010. Near duplicate | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2011. Near duplicate | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

2012. Near duplicate | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

2013. Near duplicate | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

2014. Near duplicate | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

2015. Near duplicate | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

2016. Near duplicate | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

2017. Near duplicate | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

2018. Near duplicate | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

2019. Near duplicate | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

2020. Near duplicate | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

2021. Near duplicate | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

2022. Near duplicate | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

2023. Near duplicate | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

2024. Near duplicate | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

2025. Near duplicate | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

2026. Near duplicate | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

2027. Near duplicate | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

2028. Near duplicate | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

2029. Near duplicate | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

2030. Near duplicate | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

2031. Near duplicate | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

2032. Near duplicate | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

2033. Near duplicate | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

2034. Near duplicate | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

2035. Near duplicate | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

2036. Near duplicate | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

2037. Near duplicate | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

2038. Near duplicate | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

2039. Near duplicate | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

2040. Near duplicate | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

2041. Near duplicate | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

2042. Near duplicate | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

2043. Near duplicate | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

2044. Near duplicate | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

2045. Near duplicate | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

2046. Near duplicate | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

2047. Near duplicate | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

2048. Near duplicate | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

2049. Near duplicate | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

2050. Near duplicate | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

2051. Near duplicate | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

2052. Near duplicate | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

2053. Near duplicate | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

2054. Near duplicate | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

2055. Near duplicate | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

2056. Near duplicate | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

2057. Near duplicate | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

2058. Near duplicate | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

2059. Near duplicate | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

2060. Near duplicate | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

2061. Near duplicate | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

2062. Near duplicate | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

2063. Near duplicate | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

2064. Near duplicate | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

2065. Near duplicate | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

2066. Near duplicate | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

2067. Near duplicate | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

2068. Near duplicate | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

2069. Near duplicate | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

2070. Near duplicate | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

2071. Near duplicate | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

2072. Near duplicate | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

2073. Near duplicate | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

2074. Near duplicate | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

2075. Near duplicate | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

2076. Near duplicate | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

2077. Near duplicate | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

2078. Near duplicate | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

2079. Near duplicate | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

2080. Near duplicate | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

2081. Near duplicate | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

2082. Near duplicate | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

2083. Near duplicate | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

2084. Near duplicate | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

2085. Near duplicate | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

2086. Near duplicate | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

2087. Near duplicate | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

2088. Near duplicate | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

2089. Near duplicate | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

2090. Near duplicate | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

2091. Near duplicate | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

2092. Near duplicate | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

2093. Near duplicate | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

2094. Near duplicate | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

2095. Near duplicate | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

2096. Near duplicate | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

2097. Near duplicate | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

2098. Near duplicate | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

2099. Near duplicate | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

2100. Near duplicate | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

2101. Near duplicate | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

2102. Near duplicate | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

2103. Near duplicate | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

2104. Near duplicate | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

2105. Near duplicate | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2106. Near duplicate | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

2107. Near duplicate | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

2108. Near duplicate | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

2109. Near duplicate | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

2110. Near duplicate | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2111. Near duplicate | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

2112. Near duplicate | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

2113. Near duplicate | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

2114. Near duplicate | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

2115. Near duplicate | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

2116. Near duplicate | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

2117. Near duplicate | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

2118. Near duplicate | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

2119. Near duplicate | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

2120. Near duplicate | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

2121. Near duplicate | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

2122. Near duplicate | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

2123. Near duplicate | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

2124. Near duplicate | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

2125. Near duplicate | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

2126. Near duplicate | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

2127. Near duplicate | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

2128. Near duplicate | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

2129. Near duplicate | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

2130. Near duplicate | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

2131. Near duplicate | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

2132. Near duplicate | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

2133. Near duplicate | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

2134. Near duplicate | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

2135. Near duplicate | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

2136. Near duplicate | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

2137. Near duplicate | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

2138. Near duplicate | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

2139. Near duplicate | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

2140. Near duplicate | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

2141. Near duplicate | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

2142. Near duplicate | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

2143. Near duplicate | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

2144. Near duplicate | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

2145. Near duplicate | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

2146. Near duplicate | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2147. Near duplicate | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

2148. Near duplicate | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

2149. Near duplicate | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

2150. Near duplicate | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

2151. Near duplicate | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

2152. Near duplicate | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

2153. Near duplicate | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

2154. Near duplicate | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

2155. Near duplicate | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

2156. Near duplicate | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

2157. Near duplicate | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

2158. Near duplicate | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

2159. Near duplicate | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

2160. Near duplicate | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

2161. Near duplicate | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2162. Near duplicate | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2163. Near duplicate | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

2164. Near duplicate | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

2165. Near duplicate | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2166. Near duplicate | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

2167. Near duplicate | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2168. Near duplicate | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

2169. Near duplicate | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

2170. Near duplicate | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

2171. Near duplicate | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

2172. Near duplicate | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

2173. Near duplicate | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

2174. Near duplicate | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

2175. Near duplicate | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

2176. Near duplicate | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

2177. Near duplicate | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

2178. Near duplicate | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

2179. Near duplicate | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

2180. Near duplicate | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

2181. Near duplicate | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

2182. Near duplicate | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

2183. Near duplicate | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

2184. Near duplicate | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

2185. Near duplicate | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

2186. Near duplicate | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

2187. Near duplicate | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

2188. Near duplicate | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

2189. Near duplicate | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

2190. Near duplicate | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

2191. Near duplicate | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

2192. Near duplicate | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

2193. Near duplicate | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

2194. Near duplicate | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

2195. Near duplicate | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

2196. Near duplicate | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

2197. Near duplicate | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

2198. Near duplicate | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

2199. Near duplicate | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

2200. Near duplicate | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

2201. Near duplicate | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

2202. Near duplicate | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

2203. Near duplicate | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

2204. Near duplicate | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

2205. Near duplicate | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

2206. Near duplicate | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

2207. Near duplicate | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

2208. Near duplicate | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

2209. Near duplicate | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

2210. Near duplicate | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

2211. Near duplicate | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

2212. Near duplicate | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

2213. Near duplicate | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

2214. Near duplicate | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

2215. Near duplicate | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

2216. Near duplicate | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

2217. Near duplicate | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

2218. Near duplicate | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

2219. Near duplicate | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

2220. Near duplicate | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

2221. Near duplicate | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

2222. Near duplicate | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

2223. Near duplicate | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

2224. Near duplicate | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

2225. Near duplicate | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

2226. Near duplicate | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

2227. Near duplicate | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

2228. Near duplicate | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

2229. Near duplicate | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

2230. Near duplicate | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

2231. Near duplicate | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

2232. Near duplicate | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

2233. Near duplicate | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

2234. Near duplicate | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

2235. Near duplicate | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

2236. Near duplicate | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

2237. Near duplicate | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

2238. Near duplicate | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

2239. Near duplicate | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

2240. Near duplicate | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

2241. Near duplicate | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

2242. Near duplicate | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

2243. Near duplicate | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

2244. Near duplicate | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

2245. Near duplicate | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

2246. Near duplicate | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

2247. Near duplicate | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

2248. Near duplicate | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

2249. Near duplicate | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

2250. Near duplicate | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

2251. Near duplicate | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

2252. Near duplicate | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

2253. Near duplicate | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

2254. Near duplicate | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

2255. Near duplicate | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

2256. Near duplicate | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

2257. Near duplicate | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

2258. Near duplicate | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

2259. Near duplicate | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

2260. Near duplicate | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

2261. Near duplicate | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

2262. Near duplicate | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2263. Near duplicate | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

2264. Near duplicate | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

2265. Near duplicate | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

2266. Near duplicate | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

2267. Near duplicate | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2268. Near duplicate | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

2269. Near duplicate | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

2270. Near duplicate | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

2271. Near duplicate | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

2272. Near duplicate | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

2273. Near duplicate | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

2274. Near duplicate | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

2275. Near duplicate | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

2276. Near duplicate | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

2277. Near duplicate | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

2278. Near duplicate | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

2279. Near duplicate | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

2280. Near duplicate | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

2281. Near duplicate | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

2282. Near duplicate | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

2283. Near duplicate | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

2284. Near duplicate | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

2285. Near duplicate | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

2286. Near duplicate | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

2287. Near duplicate | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

2288. Near duplicate | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

2289. Near duplicate | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

2290. Near duplicate | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

2291. Near duplicate | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

2292. Near duplicate | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

2293. Near duplicate | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

2294. Near duplicate | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

2295. Near duplicate | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

2296. Near duplicate | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

2297. Near duplicate | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

2298. Near duplicate | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

2299. Near duplicate | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

2300. Near duplicate | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

2301. Near duplicate | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

2302. Near duplicate | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

2303. Near duplicate | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2304. Near duplicate | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

2305. Near duplicate | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

2306. Near duplicate | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

2307. Near duplicate | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

2308. Near duplicate | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

2309. Near duplicate | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

2310. Near duplicate | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

2311. Near duplicate | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

2312. Near duplicate | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

2313. Near duplicate | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

2314. Near duplicate | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

2315. Near duplicate | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

2316. Near duplicate | Candidate sentence 0 | Source sentence 0

Submitted text	Source passage
StreamView: A Self-Hosted Streaming Materialized View Engine	StreamView: A Self-Hosted Streaming Materialized View Engine

2317. Near duplicate | Candidate sentence 3 | Source sentence 3

Submitted text	Source passage
Nagamani Dept.	Nagamani Dept.

2318. Near duplicate | Candidate sentence 4 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2319. Near duplicate | Candidate sentence 10 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2320. Near duplicate | Candidate sentence 12 | Source sentence 12

Submitted text	Source passage
Hyderabad, India.	Hyderabad, India.

2321. Near duplicate | Candidate sentence 14 | Source sentence 14

Submitted text	Source passage
Assis Mohanty	Assis Mohanty

2322. Near duplicate | Candidate sentence 16 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2323. Near duplicate | Candidate sentence 20 | Source sentence 20

Submitted text	Source passage
Shiva Ganesh	Shiva Ganesh

2324. Near duplicate | Candidate sentence 22 | Source sentence 22

Submitted text	Source passage
of Computer Science and Engineering	of Computer Science and Engineering

2325. Near duplicate | Candidate sentence 31 | Source sentence 31

Submitted text	Source passage
Write-ahead logging supports durability and recovery after system failures.	Write-ahead logging supports durability and recovery after system failures.

2326. Near duplicate | Candidate sentence 32 | Source sentence 32

Submitted text	Source passage
Watermarks help manage late-arriving events according to configured policies.	Watermarks help manage late-arriving events according to configured policies.

2327. Near duplicate | Candidate sentence 33 | Source sentence 33

Submitted text	Source passage
Incremental view maintenance updates affected results as new events arrive.	Incremental view maintenance updates affected results as new events arrive.

2328. Near duplicate | Candidate sentence 34 | Source sentence 34

Submitted text	Source passage
This approach aims to avoid repeated computation over historical data.	This approach aims to avoid repeated computation over historical data.

2329. Near duplicate | Candidate sentence 36 | Source sentence 36

Submitted text	Source passage
A web interface can help users configure streams, manage materialized views, and monitor system activity.	A web interface can help users configure streams, manage materialized views, and monitor system activity.

2330. Near duplicate | Candidate sentence 38 | Source sentence 38

Submitted text	Source passage
Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.	Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization.

2331. Near duplicate | Candidate sentence 42 | Source sentence 42

Submitted text	Source passage
Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.	Modern applications continuously generate large volumes of data that require efficient processing and timely analysis.

2332. Near duplicate | Candidate sentence 47 | Source sentence 47

Submitted text	Source passage
Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.	Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation.

2333. Near duplicate | Candidate sentence 51 | Source sentence 51

Submitted text	Source passage
The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.	The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources.

2334. Near duplicate | Candidate sentence 53 | Source sentence 53

Submitted text	Source passage
LITERATURE SURVEY	LITERATURE SURVEY

2335. Near duplicate | Candidate sentence 55 | Source sentence 55

Submitted text	Source passage
Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.	Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay.

2336. Near duplicate | Candidate sentence 58 | Source sentence 58

Submitted text	Source passage
Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.	Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing.

2337. Near duplicate | Candidate sentence 61 | Source sentence 61

Submitted text	Source passage
Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.	Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers.

2338. Near duplicate | Candidate sentence 70 | Source sentence 70

Submitted text	Source passage
Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.	Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state.

2339. Near duplicate | Candidate sentence 73 | Source sentence 73

Submitted text	Source passage
Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.	Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording.

2340. Near duplicate | Candidate sentence 83 | Source sentence 83

Submitted text	Source passage
However, combining semantic search with live messaging can increase server memory usage under heavy workloads.	However, combining semantic search with live messaging can increase server memory usage under heavy workloads.

2341. Near duplicate | Candidate sentence 88 | Source sentence 88

Submitted text	Source passage
Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.	Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration.

2342. Near duplicate | Candidate sentence 91 | Source sentence 91

Submitted text	Source passage
Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.	Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding.

2343. Near duplicate | Candidate sentence 100 | Source sentence 100

Submitted text	Source passage
Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.	Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments.

2344. Near duplicate | Candidate sentence 112 | Source sentence 112

Submitted text	Source passage
Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical	Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical

2345. Near duplicate | Candidate sentence 113 | Source sentence 113

Submitted text	Source passage
documentation can improve cross-platform knowledge discovery.	documentation can improve cross-platform knowledge discovery.

2346. Near duplicate | Candidate sentence 116 | Source sentence 116

Submitted text	Source passage
Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.	Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings.

2347. Near duplicate | Candidate sentence 118 | Source sentence 118

Submitted text	Source passage
GAPS IDENTIFIED	GAPS IDENTIFIED

2348. Near duplicate | Candidate sentence 122 | Source sentence 122

Submitted text	Source passage
Repeatedly recomputing query results can increase processing overhead.	Repeatedly recomputing query results can increase processing overhead.

2349. Near duplicate | Candidate sentence 123 | Source sentence 123

Submitted text	Source passage
Efficient incremental maintenance of materialized views remains an important challenge.	Efficient incremental maintenance of materialized views remains an important challenge.

2350. Near duplicate | Candidate sentence 127 | Source sentence 127

Submitted text	Source passage
Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.	Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations.

2351. Near duplicate | Candidate sentence 129 | Source sentence 129

Submitted text	Source passage
Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.	Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult.

2352. Near duplicate | Candidate sentence 131 | Source sentence 131

Submitted text	Source passage
System failures can interrupt stream processing and affect stored results.	System failures can interrupt stream processing and affect stored results.

2353. Near duplicate | Candidate sentence 134 | Source sentence 134

Submitted text	Source passage
Increasing event volumes can raise CPU, memory, and storage requirements.	Increasing event volumes can raise CPU, memory, and storage requirements.

2354. Near duplicate | Candidate sentence 136 | Source sentence 136

Submitted text	Source passage
Handling High-Volume Event Streams	Handling High-Volume Event Streams

2355. Near duplicate | Candidate sentence 139 | Source sentence 139

Submitted text	Source passage
Between Events and Materialized Views	Between Events and Materialized Views

2356. Near duplicate | Candidate sentence 141 | Source sentence 141

Submitted text	Source passage
Reliable update mechanisms are needed to maintain accurate materialized views.	Reliable update mechanisms are needed to maintain accurate materialized views.

2357. Near duplicate | Candidate sentence 145 | Source sentence 145

Submitted text	Source passage
PROBLEM STATEMENT	PROBLEM STATEMENT

2358. Near duplicate | Candidate sentence 150 | Source sentence 150

Submitted text	Source passage
StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.	StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform.

2359. Near duplicate | Candidate sentence 155 | Source sentence 155

Submitted text	Source passage
Modern applications require continuous data processing to support timely decision-making and real-time analytics.	Modern applications require continuous data processing to support timely decision-making and real-time analytics.

2360. Near duplicate | Candidate sentence 158 | Source sentence 158

Submitted text	Source passage
These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.	These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation.

2361. Near duplicate | Candidate sentence 164 | Source sentence 164

Submitted text	Source passage
RESEARCH OBJECTIVES	RESEARCH OBJECTIVES

2362. Near duplicate | Candidate sentence 171 | Source sentence 171

Submitted text	Source passage
Achieve Incremental Materialized View Maintenance	Achieve Incremental Materialized View Maintenance

2363. Near duplicate | Candidate sentence 172 | Source sentence 172

Submitted text	Source passage
Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.	Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive.

2364. Near duplicate | Candidate sentence 174 | Source sentence 174

Submitted text	Source passage
Improve State Persistence and Recovery	Improve State Persistence and Recovery

2365. Near duplicate | Candidate sentence 175 | Source sentence 175

Submitted text	Source passage
Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.	Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures.

2366. Near duplicate | Candidate sentence 181 | Source sentence 181

Submitted text	Source passage
Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.	Reprocessing entire datasets for simple updates can waste computational resources and increase query execution time.

2367. Near duplicate | Candidate sentence 182 | Source sentence 182

Submitted text	Source passage
These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.	These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently.

2368. Near duplicate | Candidate sentence 184 | Source sentence 184

Submitted text	Source passage
EXSISTING SYSTEM	EXSISTING SYSTEM

2369. Near duplicate | Candidate sentence 185 | Source sentence 185

Submitted text	Source passage
FIG 7.1 Existing System	FIG 7.1 Existing System

2370. Near duplicate | Candidate sentence 188 | Source sentence 188

Submitted text	Source passage
PROPOSED METHODOLOGY	PROPOSED METHODOLOGY

2371. Near duplicate | Candidate sentence 193 | Source sentence 193

Submitted text	Source passage
The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.	The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components.

2372. Near duplicate | Candidate sentence 201 | Source sentence 201

Submitted text	Source passage
The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.	The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization.

2373. Near duplicate | Candidate sentence 206 | Source sentence 206

Submitted text	Source passage
StreamView consumes the incoming events for continuous processing.	StreamView consumes the incoming events for continuous processing.

2374. Near duplicate | Candidate sentence 208 | Source sentence 208

Submitted text	Source passage
Step 5: Create a Materialized View – Define a view based on the required query or aggregation.	Step 5: Create a Materialized View – Define a view based on the required query or aggregation.

2375. Near duplicate | Candidate sentence 214 | Source sentence 214

Submitted text	Source passage
Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.	Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project.

2376. Near duplicate | Candidate sentence 216 | Source sentence 216

Submitted text	Source passage
Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.	Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard.

2377. Near duplicate | Candidate sentence 226 | Source sentence 226

Submitted text	Source passage
Output: Updated Materialized Views \(V'\), Query Results \(R\)	Output: Updated Materialized Views \(V'\), Query Results \(R\)

2378. Near duplicate | Candidate sentence 232 | Source sentence 232

Submitted text	Source passage
For each incoming event \(e\) in \(E\), do:	For each incoming event \(e\) in \(E\), do:

2379. Near duplicate | Candidate sentence 234 | Source sentence 234

Submitted text	Source passage
Receive event \(e\) from the message broker.	Receive event \(e\) from the message broker.

2380. Near duplicate | Candidate sentence 236 | Source sentence 236

Submitted text	Source passage
Validate the event structure and required fields.	Validate the event structure and required fields.

2381. Near duplicate | Candidate sentence 240 | Source sentence 240

Submitted text	Source passage
If the event satisfies the processing conditions, compute the required aggregation or state changes.	If the event satisfies the processing conditions, compute the required aggregation or state changes.

2382. Near duplicate | Candidate sentence 244 | Source sentence 244

Submitted text	Source passage
Incrementally update the affected view results using the new event and existing state.	Incrementally update the affected view results using the new event and existing state.

2383. Near duplicate | Candidate sentence 248 | Source sentence 248

Submitted text	Source passage
Continue processing incoming events while updating the relevant materialized views.	Continue processing incoming events while updating the relevant materialized views.

2384. Near duplicate | Candidate sentence 252 | Source sentence 252

Submitted text	Source passage
Identify the corresponding materialized view and retrieve the required results from the state store.	Identify the corresponding materialized view and retrieve the required results from the state store.

2385. Near duplicate | Candidate sentence 256 | Source sentence 256

Submitted text	Source passage
Monitor processing activity, query latency, event throughput, and resource utilization.	Monitor processing activity, query latency, event throughput, and resource utilization.

2386. Near duplicate | Candidate sentence 259 | Source sentence 259

Submitted text	Source passage
Note: This algorithm describes the intended StreamView workflow.	Note: This algorithm describes the intended StreamView workflow.

2387. Near duplicate | Candidate sentence 261 | Source sentence 261

Submitted text	Source passage
SYSTEM OVERVIEW	SYSTEM OVERVIEW

2388. Near duplicate | Candidate sentence 267 | Source sentence 267

Submitted text	Source passage
Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.	Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring.

2389. Near duplicate | Candidate sentence 268 | Source sentence 268

Submitted text	Source passage
The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.	The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources.

2390. Near duplicate | Candidate sentence 272 | Source sentence 272

Submitted text	Source passage
Its storage and query components work together to make analytical results available through backend APIs.	Its storage and query components work together to make analytical results available through backend APIs.

2391. Near duplicate | Candidate sentence 279 | Source sentence 279

Submitted text	Source passage
SYSTEM ARCHITECTURE	SYSTEM ARCHITECTURE

2392. Near duplicate | Candidate sentence 281 | Source sentence 281

Submitted text	Source passage
The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.	The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity.

2393. Near duplicate | Candidate sentence 285 | Source sentence 285

Submitted text	Source passage
Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.	Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization.

2394. Near duplicate | Candidate sentence 287 | Source sentence 287

Submitted text	Source passage
FIG 11.1 System Architecture	FIG 11.1 System Architecture

2395. Near duplicate | Candidate sentence 289 | Source sentence 289

Submitted text	Source passage
The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.	The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine.

2396. Near duplicate | Candidate sentence 290 | Source sentence 290

Submitted text	Source passage
The dashboard provides an overview of active streams, materialized views, event throughput, and system health.	The dashboard provides an overview of active streams, materialized views, event throughput, and system health.

2397. Near duplicate | Candidate sentence 299 | Source sentence 299

Submitted text	Source passage
Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.	Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values.

2398. Near duplicate | Candidate sentence 302 | Source sentence 302

Submitted text	Source passage
By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.	By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself.

2399. Near duplicate | Candidate sentence 303 | Source sentence 303

Submitted text	Source passage
FUTURE ENHANCEMENTS	FUTURE ENHANCEMENTS

2400. Near duplicate | Candidate sentence 312 | Source sentence 312

Submitted text	Source passage
The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.	The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform.

2401. Near duplicate | Candidate sentence 317 | Source sentence 317

Submitted text	Source passage
By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.	By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads.

2402. Near duplicate | Candidate sentence 319 | Source sentence 319

Submitted text	Source passage
[1] M.	[1] M.

2403. Near duplicate | Candidate sentence 320 | Source sentence 320

Submitted text	Source passage
Fragkoulis, P.	Fragkoulis, P.

2404. Near duplicate | Candidate sentence 321 | Source sentence 321

Submitted text	Source passage
Carbone, V.	Carbone, V.

2405. Near duplicate | Candidate sentence 324 | Source sentence 324

Submitted text	Source passage
33, pp.	33, pp.

2406. Near duplicate | Candidate sentence 327 | Source sentence 327

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-023-00819-8	Available: https://doi.org/10.1007/s00778-023-00819-8

2407. Near duplicate | Candidate sentence 328 | Source sentence 328

Submitted text	Source passage
[2] Y.	[2] Y.

2408. Near duplicate | Candidate sentence 329 | Source sentence 329

Submitted text	Source passage
Yang, L.	Yang, L.

2409. Near duplicate | Candidate sentence 333 | Source sentence 333

Submitted text	Source passage
68, pp.	68, pp.

2410. Near duplicate | Candidate sentence 337 | Source sentence 337

Submitted text	Source passage
[3] M.	[3] M.

2411. Near duplicate | Candidate sentence 338 | Source sentence 338

Submitted text	Source passage
Budiu, T.	Budiu, T.

2412. Near duplicate | Candidate sentence 339 | Source sentence 339

Submitted text	Source passage
Chajed, F.	Chajed, F.

2413. Near duplicate | Candidate sentence 340 | Source sentence 340

Submitted text	Source passage
McSherry, L.	McSherry, L.

2414. Near duplicate | Candidate sentence 343 | Source sentence 343

Submitted text	Source passage
16, no.	16, no.

2415. Near duplicate | Candidate sentence 344 | Source sentence 344

Submitted text	Source passage
7, pp.	7, pp.

2416. Near duplicate | Candidate sentence 347 | Source sentence 347

Submitted text	Source passage
Available: https://doi.org/10.14778/3587136.3587137	Available: https://doi.org/10.14778/3587136.3587137

2417. Near duplicate | Candidate sentence 348 | Source sentence 348

Submitted text	Source passage
[4] A.	[4] A.

2418. Near duplicate | Candidate sentence 352 | Source sentence 352

Submitted text	Source passage
18, no.	18, no.

2419. Near duplicate | Candidate sentence 353 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2420. Near duplicate | Candidate sentence 355 | Source sentence 355

Submitted text	Source passage
[5] A.	[5] A.

2421. Near duplicate | Candidate sentence 356 | Source sentence 356

Submitted text	Source passage
Gupta, I.	Gupta, I.

2422. Near duplicate | Candidate sentence 360 | Source sentence 360

Submitted text	Source passage
Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.	Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol.

2423. Near duplicate | Candidate sentence 361 | Source sentence 361

Submitted text	Source passage
22, no.	22, no.

2424. Near duplicate | Candidate sentence 362 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2425. Near duplicate | Candidate sentence 365 | Source sentence 365

Submitted text	Source passage
Available: https://doi.org/10.1145/170036.170066	Available: https://doi.org/10.1145/170036.170066

2426. Near duplicate | Candidate sentence 366 | Source sentence 366

Submitted text	Source passage
[6] K.	[6] K.

2427. Near duplicate | Candidate sentence 368 | Source sentence 368

Submitted text	Source passage
Ross, D.	Ross, D.

2428. Near duplicate | Candidate sentence 373 | Source sentence 373

Submitted text	Source passage
Management of Data, 1996, pp.	Management of Data, 1996, pp.

2429. Near duplicate | Candidate sentence 374 | Source sentence 374

Submitted text	Source passage
447–458.	447–458.

2430. Near duplicate | Candidate sentence 375 | Source sentence 375

Submitted text	Source passage
[7] S.	[7] S.

2431. Near duplicate | Candidate sentence 377 | Source sentence 377

Submitted text	Source passage
Widom, “Deriving production rules for incremental view maintenance,” in Proc.	Widom, “Deriving production rules for incremental view maintenance,” in Proc.

2432. Near duplicate | Candidate sentence 378 | Source sentence 378

Submitted text	Source passage
17th Int.	17th Int.

2433. Near duplicate | Candidate sentence 380 | Source sentence 380

Submitted text	Source passage
Very Large Data Bases (VLDB), 1991, pp.	Very Large Data Bases (VLDB), 1991, pp.

2434. Near duplicate | Candidate sentence 381 | Source sentence 381

Submitted text	Source passage
577–589.	577–589.

2435. Near duplicate | Candidate sentence 382 | Source sentence 382

Submitted text	Source passage
[8] J.	[8] J.

2436. Near duplicate | Candidate sentence 387 | Source sentence 387

Submitted text	Source passage
Tompa, “Efficiently updating materialized views,” in Proc.	Tompa, “Efficiently updating materialized views,” in Proc.

2437. Near duplicate | Candidate sentence 390 | Source sentence 390

Submitted text	Source passage
Management of Data, 1986, pp.	Management of Data, 1986, pp.

2438. Near duplicate | Candidate sentence 391 | Source sentence 391

Submitted text	Source passage
61–71.	61–71.

2439. Near duplicate | Candidate sentence 392 | Source sentence 392

Submitted text	Source passage
[9] T.	[9] T.

2440. Near duplicate | Candidate sentence 394 | Source sentence 394

Submitted text	Source passage
8, no.	8, no.

2441. Near duplicate | Candidate sentence 395 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

2442. Near duplicate | Candidate sentence 397 | Source sentence 397

Submitted text	Source passage
[10] T.	[10] T.

2443. Near duplicate | Candidate sentence 399 | Source sentence 399

Submitted text	Source passage
39th Int.	39th Int.

2444. Near duplicate | Candidate sentence 402 | Source sentence 402

Submitted text	Source passage
[11] P.	[11] P.

2445. Near duplicate | Candidate sentence 404 | Source sentence 404

Submitted text	Source passage
38, no.	38, no.

2446. Near duplicate | Candidate sentence 405 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

2447. Near duplicate | Candidate sentence 407 | Source sentence 407

Submitted text	Source passage
[12] M.	[12] M.

2448. Near duplicate | Candidate sentence 409 | Source sentence 409

Submitted text	Source passage
ACM Symp.	ACM Symp.

2449. Near duplicate | Candidate sentence 411 | Source sentence 411

Submitted text	Source passage
423–438.	423–438.

2450. Near duplicate | Candidate sentence 412 | Source sentence 412

Submitted text	Source passage
[13] T.	[13] T.

2451. Near duplicate | Candidate sentence 414 | Source sentence 414

Submitted text	Source passage
14, no.	14, no.

2452. Near duplicate | Candidate sentence 415 | Source sentence 415

Submitted text	Source passage
12, pp.	12, pp.

2453. Near duplicate | Candidate sentence 418 | Source sentence 418

Submitted text	Source passage
Available: https://doi.org/10.14778/3476311.3476389	Available: https://doi.org/10.14778/3476311.3476389

2454. Near duplicate | Candidate sentence 419 | Source sentence 419

Submitted text	Source passage
[14] M.	[14] M.

2455. Near duplicate | Candidate sentence 421 | Source sentence 421

Submitted text	Source passage
Shah, J.	Shah, J.

2456. Near duplicate | Candidate sentence 425 | Source sentence 425

Submitted text	Source passage
19th Int.	19th Int.

2457. Near duplicate | Candidate sentence 428 | Source sentence 428

Submitted text	Source passage
[15] S.	[15] S.

2458. Near duplicate | Candidate sentence 432 | Source sentence 432

Submitted text	Source passage
[16] D.	[16] D.

2459. Near duplicate | Candidate sentence 435 | Source sentence 435

Submitted text	Source passage
12, no.	12, no.

2460. Near duplicate | Candidate sentence 436 | Source sentence 436

Submitted text	Source passage
2, pp.	2, pp.

2461. Near duplicate | Candidate sentence 438 | Source sentence 438

Submitted text	Source passage
[17] A.	[17] A.

2462. Near duplicate | Candidate sentence 439 | Source sentence 439

Submitted text	Source passage
Arasu, S.	Arasu, S.

2463. Near duplicate | Candidate sentence 441 | Source sentence 441

Submitted text	Source passage
Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.	Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol.

2464. Near duplicate | Candidate sentence 442 | Source sentence 442

Submitted text	Source passage
15, pp.	15, pp.

2465. Near duplicate | Candidate sentence 444 | Source sentence 444

Submitted text	Source passage
[18] M.	[18] M.

2466. Near duplicate | Candidate sentence 445 | Source sentence 445

Submitted text	Source passage
Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.	Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol.

2467. Near duplicate | Candidate sentence 446 | Source sentence 446

Submitted text	Source passage
34, no.	34, no.

2468. Near duplicate | Candidate sentence 447 | Source sentence 447

Submitted text	Source passage
4, pp.	4, pp.

2469. Near duplicate | Candidate sentence 449 | Source sentence 449

Submitted text	Source passage
[19] M.	[19] M.

2470. Near duplicate | Candidate sentence 450 | Source sentence 450

Submitted text	Source passage
Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.	Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc.

2471. Near duplicate | Candidate sentence 452 | Source sentence 452

Submitted text	Source passage
[20] M.	[20] M.

2472. Near duplicate | Candidate sentence 456 | Source sentence 456

Submitted text	Source passage
Available: https://doi.org/10.1007/s00778-025-00922-y	Available: https://doi.org/10.1007/s00778-025-00922-y

Analysed document

The extracted document text is included for traceability. Highlighted match spans are represented in the detailed matching section above; no unverified text is highlighted.

StreamView: A Self-Hosted Streaming Materialized View Engine Mrs . K . Nagamani Dept. of Computer Science and Engineering MLR Institute of Technology Hyderabad, India. astakalanagamani78@gmail.com P. Srinath Dept. of Computer Science and Engineering MLR Institute of Technology Hyderabad, India. 23r21a05y0@mlrit.ac.in Assis Mohanty Dept. of Computer Science and Engineering MLR Institute of Technology Hyderabad, India. 23r21a05t0@mlrit.ac.in V. Shiva Ganesh Dept. of Computer Science and Engineering MLR Institute of Technology Hyderabad, India. 24r21a0532@mlrit.ac.in ABSTRACT Real-time analytics over continuously generated data requires efficient event processing and timely access to aggregated results. StreamView is a self-hosted streaming materialized-view engine designed to reduce dependence on managed cloud services. It combines message-based event ingestion, stream processing, and persistent state management. The processing pipeline supports filtering, transformation, aggregation, and event-time operations. An LSM-tree-based state store manages persistent processing state. Write-ahead logging supports durability and recovery after system failures. Watermarks help manage late-arriving events according to configured policies. Incremental view maintenance updates affected results as new events arrive. This approach aims to avoid repeated computation over historical data. The architecture is intended to support low-latency analytical queries and continuous data processing. A web interface can help users configure streams, manage materialized views, and monitor system activity. StreamView also provides a self-hosted alternative that gives developers greater control over deployment and resources. Its performance can be evaluated through throughput, query latency, view-update latency, recovery behavior, and resource utilization. The project provides a foundation for studying practical trade-offs in self-hosted real-time analytics infrastructure. Keywords: Stream Processing, Materialized Views, Incremental Computation, Real-Time Analytics, Event Streaming, Message Broker, LSM-Tree, Write-Ahead Logging, Event-Time Processing, Self-Hosted Infrastructure, Query Optimization, Distributed Systems. INTRODUCTION Modern applications continuously generate large volumes of data that require efficient processing and timely analysis. However, traditional batch-processing approaches can delay analytical results, while repeated query execution increases computational overhead. These challenges highlight the need for systems that can process incoming events continuously and maintain updated analytical results with minimal redundant computation. Distributed systems and incremental processing techniques provide opportunities to improve data processing efficiency and query performance [1], [2], [3]. A streaming materialized-view engine can maintain precomputed query results as new events arrive. Unlike conventional approaches that repeatedly process historical data, incremental view maintenance updates only the affected results, reducing unnecessary computation. Event-time processing and watermark mechanisms help manage late-arriving events, while persistent state management supports reliable processing and recovery [4], [5], [6]. This paper proposes StreamView, a self-hosted streaming materialized-view engine integrating event ingestion, continuous stream processing, and incremental view maintenance. The framework combines message-broker-based ingestion, filtering and aggregation operations, persistent state storage, and query interfaces to support real-time analytics. The proposed architecture aims to improve query efficiency and provide greater control over deployment and infrastructure resources. Its effectiveness can be evaluated using event-processing throughput, query latency, view-update latency, recovery behavior, and resource utilization under different workloads. LITERATURE SURVEY 1. Real-Time Stream Processing Systems (2024) – Continuous event processing enables applications to analyze incoming data with minimal delay. However, maintaining consistent analytical results under high event volumes remains a challenge. 2. Incremental Materialized View Maintenance (2023) – Incremental computation updates query results when source data changes, reducing repeated processing. However, complex aggregations and large-scale workloads can increase update overhead. 3. Distributed Stream Processing Architectures (2024) – Distributed processing frameworks improve scalability by dividing workloads across multiple workers. However, coordination, network communication, and fault recovery can introduce additional overhead. 4. Event-Time Processing in Streaming Applications (2023) – Event-time processing and watermarks help systems handle delayed and out-of-order events. However, selecting appropriate watermark policies can affect result completeness and processing latency. 5. Persistent State Management for Stream Processing (2024) – Persistent state stores enable streaming applications to retain intermediate results and recover after failures. However, storage performance and recovery mechanisms can influence overall system efficiency. 6. Write-Ahead Logging for Reliable Data Systems (2023) – Write-ahead logging improves durability by recording changes before applying them to persistent state. However, log management and recovery procedures can increase storage and processing overhead. 7. Embedding Models for Technical Information Retrieval (2023) – Embedding-based search helps developers retrieve relevant documentation even when search terms differ from the original wording. However, continuous index updates can increase memory consumption and computational costs. 8. Third-Party Integrations in Collaboration Platforms (2022) – Integrating services such as GitHub, Jira, and Slack reduces application switching. However, maintaining reliable integrations requires ongoing API updates and webhook management. 9. AI-Based Task Recommendations for Software Teams (2024) – Machine-learning models can recommend task assignments using historical work patterns. However, recommendations may become outdated when priorities and team workloads change. 10. Semantic Search for Developer Communication (2024) – Vector-based retrieval helps developers locate previous discussions and technical decisions. However, combining semantic search with live messaging can increase server memory usage under heavy workloads. 11. Enterprise Knowledge Management Using AI (2024) – AI-assisted knowledge management can improve access to technical information and automate routine workflows. However, integrating documentation with daily development activities remains challenging. 12. Cloud Collaboration for Distributed Engineering Teams (2023) – Cloud platforms simplify deployment and support remote collaboration. However, organizations must address data privacy, access control, and data-residency requirements. 13. Conversational AI for Software Development (2024) – AI chatbots can answer common development questions and support onboarding. However, complex debugging tasks involving multiple codebases may require additional contextual information. 14. Vector Databases for Enterprise Search (2024) – Vector databases support semantic retrieval by matching contextual meaning rather than exact keywords. However, high query volumes can increase memory consumption and affect write performance. 15. Machine Learning for Software Development Workflows (2023) – Machine-learning techniques can assist with bug classification, issue prioritization, and

workflow automation. However, their effectiveness depends on the availability of accurate historical data. 16. Recommendation Systems for Team Productivity (2024) – Historical commits and task records can help identify developers suited to particular assignments. However, limited historical activity can reduce recommendation accuracy for new team members. 17. Large Language Models for Context-Aware Collaboration (2024) – Large language models can summarize meetings and technical discussions. However, generated summaries may contain inaccuracies when project context or code versions are outdated. 18. AI-Enabled Enterprise Collaboration Platforms (2024) – Combining communication, task management, and search in one platform improves information accessibility. However, integrating different permission models and data structures introduces administrative complexity. 19. Semantic Search for Software Engineering Documentation (2024) – Vector search helps developers discover relevant technical documents using natural-language queries. However, retrieving information distributed across issue trackers, pull requests, and discussions remains challenging. 20. Cross-Platform Knowledge Discovery for Engineering Teams (2024) – Indexing pull requests, chat messages, and technical documentation can improve cross-platform knowledge discovery. However, maintaining contextual relevance requires regular indexing updates and efficient management of computational resources. Important: These are rewritten literature-review summaries based on the descriptions you provided, not verified publications. Before submitting your IEEE paper, verify each paper's actual title, authors, publication venue, year, and findings. Do not present unverified titles or numerical claims as published research. GAPS IDENTIFIED Fragmented Data Processing Traditional data-processing systems often rely on separate tools for event ingestion, processing, storage, and querying, making real-time analytics difficult to manage efficiently. Limited Incremental Computation Repeatedly recomputing query results can increase processing overhead. Efficient incremental maintenance of materialized views remains an important challenge. Inefficient Query Execution Traditional query execution may require processing large amounts of historical data, increasing response times for continuously changing datasets. Complex Infrastructure Management Existing streaming platforms may require complex configuration, deployment, and maintenance, creating challenges for developers and smaller organizations. Limited Processing Visibility Monitoring event ingestion, processing performance, materialized views, and query execution across separate tools can make system management more difficult. Challenges in State Recovery System failures can interrupt stream processing and affect stored results. Reliable state persistence and recovery mechanisms are necessary to maintain processing continuity. Resource Utilization and Scalability Increasing event volumes can raise CPU, memory, and storage requirements. Efficient resource utilization and scalable processing are essential for maintaining performance. Handling High-Volume Event Streams Processing a continuously increasing volume of events while maintaining low latency is challenging. Efficient ingestion and processing mechanisms are required to handle high-throughput workloads. Between Events and Materialized Views Delayed, duplicated, or out-of-order events can cause discrepancies in analytical results. Reliable update mechanisms are needed to maintain accurate materialized views. Limited Benchmarking of Self-Hosted Solutions The performance of self-hosted streaming materialized-view engines may vary across workloads and hardware configurations. Systematic evaluation of throughput, query latency, recovery time, and resource consumption is needed to understand their practical limitations. PROBLEM STATEMENT Modern applications continuously generate large volumes of data that require real-time processing and efficient analytical access. Traditional batch-processing systems often introduce delays, while repeatedly executing queries over large datasets increases computational overhead. These limitations make it challenging to obtain up-to-date analytical results as new events arrive. Additionally, managing separate tools for event ingestion, stream processing, state storage, and querying can increase infrastructure complexity and operational costs. StreamView addresses these challenges by providing a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, and incremental view maintenance into a unified platform. It processes incoming events through filtering, transformation, and aggregation operations while updating materialized views as relevant data changes. Instead of recomputing query results over the entire dataset, the system aims to update only the affected results, reducing redundant computation. By providing query interfaces and a monitoring dashboard, StreamView aims to simplify system management, improve analytical query efficiency, and give developers greater control over deployment and resource utilization. MOTIVATION Modern applications require continuous data processing to support timely decision-making and real-time analytics. However, traditional data-processing systems often depend on batch execution or repeated query computation, which can increase response times and resource consumption. As data volumes grow, maintaining up-to-date analytical results efficiently becomes increasingly important. These challenges motivate the development of a unified streaming materialized-view engine that simplifies event processing and reduces redundant computation. StreamView is motivated by the need for a self-hosted platform that combines event ingestion, stream processing, persistent state management, and incremental materialized-view maintenance. By processing incoming events continuously and updating only the affected view results, the system aims to improve query efficiency and reduce unnecessary processing overhead. Its monitoring dashboard and query interfaces provide visibility into stream activity, view updates, and system performance. Furthermore, a self-hosted deployment gives developers greater control over infrastructure, resource allocation, and operational costs. StreamView therefore provides a foundation for exploring efficient, manageable, and cost-conscious real-time analytics without relying entirely on managed cloud services. RESEARCH OBJECTIVES Develop a Self-Hosted Streaming Engine Design and develop StreamView as a self-hosted streaming materialized-view engine that integrates event ingestion, stream processing, persistent state management, and analytical querying. The system aims to simplify real-time data processing through a unified architecture. Implement Continuous Stream Processing Enable continuous ingestion and processing of incoming events through operations such as filtering, transformation, and aggregation. The objective is to process data as it arrives and reduce the delays associated with traditional batch-processing approaches. Achieve Incremental Materialized View Maintenance Develop mechanisms to maintain materialized views by updating affected query results whenever new events arrive. This approach aims to minimize redundant computation and improve the efficiency of analytical queries over continuously changing datasets. Improve State Persistence and Recovery Explore persistent state storage and recovery mechanisms to preserve processing state and restore operations after failures. Where implemented, techniques such as write-ahead logging and event-time watermarks can support durability and the handling of late-arriving events. Traditional data-processing systems often depend on scheduled jobs and repeated queries to generate analytical results. As data volumes increase, these approaches may lead to slower responses, unnecessary computation, and higher resource consumption. Managing separate processing and storage components can also increase system complexity, making continuous data analysis more difficult. Periodic refreshes may prevent users from accessing the latest information immediately. Reprocessing entire datasets for simple updates can waste computational resources

and increase query execution time. These limitations highlight the need for a unified system that processes incoming events continuously and maintains analytical results efficiently. StreamView aims to address these challenges through stream processing and incremental materialized-view maintenance. EXSISTING SYSTEM FIG 7.1 Existing System The modern software team uses all sorts of different tools for its work: chats are in Slack, tasks – in Jira, documentation – in Notion, code reviews – in GitHub. Every tool is great on its own, but they all require a separate set of notifications that clutter up the user's time and confuse the context. PROPOSED METHODOLOGY StreamView is designed as a self-hosted streaming materialized-view engine that integrates continuous event ingestion, stream processing, persistent state management, and analytical querying within a unified platform. Incoming events are delivered through a message broker such as Redpanda, which provides Kafka-compatible event ingestion. The processing engine consumes these events and performs operations such as filtering, transformation, and aggregation to generate continuously updated results . The backend manages stream configurations, materialized-view definitions, query requests, and communication between system components. A persistent state store maintains the data required for processing and view updates. Where implemented, write-ahead logging (WAL) can support recovery after failures, while event-time processing and watermarks can help manage late-arriving events. Materialized views are updated incrementally as new events arrive, reducing the need to repeatedly recompute analytical results over the complete dataset. The web dashboard provides an interface for creating and managing streams, configuring materialized views, executing queries, and monitoring system activity. Backend APIs connect the dashboard with the processing engine and state store. REST APIs support configuration and query operations, while WebSocket connections can provide real-time updates about stream activity and processing status. Authentication, authorization, and organization-level access controls can be incorporated to protect configurations and analytical data. The system can be evaluated using event throughput, query latency, view-update latency, recovery behavior, and resource utilization. The following steps should be taken while using StreamView: Step 1: Access the Dashboard – Open the StreamView web application and access the dashboard to view stream activity, materialized views, query information, and system performance. Step 2: Configure an Event Stream – Create or select a stream and configure its required properties, including the event source, topic, and relevant processing settings. Step 3: Ingest Events – Send events from a producer application to the configured message broker. StreamView consumes the incoming events for continuous processing. Step 4: Configure Stream Processing – Define the required operations, such as filtering, transformation, and aggregation, to process events according to the analytical requirements. Step 5: Create a Materialized View – Define a view based on the required query or aggregation. The engine processes incoming events and maintains the corresponding results. Step 6: Query Materialized Results – Execute queries through the query interface or API to retrieve the latest available results without repeatedly processing the entire underlying dataset. Step 7: Monitor Performance and Processing – Use the dashboard to observe stream activity, view updates, query latency, event throughput, and available resource metrics. Step 8: Manage Persistence and Recovery – Monitor stored processing state and use the recovery mechanisms supported by the implementation to restore operations following a failure. Step 9: Evaluate System Performance – Test the engine under different event volumes and query workloads, recording throughput, query latency, view-update latency, and resource consumption to assess its effectiveness. Note: Keep claims about WAL, watermarks, WebSockets, authentication, and recovery aligned with the features actually implemented in your StreamView project. FIG 8.1 Proposed workflow Diagram Step 10: Real-Time Monitoring – StreamView monitors event ingestion, stream-processing activity, and materialized-view updates through the dashboard. Users can observe system performance and identify processing delays or potential bottlenecks. Step 11: Update Stream and View Configurations – Users can modify stream settings, processing operations, and materialized-view definitions according to changing analytical requirements. The system applies the updated configurations based on its supported configuration and execution mechanisms. Step 12: Performance Analysis and Optimization – Evaluate StreamView under different workloads by measuring event throughput, query latency, view-update latency, CPU usage, and memory consumption. These measurements help identify performance bottlenecks and guide future improvements to processing efficiency and resource utilization. ALGORITHM The following steps describe the processing workflow of the StreamView application. Algorithm 1: Streaming Event Ingestion and Incremental Materialized View Maintenance Pipeline Input: Incoming Events \(E\), Stream Configuration \(C\), Materialized View Definitions \(V\) Output: Updated Materialized Views \(V'\), Query Results \(R\) 1. Initialize the message broker, stream-processing engine, persistent state store, and query interface. 2. Configure the event stream and subscribe to the required message topic. 3. For each incoming event \(e\) in \(E\), do: 4. Receive event \(e\) from the message broker. 5. Validate the event structure and required fields. 6. Apply the configured filtering and transformation operations to \(e\). 7. If the event satisfies the processing conditions, compute the required aggregation or state changes. 8. Identify the materialized views affected by the processed event. 9. Incrementally update the affected view results using the new event and existing state. 10. Persist the updated state and view data using the supported storage mechanism. 11. Continue processing incoming events while updating the relevant materialized views. 12. On receiving a query \(Q\) from the client interface: 13. Identify the corresponding materialized view and retrieve the required results from the state store. 14. Return the query results \(R\) to the client interface. 15. Monitor processing activity, query latency, event throughput, and resource utilization. 16. Repeat the process until the stream is stopped or the system is shut down. Note: This algorithm describes the intended StreamView workflow. Ensure that validation, persistence, incremental updates, and monitoring steps match your actual implementation. SYSTEM OVERVIEW StreamView is a self-hosted streaming materialized-view engine designed to process continuously generated events and provide efficient access to analytical results. The system enables users to configure event streams, define processing operations, create materialized views, and execute queries through a unified web interface. By integrating event ingestion, stream processing, persistent state management, and querying, StreamView aims to simplify real-time analytics while reducing unnecessary computation and dependence on managed cloud services. From an architectural perspective, StreamView uses a message broker such as Redpanda for event ingestion, a stream-processing engine for filtering, transformation, and aggregation, and a persistent state store for maintaining processing data and materialized-view results. The engine updates affected views as new events arrive, allowing queries to retrieve precomputed results rather than repeatedly processing the entire dataset. Backend APIs connect these components with the web dashboard, enabling stream configuration, view management, query execution, and performance monitoring. The system is intended to support real-time analytical workloads while providing greater control over deployment and infrastructure resources. Its effectiveness can be evaluated through event-processing throughput, query latency, view-update latency, recovery behavior, and CPU and memory utilization. This architecture provides a foundation for developing and evaluating a practical, self-hosted solution for continuous data processing and incremental analytics. The

system maintains precomputed query results and updates affected data when new events arrive, reducing the need to repeatedly process the entire dataset. Its storage and query components work together to make analytical results available through backend APIs. Performance can be assessed using event throughput, query latency, view-update latency, and resource utilization. By combining continuous event processing with incremental view maintenance, StreamView aims to improve query efficiency, simplify data management, and provide greater control over deployment and infrastructure costs. StreamView is a self-hosted platform designed to process continuous data streams and maintain updated analytical results. It uses a message broker to receive events and a processing engine to perform filtering, transformation, and aggregation. Materialized views are updated incrementally as new events arrive, reducing repeated computation. The dashboard enables users to manage streams, execute queries, and monitor system performance. SYSTEM ARCHITECTURE The StreamView system architecture consists of five major layers: the client interface, application backend, event ingestion and stream processing, persistent storage, and query and monitoring. The client interface provides a web dashboard for managing streams, configuring materialized views, executing queries, and monitoring system activity. The application backend manages business logic, API requests, stream configurations, and communication between components. The event ingestion layer uses a message broker such as Redpanda to receive continuous data streams, while the processing engine performs filtering, transformation, and aggregation. The persistent storage layer maintains processing state and materialized-view results for efficient retrieval. Finally, the query and monitoring layer enables users to access analytical results and observe system performance through metrics such as event throughput, query latency, view-update latency, and resource utilization. These layers work together to support continuous data processing, incremental materialized-view maintenance, and efficient real-time analytics in a self-hosted environment. FIG 11.1 System Architecture RESULTS The results of the StreamView system demonstrate the major functional components of the proposed streaming materialized-view engine. The dashboard provides an overview of active streams, materialized views, event throughput, and system health. The stream management module enables users to configure and monitor event streams, while the materialized-view module displays view definitions, execution status, and update information. The query interface allows users to retrieve analytical results from materialized views using SQL-like queries. These components work together to provide a unified environment for managing continuous data processing and accessing analytical information. The monitoring and event ingestion modules provide additional visibility into system operations. Incoming events are received through the message broker and processed by the stream-processing engine before the relevant materialized views are updated. The monitoring interface can display metrics such as event throughput, query latency, view-update latency, and CPU and memory utilization. These measurements help evaluate system performance and identify potential processing bottlenecks. Overall, StreamView aims to reduce repeated computation and provide timely analytical results through incremental view maintenance. Actual performance improvements should be established through experiments using measured results from the implemented system rather than illustrative dashboard values. FIG 12.1: Result By using Weaviate's semantic search and Google Gemini, this workspace is able to address the collaborative and operational shortcomings faced by teams when it comes to their project knowledge. By directly integrating with GitHub, Notion and Google Meet, it is able to extract mentions, pull requests, notes, and documentation in a single pane to avoid the chaos of multiple tools that dilutes team focus on the work itself. FUTURE ENHANCEMENTS In the future, StreamView can be enhanced to support distributed stream processing and horizontal scalability to handle increasing event volumes and large-scale analytical workloads. Advanced query optimization techniques and adaptive materialized-view maintenance can be introduced to improve processing efficiency and reduce query latency. Additional features such as exactly-once processing, fault tolerance, automated recovery, and improved handling of late-arriving events can strengthen system reliability. Integration with multiple data sources and message brokers can expand its flexibility, while advanced monitoring, alerting, and resource management can help users identify performance bottlenecks. Furthermore, container orchestration using Docker and Kubernetes can simplify deployment and scaling across different environments. Security enhancements, including role-based access control, encrypted communication, and audit logging, can improve data protection. Finally, machine learning techniques could be explored to predict workloads, optimize resource allocation, and recommend materialized views based on query patterns, making StreamView a more scalable, efficient, and intelligent real-time analytics platform. CONCLUSION The proposed StreamView architecture integrates event ingestion, stream processing, incremental materialized-view maintenance, and analytical querying into a unified platform. Instead of repeatedly processing entire datasets, the system continuously consumes events through a message broker such as Redpanda and updates relevant materialized views as new data arrives. The stream-processing engine performs filtering, transformation, and aggregation, while persistent storage maintains processing state and view results for efficient retrieval. Backend APIs connect these components to a web dashboard, enabling users to configure streams, manage materialized views, execute queries, and monitor system performance. This architecture aims to reduce redundant computation, improve analytical query efficiency, and provide timely access to continuously updated results. By supporting self-hosted deployment and centralized monitoring, StreamView gives developers greater control over infrastructure, resource utilization, and the management of real-time data-processing workloads. REFERENCES [1] M. Fragkoulis, P. Carbone, V. Kalavri, and A. Katsifodimos, “A survey on the evolution of stream processing systems,” The VLDB Journal, vol. 33, pp. 507–541, 2024. [Online]. Available: https://doi.org/10.1007/s00778-023-00819-8 [2] Y. Yang, L. Golab, and M. T. Özsu, “ViewDF: Declarative incremental view maintenance for streaming data,” Information Systems, vol. 68, pp. 55–67, 2017. [Online]. Available: https://doi.org/10.1016/j.is.2017.07.002 [3] M. Budiu, T. Chajed, F. McSherry, L. Ryzhyk, and V. Tannen, “DBSP: Automatic incremental view maintenance for rich query languages,” Proceedings of the VLDB Endowment, vol. 16, no. 7, pp. 1601–1614, 2023. [Online]. Available: https://doi.org/10.14778/3587136.3587137 [4] A. Gupta and I. S. Mumick, “Maintenance of materialized views: Problems, techniques, and applications,” IEEE Data Engineering Bulletin, vol. 18, no. 2, pp. 3–18, 1995. [5] A. Gupta, I. S. Mumick, and V. S. Subrahmanian, “Maintaining views incrementally,” ACM SIGMOD Record, vol. 22, no. 2, pp. 157–166, 1993. [Online]. Available: https://doi.org/10.1145/170036.170066 [6] K. A. Ross, D. Srivastava, and S. Sudarshan, “Materialized view maintenance and integrity constraint checking: Trading space for time,” in Proc. ACM SIGMOD Int. Conf. Management of Data, 1996, pp. 447–458. [7] S. Ceri and J. Widom, “Deriving production rules for incremental view maintenance,” in Proc. 17th Int. Conf. Very Large Data Bases (VLDB), 1991, pp. 577–589. [8] J. A. Blakeley, P.-Å. Larson, and F. W. Tompa, “Efficiently updating materialized views,” in Proc. ACM SIGMOD Int. Conf. Management of Data, 1986, pp. 61–71. [9] T. Akidau et al., “The Dataflow Model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-, vol. 8, no. 12, pp. 1792–1803, 2015. [10] T. Akidau et al., “MillWheel: Fault-tolerant stream processing at internet scale,” in Proc. 39th Int. Conf. Very Large Data Bases (VLDB), 2013. [11] P.

Carbone et al., “Apache Flink: Stream and batch processing in a single engine,” IEEE Data Engineering Bulletin, vol. 38, no. 4, pp. 28–38, 2015. [12] M. Zaharia et al., “Discretized streams: Fault-tolerant streaming computation at scale,” in Proc. ACM Symp. Operating Systems Principles (SOSP), 2013, pp. 423–438. [13] T. Akidau et al., “Watermarks in stream processing systems: Semantics Dataflow,” Proceedings of the VLDB Endowment, vol. 14, no. 12, pp. 3135–3147, 2021. [Online]. Available: https://doi.org/10.14778/3476311.3476389 [14] M. A. Shah, J. M. Hellerstein, and S. Chandrasekaran, “Flux: An adaptive partitioning operator for continuous query systems,” in Proc. 19th Int. Conf. Data Engineering (ICDE), 2003. [15] S. Chandrasekaran et al., “TelegraphCQ: Continuous dataflow processing for an uncertain world,” in Proc. 1st Biennial Conf. Innovative Data Systems Research (CIDR), 2003. [16] D. J. Abadi et al., “Aurora: A new model and architecture for data stream management,” The VLDB Journal, vol. 12, no. 2, pp. 120–139, 2003. [17] A. Arasu, S. Babu, and J. Widom, “The CQL continuous query language: Semantic foundations and query execution,” The VLDB Journal, vol. 15, pp. 121–142, 2006. [18] M. Stonebraker et al., “The 8 requirements of real-time stream processing,” ACM SIGMOD Record, vol. 34, no. 4, pp. 42–47, 2005. [19] M. Zaharia et al., “Discretized streams: An efficient and fault-tolerant model for stream processing on large clusters,” in Proc. 4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2012. [20] M. Budiu et al., “DBSP: Automatic incremental view maintenance for rich query languages,” The VLDB Journal, vol. 34, article 39, 2025. [Online]. Available: https://doi.org/10.1007/s00778-025-00922-y

Integrity observations

Flags are evidence for review, not findings of misconduct.

No evidence-based integrity flags were detected by the configured extraction checks.

Methodology and limitations

Unique candidate sentence positions are counted once. Exact matches count fully; near duplicates contribute 0.6 of a sentence. Overlapping source matches are not double-counted.

Source coverage is limited to the authorized uploaded-document corpus. Internet sources and indexed publications are not searched unless configured. Similarity is not a conclusion of plagiarism, and no-match results do not certify originality.